IEOR-E4709: Data Analysis for Financial Engineering

Instructor: Dr. Agostino Capponi

Instructor: Dr. Agostino Capponi (who am I?)

Textbook: No required textbook. Recommended textbooks are listed in the syllabus. The book by John Rice is a good source for the first part of the course (MLE, hypothesis testing, confidence intervals). Montgomery et al. is a good reference for linear regression. The books by Tsay and Hamilton are good textbooks for the time series analysis part. The lecture notes by John Cochrane on Time series for macroeconomics and Finance are an excellent resource for time series analysis. All these resources are good complements to the material covered in class. However, a thorough knowledge and understanding of my lecture notes would suffice to perform well in the class.

Course will be administered on Courseworks. All teaching material will be posted on Courseworks (keep a close eye on Courseworks daily). The syllabus is also on Courseworks (please read it carefully).

Attendance is mandatory. My lecture notes will be posted on Courseworks but they are a complement, rather than a substitute, of class attendance.

Grade distribution:

- Homework (25%). Will cover class material. Biweekly homework assignments. No late homework accepted. Collaboration allowed, but write your solution independently.
- Midterm Exam (35%). Closed book, closed notes, no collaboration allowed.
- Final Exam (40%). Closed book, closed notes, no collaboration allowed. It will be cumulative and include all material covered since the beginning of the semester.

Final Grade:

Based on total weighted points earned during semester: 89% → A-, 79%→ B-, 69%→ C- (guaranteed). Discretion will be applied and final grade may depend on the overall performance of the class.

Course topics: (see syllabus). Feel free to ask questions in class.

LECTURE 1: Data Analysis for Financial Engineering

- What is Data Analysis for Financial Engineering?

  Study of quantitative methods to make statistical inferences from data, provide low-dimensional representation of data, and analyze time series of data. It develops mathematical and statistical tools to accomplish these tasks with a focus on financial data (stock returns, transaction costs, bond prices, etc…).

  Aims at answering questions such as

  - How to perform point estimation, for instance how to estimate the mean and variance of a stock price? How to perform interval estimation and provide a range of values which contains the average stock return with high probability?
  - Can we estimate the average returns of US hedge funds?
  - How to describe the dynamics of stock volatility over time?
  - How to model the dynamics of a stock price, taking into account its dependence on past prices, and which econometric models can be best used to account for correlation between volatility and asset prices?
  - Can we filter out noise from prices? How to use stochastic filtering techniques to estimate hidden states from noisy observations?

- Course objectives: Develop analytical tools to

  - Understand the pros and cons of different estimators for the mean and variance of the stock price process (maximum likelihood, methods of moments).
  - Test hypotheses such as "the return process of stock prices is Gaussian with zero mean", or "the return price of a stock price comes from a *t*-distribution with *n* degrees of freedom".
  - Use principal component analysis to extract the leading risk factors of interest rate models or stock prices in the S&P500.
  - Analyze financial time series, and use statistical tools to estimate autocorrelation and intertemporal dependence.
  - Find the model which best fits a time series of stock price data, and use it to price options.
  - Construct parsimonious linear models, in which a set of explanatory variables (stock prices, stock volatility, GDP, etc…) is used to predict default risk of an institution.
  - Kalman filtering techniques for estimating hidden states from noisy signals.

- Which is the target audience?

    - Engineers and applied mathematicians who want to develop sophisticated quantitative skills for solving data analysis problems arising in the financial industry.
    - Students interested in pursuing Wall-Street careers in risk management, data science, or quantitative trading
- Ph.D. students interested in research. This class would be foundational.

Lecture 1: Overview

1 Time Series, Returns, and Data Visualization

A sequence of observations taken over time is called a *time series*. The frequency of the data often depends on the specific financial application. For instance, we can have daily data (open or close stock price data), annual data (inflation), quarterly data (unemployment), millisecond data (tick by tick stock prices).

Consider returns from investing in stocks. If you buy a stock at time t at price S_t, and the price of the stock at time $t+1$ is S_{t+1}, then the return on the investment is $\frac{S_{t+1}-S_t}{S_t}$. Table 1 displays the returns on the IBM stock price data from November, 30, 2016 through December 30, 2016. The Excel spreadsheet on Courseworks reports a longer sequence of returns for IBM stock price data. We can also do a scatterplot of the returns using the following set of R instructions:

Table 1: IBM Returns

Date	Returns
12/30/2016	-0.003
12/29/2016	-0.002
12/28/2016	0.008
12/27/2016	-0.002
12/23/2016	0
12/22/2016	0.002
12/21/2016	-0.007
12/20/2016	0.007
12/19/2016	-0.004
12/16/2016	0.013
12/15/2016	-0.006
12/14/2016	0.002
12/13/2016	-0.016
12/12/2016	0.006
12/9/2016	-0.009
12/8/2016	-0.002
12/7/2016	-0.026
12/6/2016	-0.003
12/5/2016	0.004
12/2/2016	-0.012
12/1/2016	0.019
11/30/2016	0.008

```
df = read.table("IBMreturns.csv", header = TRUE, sep = ",");
df$Date ← as.Date(df$Date,"%m/%d/%Y");
ggplot(data = df, aes(Date,Open)) + geom_point() + labs(x="Date",y="Returns",title="IBM Returns")).
```

The resulting output is provided in Figure 1. Similarly, we can plot the time series of Microsoft

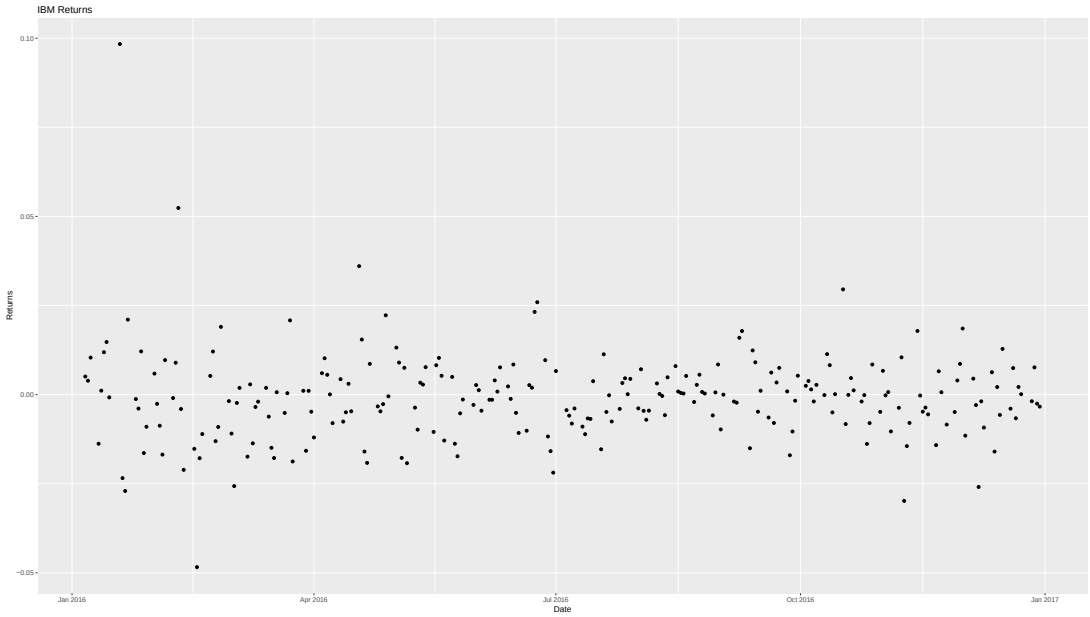


Figure 1: Scatterplot of IBM returns, computed using the time series of returns from January 6, 2016 through December 30, 2016

returns (msdf = read.table("msftreturns.csv", header = TRUE, sep = ",")). We use the same instructions as above for plotting, and plot the data in Figure 2.

An alternative means of representing data are histograms. A frequency histogram for both IBM and Microsoft returns is reported in Figure 3. The corresponding R instructions to plot histograms are given below.

```
hist(df$Open, breaks= 50, main="IBM");
hist(msdf$Open, breaks= 50, main="MSFT").
```

The histograms suggest that both time series seem to be centered around zero.

1.1 Summarizing data with a single numeric variable

In the previous section, we have analyzed data only visually through graphs. Suppose we are now interested in having a numerical summary of the data rather than just a graphical representation.

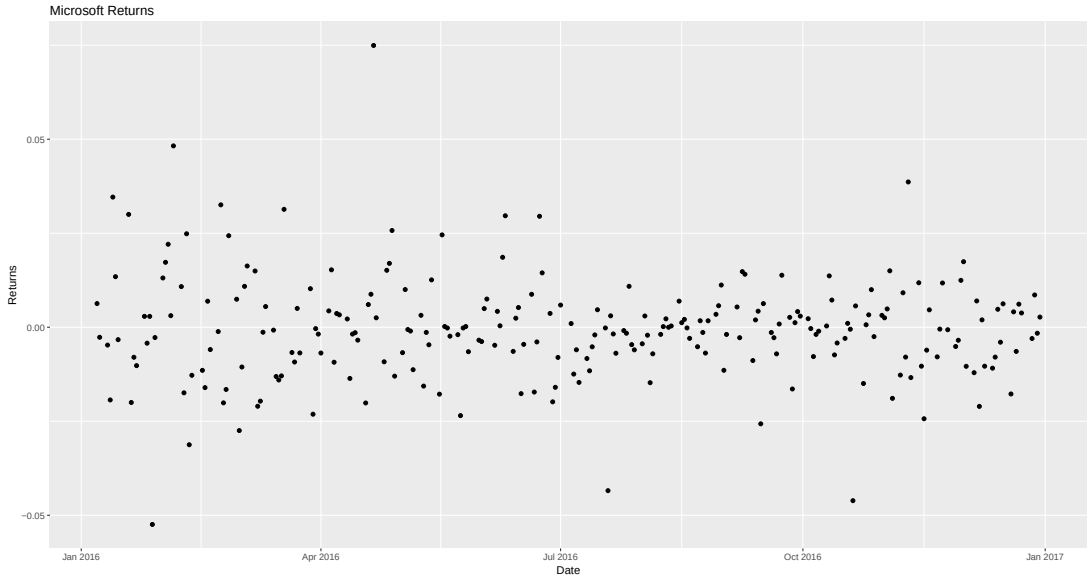


Figure 2: Scatterplot of Microsoft returns, computed using the time series of returns from January 6, 2016 through December 30, 2016

Two important summary statistics of any data set are: (1) the average value, and (2) how spread out the values of this variable are.

1.1.1 Sample Mean

A basic statistics which allows to summarize a time series with a single number is the sample mean. The sample mean of the Microsoft returns is –0.0006082268 (R instruction is mean(msdf$Open)). The sample mean of the IBM returns is -0.0007897998 (R instruction is mean(df$Open)). Despite the difference between these sample means being small, there is still a difference. It was hard to quantify this difference from the plots, because the difference is small compared to the variation in the data.

1.2 The Median

The median is a measure of central tendency. Arrange the data in ascending order. After this ranking, the middle number in the dataset is the median. If there is an even number of data points, then the median is the average of the two middle numbers. For example, consider the series 1,4,7,8,10. Then the median is 7. Then, consider the series 1,4,5,8. Then the median is 4.5.

Unlike the mean, the median is not affected by outliers. Next, let us see briefly why this is the case. Consider the dataset: 1,4,7,8,10. Then the median is 7, and the mean is 6. Consider the dataset 1,4,7,8,100. Then the median is 7, and the mean is 24. It is always good to report the median when there are extreme values in the data. For instance, think about the US household income in 2004. The median was $44,334. Of course, Bill Gates' income was around $50 billions

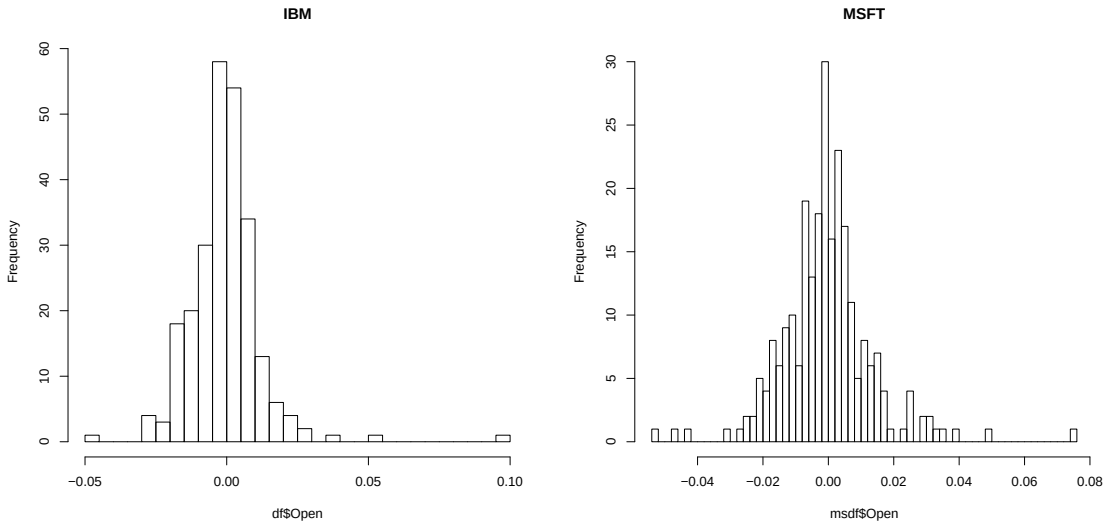


Figure 3: Histogram of IBM and Microsoft returns, computed using the time series of returns from January 6, 2016 through December 30, 2016

USD. So, Bill Gates' income is an outlier. The R command to compute the median of a data vector x is median(x). For instance, the median of MSFT returns, is median(msdf$Open).

1.3 Sample Variance

The sample mean provides a point estimate for the average entry of the vector x. Next, we want a numerical measure of variation or spread. Such a measure can be computed by averaging the differences $x_i - \bar{x}$, capturing the deviation of each entry from the sample mean. We need to be careful, and cannot just sum up these differences because the negative differences would offset the positive differences. To avoid this problem, we can square the differences and thus obtain the measure $\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$. For technical reasons (you will see why in the homework problem), the sample variance of the data is defined to be

$$s_x^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2,$$

i.e. dividing by $n-1$ and not by n. If n is large, there is little difference in dividing by n or by $n-1$. It is also helpful to have a measure of spread which is in the same units as the data. This is given by the sample standard deviation: $s_x = \sqrt{s_x^2}$. The units of the standard deviation are the same as those of the original data.

For our time series of IBM and Microsoft returns, a visual inspection appears to indicate that IBM returns are slightly less spread out than Microsoft returns. Computing the standard deviation for both series of returns (R instructions are sd(msdf$Open, na.rm=T), and sd(df$Open, na.rm=T),

respectively for Microsoft and IBM returns, we obtain respectively 0.01384712 and 0.01269863.[1] Hence, IBM returns are indeed less spread out than Microsoft returns.

1.3.1 Covariance and Correlation

The mean and standard deviation help us summarize numbers which are measurements of just one thing. A fundamentally different question is how one time series relates to another. For instance, how Microsoft returns relate to the DJIA returns? Are the returns on a mutual fund related to market returns? The relationship between two data vectors can be captured by the covariance and correlation. The sample covariance between x and y is

$$s_{xy} := \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x}) (y_i - \bar{y}) .$$

A positive covariance means that when one variable is above its average, the other one tends to be above its own average as well, and viceversa, when one variable is below its average, the other one tends to be below its own average as well. They move up and down together. A negative covariance means that when one moves up, the other tends to move down, i.e. they move in opposite directions. The R command to compute the covariance between Microsoft and IBM returns is cov(msdf$Open, df$Open). If we want to remove missing values, then we can do the following completedata <- na.omit(cbind(msdf$Open, df$Open)), and then cov(completedata)

The sample correlation between x and y is $r_{xy} = \frac{s_{xy}}{s_x s_y}$, so the correlation is just the covariance of the two vectors divided by the product of their standard deviations. Let us state some facts: the closer r_{xy} is to 1, the stronger the linear relationship is with a positive slope. That is, the largest values of y tend to be associated with the largest values of x (and vice-versa). The closer r_{xy} is to -1, the stronger the linear relationship is with a negative slope. That is, large values of y tend to be associated with small values of x (and vice-versa). The command to compute the Pearson correlation is cor(msdf$Open, df$Open).

[1] na.rm = TRUE indicates that potential missing values in the time series need to be ignored in the computation of the statistics, which in this case is the standard deviation.

Lecture 2: Basic Statistics and Review of Probability Results

1 Announcements

- Homework 1 has been posted. It is due on January 31st.

2 What is Data Analysis?

Data analysis is a process which can be split into four stages: (1) Data Collection, (2) Data Cleaning, (3) Data visualization, and (4) Statistical Data Analysis. Our focus will be on the stage (4). We begin by studying statistical techniques used to estimate mean and variance of random variables, to infer the parameter of a model from observations, and to create confidence intervals around our estimates. Before doing so, we recall some notion of convergence of random variables, as well as two fundamental results in probability (law of large numbers and central limit theorem).

2.1 Convergence of Random Variables

This section provides a very basic introduction to the various convergence notions for sequences of random variables. Students who would like to see the proofs of these results and additional examples are referred to `http://www.princeton.edu/~rvan/acm217/ACM217.pdf`, Section 1.5.

Let $(\Omega, \mathbb{F}, \mathbb{P})$ be a probability space. A sequence of random variables $\{X_n\}_{n\geq 1}$ *converges almost surely* to a random variable X in $(\Omega, \mathbb{F}, \mathbb{P})$ if

$$\lim_{n\to\infty} X_n(\omega) = X(\omega),$$

with probability 1, i.e. $\mathbb{P}$-almost surely. We will use the notation $X_n \to X$ to denote this a.s. convergence.

A sequence of random variables $\{X_n\}_{n\geq 1}$ *converges in probability* to a random variable X in $(\Omega, \mathbb{F}, \mathbb{P})$ if, for any $\delta > 0$,

$$\lim_{n\to\infty} \mathbb{P}\left(|X_n - X| > \delta\right) = 0.$$

We will use the notation $X_n \xrightarrow{\mathbb{P}} X$ to denote convergence in probability.

Result 1: Almost sure convergence implies convergence in probability. The converse is not true, however. To convince yourself, consider the sequence $X_n = 1$ with probability $1/n$, and zero otherwise. Suppose that X_n's are independent. Then $\lim_{n\to\infty} \mathbb{P}\left(|X_n| > \delta\right) = \lim_{n\to\infty} \frac{1}{n} = 0$. This means that the sequence $\{X_n\}$ converges to zero in probability. However, because $\sum_{n\geq 1} \mathbb{P}(X_n = 1) = \infty$ and the events $\{X_n\}$ are independent, the second Borel Cantelli lemma guarantees that for any ω and N, there exists $m, n \geq N$ such that $X_n(\omega) = 1$ and $X_m(\omega) = 0$.

Remark 2.1. *The second Borel-Cantelli lemma states: If the events E_n are independent and the sum of the probabilities of the E_n diverges to infinity, then the probability that infinitely many of them occur is 1. Indeed, in our case, set $E_n := \{X_n = 1\}$. Then, $\sum_{n=1}^{\infty} \mathbb{P}(E_n) = \infty$, and hence the event $\{X_n = 1\}$ occurs infinitely often.*

A sequence of random variables $\{X_n\}_{n\geq 1}$ with cumulative probability distribution functions $\{F_n\}_{n\geq 1}$ *converges in distribution* to a random variable X with distribution F if for any bounded continuous function g,

$$\lim_{n\to\infty} \int g(x)dF_n(x) = \int g(x)dF(x)$$

We will use the notation $X_n \xrightarrow{d} X$, or $F_n \to F$ to denote this convergence.

Result 2: Convergence in probability implies convergence in distribution. The converse is not true, however. Consider the random variables X, X_n to be i.i.d. and nonconstant random variables, in which case $X_n \xrightarrow{d} X$. Fix some ϵ. Then, $\mathbb{P}(|X_n - X| \geq \epsilon)$ is positive and the same for all n, showing that X_n does not converge to X in probability.

Theorem 2.2 (Slutsky's theorem). *Suppose $X_n \xrightarrow{d} X$, $Y_n \xrightarrow{\mathbb{P}} c$ for some constant c. Then, $X_n + Y_n \xrightarrow{d} X + c$, and $X_n Y_n \xrightarrow{d} cX$. Furthermore, if $c \neq 0$, then $X_n/Y_n \xrightarrow{d} X/c$.*

3 LLN and Central Limit Theorem

We give two fundamental results in probability, the law of large numbers and the central limit theorem. The law of large numbers characterizes the average behavior of a large sequence of independent identically distributed random variables. The central limit theorem characterizes the fluctuation of these averages.

Strong Law of Large numbers: Suppose $\{X_n\}_{n\geq 1}$ is a sequence of independent and identically distributed random variables such that $\mathbb{E}[X_i] = \mu$, and $\mathbb{E}[|X_i|] < \infty$. Then

$$\frac{1}{n}\sum_{i=1}^{n} X_i \to \mu,$$

i.e., we have almost sure convergence of the sample mean $\frac{1}{n}\sum_{i=1}^{n} X_i$ to its actual mean μ.

Central Limit Theorem: Suppose $\{X_n\}_{n\geq 1}$ is a sequence of independent and identically distributed random variables with mean $\mathbb{E}[X_i] = \mu < \infty$ and variance $Var(X_i) = \sigma^2 < \infty$. Then

$$\frac{\frac{1}{n}\sum_{i=1}^{n} X_i - \mu}{\frac{\sigma}{\sqrt{n}}}$$

converges in distribution to the standard Gaussian distribution (zero mean and unit variance).

4 Estimators and Estimates

The central objective of statistical inference is to make valid generalization from *samples*. The typical setup is as follows: we have a collection of items (subjects, measurements) about which we want to have information. For instance, the *population* can consist of US voters and we are interested in the question: what fraction of the population is in favor of candidate A? Or, the population can consist of US employees and we are interested in the question: what is the average income? In order to answer these questions, we collect items from the large population, also referred to as samples (e.g. collection by sample surveys). The fraction of the population in favor of A, and the average income, are the unknown *parameters*. An estimator is a function of the sample. It is used to estimate the parameter of interest.

Example: The estimator of the average income is the sample mean $\frac{1}{n}\sum_{i=1}^{n} X_i$, where X_i is the income of the ith employee. The sample mean is basically the average of the data.

An important question is: what is the distribution of the statistics? Before samples are taken, they are unknown and random, so the estimator can be viewed as a random variable, and consequently admits a distribution. This is the distribution of the values of the statistics in all possible samples of a *fixed size* from the same population.

4.1 Consistency and Unbiasedness

An estimator $\hat{\theta} := f(X_1, X_2, \ldots, X_n)$ of a parameter θ is *unbiased* if $\mathbb{E}\left[f(X_1, \ldots, X_n)\right] = \theta$. The *bias* of an estimator is $Bias(\hat{\theta}) := \mathbb{E}\left[f(X_1, \ldots, X_n)\right] - \theta$. An estimator $f(X_1, X_2, \ldots, X_n)$ of a parameter θ is *consistent* if it converges to θ in probability. For instance, the sample mean $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ is an unbiased and consistent estimator of the population mean. Indeed, let X_i, $i \geq 0$, be i.i.d. random variables with mean μ and variance σ^2. Then $\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \mu$, i.e. it is unbiased. To prove it is consistent, recall that almost sure convergence implies convergence in probability. The strong law of large numbers implies what is referred to as the *weak law* of large numbers, i.e. $\bar{X} := \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{\mathbb{P}} \mu$, i.e. that the sample mean is a consistent estimator of the population mean μ.

We can also ask other questions: for instance, what is the variance of $\bar{X}$? It is clear that $Var\left(\bar{X}\right) = \frac{1}{n^2}\sum_{i=1}^{n} Var(X_i) = \frac{\sigma^2}{n}$, where we have assumed that X_i's are i.i.d.

Consider now the *sample variance*, defined by

$$s_X^2 := \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2,$$

where $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$. This is a consistent and unbiased estimator of σ^2. (homework problem).

Lecture 3: Mean-Variance Tradeoff, Method of Moment Estimator

IEOR E4709: Statistic Analysis and Time Series
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1 Bias-Variance Tradeoff

Let $\hat{\theta}$ be an estimator of the (unknown) parameter θ. The *mean squared error* of the estimator $\hat{\theta}$ is

$$MSE(\hat{\theta}) := \mathbb{E}\left[\left(\hat{\theta}-\theta\right)^2\right].$$

Result: $\mathbb{E}\left[\left(\hat{\theta}-\theta\right)^2\right] = Var\left(\hat{\theta}\right) + \left(Bias(\hat{\theta})\right)^2$. To see this, notice that

$$\begin{aligned}
\mathbb{E}\left[\left(\hat{\theta}-\theta\right)^2\right] &= \mathbb{E}\left[\left(\hat{\theta}-\mathbb{E}\left[\hat{\theta}\right]+\mathbb{E}\left[\hat{\theta}\right]-\theta\right)^2\right] \\
&= Var\left(\hat{\theta}\right) + (Bias(\hat{\theta}))^2 + 2\mathbb{E}\left[\left(\hat{\theta}-\mathbb{E}\left[\hat{\theta}\right]\right)\left(\mathbb{E}\left[\hat{\theta}\right]-\theta\right)\right] \\
&= Var\left(\hat{\theta}\right) + (Bias(\hat{\theta}))^2
\end{aligned}$$

Since the MSE decomposes into a sum of the bias and variance of the estimator, both quantities need to be as small as possible to achieve a good estimation performance. Typically, one trades off some increase in bias for a decrease in variance, and vice-versa. Assume we have a sequence of i.i.d.. random variables $X_1, X_2, \ldots, X_n$, where each X_i has mean μ and variance σ^2. Why do we estimate the population mean using the sample mean $\bar{X}$ as opposed using a single random variable X_1? Because there is less variability in averages than in individual observations. Indeed, both $\bar{X}$ and X_1 are unbiased estimators of the population mean μ. However, $MSE(\bar{X}) = \frac{\sigma^2}{n} \leq \sigma^2 := MSE(X_1)$.

Let us next provide an example to show that different unbiased estimators may have different mean-squared errors.

Example: Consider the uniform distribution on the interval $(0, 1)$. The sample mean estimator is $\bar{U} = \frac{1}{n}\sum_{i=1}^{n} U_i$. We know from our previous arguments that this is an unbiased estimator of the

mean $\frac{1}{2}$. The variance of $\bar{U}$ is $Var(\bar{U}) = \frac{1}{n}Var(U_1)$. The variance of the uniform distribution in the interval (a, b) is $\frac{(b-a)^3}{12}$. So, in our specific case, it is 1/12. Hence, $MSE(\bar{U}) = \frac{1}{12n}$.

Consider now the estimator $\hat{\mu} := \frac{n+1}{2n} \max_{1 \leq i \leq n} U_i$. Notice that the maximum of n uniform random variables $U_1, U_2, \ldots, U_n$ has mean given by $\frac{n}{n+1}$ and standard deviation $\frac{1}{n+1}\sqrt{n/(n+2)}$.[1] Then $\mathbb{E}[\hat{\mu}] = \frac{n+1}{2n}\frac{n}{n+1} = \frac{1}{2}$. So, $\hat{\mu}$ is also an unbiased estimator of the mean of the uniform distribution. We have that $MSE(\hat{\mu}) = \frac{(n+1)^2}{4n^2}\frac{1}{(n+1)^2}\frac{n}{n+2} = \frac{1}{4n(n+2)}$. Comparing the MSE of $\bar{U}$ and of $\hat{\mu}$, we have that $MSE(\hat{\mu}) \leq MSE(\bar{U})$ if and only if $n > 1$. Hence, the nonlinear estimator $\hat{\mu}$ has a smaller mean squared error than the sample mean estimator $\bar{U}$. It is, however, well known that the sample mean estimator has the smallest MSE among all unbiased *linear estimators*.

2 Method of Moments

The method of moments is a parameter estimation technique which is only based on the law of large numbers. Let us first give a general description of the technique, and then provide an example. Consider independent and identically distributed random variables $X_1, X_2, \ldots$ chosen according to a parametric probability distribution depending on the parameter θ. Let f be a real valued function. If $k(\theta) = \mathbb{E}_\theta[f(X_1)]$, where $\mathbb{E}_\theta$ indicates that the expectation is being taken with respect to the probability distribution parameterized by θ, then

$$\frac{1}{n}\sum_{i=1}^{n} f(X_i) \to k(\theta), \tag{1}$$

with probability one, as $n \to \infty$, by the strong law of large numbers. The method of moments amounts to equating the population moment to the sample moment estimates. Concretely, it goes as follows. The estimation procedure consists of four steps. Denote by $\mu_m := \mathbb{E}[X^m] = k_m(\theta)$ the m-th moment. Suppose that the model has d parameters, i.e. $\theta = (\theta_1, \theta_2, \ldots, \theta_d)$.

- **Step 1.** Obtain the expressions $k_m(\theta)$ for the first d moments, i.e.,

  $$\mu_1 = k_1(\theta_1, \theta_2, \ldots, \theta_d), \qquad \mu_2 = k_2(\theta_1, \theta_2, \ldots, \theta_d), \qquad \ldots \qquad \mu_d = k_d(\theta_1, \theta_2, \ldots, \theta_d),$$

  yielding d equations in d unknowns (the unknowns being the parameters $\theta_1, \theta_2, \ldots, \theta_d$).

- **Step 2.** We solve for the d parameters as a function of the moments, i.e.,

  $$\theta_1 = g_1(\mu_1, \ldots, \mu_d), \qquad \theta_2 = g_2(\mu_1, \ldots, \mu_d), \qquad \ldots, \qquad \theta_d = g_d(\mu_1, \ldots, \mu_d) \tag{2}$$

- **Step 3.** Based on the samples $X := (X_1, X_2, \ldots, X_n)$, compute the first d sample moments

  $$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad \overline{X^2} = \frac{1}{n}\sum_{i=1}^{n} X_i^2, \qquad \ldots \qquad \overline{X^d} = \frac{1}{n}\sum_{i=1}^{n} X_i^d.$$

  By the strong law of large numbers, for each moment, $m = 1, \ldots, d$, we have that $\mu_m \approx \overline{X^m}$.

[1] $\mathbb{P}(\max_{i=1,\ldots,n} U_i <= x) = \mathbb{P}(U_1 \leq x, U_2 \leq x, \ldots U_n \leq x) = x^n$. Hence, the probability density is $f(x) = \frac{\partial \mathbb{P}(\max_{i=1,\ldots,n} U_i <= x)}{\partial x} = nx^{n-1}$. The mean is then equal to $\int_0^1 nx^{n-1}xdx = \frac{n}{n+1}$. For the standard deviation, verify it yourself with a direct calculation.

- **Step 4.** Replace the distributional moments μ_m by the sample moments $\overline{X^m}$. Then the equations in (2) give us formulas for the estimators $(\hat{\theta}_1, \ldots, \hat{\theta}_d)$. Calculating these estimators on the samples X, we obtain the estimates

$$\hat{\theta}_1(X) = g_1(\bar{X}, \overline{X^2}, \ldots, \overline{X^d}), \hat{\theta}_2(X) = g_2(\bar{X}, \overline{X^2}, \ldots, \overline{X^d}), \ldots\ldots, \hat{\theta}_d(X) = g_d(\bar{X}, \overline{X^2}, \ldots, \overline{X^d}).$$

Example: Let $X_1, X_2, \ldots, X_n$ be a random sample of exponential distributed random variables with density

$$f_\beta(x) = \beta e^{-\beta x}, \qquad x > 0.$$

The cumulative distribution function is

$$F(x) = 1 - e^{-\beta x}, \qquad x > 0.$$

How many parameters do we need to estimate, i.e., what is d? In this case, $d = 1$, as we only need to estimate β. Hence, in the first step of the above procedure, we compute $\mu = k_1(\beta) = \frac{1}{\beta}$. In step 2, we solve for β as a function of μ. This yields $\beta = g(\mu) = \frac{1}{\mu}$. In step 4, we obtain $\hat{\beta} = \frac{1}{\bar{X}}$, i.e., replacing the (unknown) population mean with its sample mean.

Maximum Likelihood Estimator (MLE)

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1 Beyond MM Estimators

Consider the previous example of the exponential distributed random variable, which we have seen last time. We could have used two moments, instead of only one, to estimate the parameter β. More generally, we could use only d moments to estimate a d dimensional parameter vector, but by dismissing the $q - d > 0$ additional moments, we would loose information. The remedy for this situation was introduced in the econometrics literature by Hansen (1982) in his famous article, and it is called the *Generalized Method of Moments* (GMM). The idea behind GMM estimation is that even though it is impossible to solve q system of equations provided by the sample moment conditions to estimate $d < q$ parameters, we can still find an estimate of the parameter vector θ that brings the sample moments as close as possible to the population moments. We estimate the parameter vector by minimizing the sum of squares of the differences between the population moments and the sample moments. This yields the minimum variance estimator in the class of estimators that use these moment conditions.

2 Maximum Likelihood Estimator

The MLE estimator is a point estimator, like the method of moments. It was first proposed by the geneticist/statistician Ronald A. Fisher in 1922. The basic idea is to find the values of the parameters which are more plausible to produce the data that we observe. The method assumes that *we know* the structural form of the probability density function of the samples, but *we do not know* the parameters of this density function.

Definition 2.1. *Let $f(x_1, x_2, \ldots, x_n; \theta)$, where $\theta \in \Theta \subseteq \mathbb{R}^k$, be the joint probability density function of n random variables $X_1, \ldots, X_n$ with sample values $x_1, \ldots, x_n$. The* ***likelihood function*** *of the sample is given by*

$$L(\theta | x_1, x_2, \ldots, x_n) = f(x_1, x_2, \ldots, x_n; \theta).$$

Hence, the likelihood of $\theta = 1$ is the probability of observing $X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n$ when $\theta = 1$. More generally, the likelihood of θ given an observed sample $x = (x_1, \ldots, x_n)$, is the probability of a random sample taking value x under the parameter θ. If we fix θ, the likelihood is the probability density function of the random variable X. The maximum likelihood estimator

(MLE) of θ is defined as the maximizer of the likelihood, over θ.

Example 1: Consider the estimation of the mean μ and variance σ^2 of a Gaussian distribution from i.i.d. sample observations $x_1, x_2, \ldots, x_n$. The parameter vector is $\theta := (\mu, \sigma)$. The likelihood function is

$$L(\theta|x) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}}$$

To find the MLE estimates of μ and σ, we need to maximize $L(x\big|\theta)$ over μ and σ. This is equivalent to maximizing the log-likelihood function, given by

$$\log\left(L(\theta|x)\right) = -n\log\left(\sqrt{2\pi}\right) - \frac{n}{2}\log\left(\sigma^2\right) - \frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2\sigma^2}$$

We can then find the values of μ and σ that maximize the log likelihood by taking first-order derivatives and setting them equal to zero. By doing so, we obtain

$$\partial_\mu \sum_{i=1}^{n}(x_i-\mu)^2 = 0 \leftrightarrow \hat{\mu}_{MLE} = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{1}$$

$$-\frac{n}{2}\partial_{\sigma^2}\log(\sigma^2) - \frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2}\partial_{\sigma^2}\frac{1}{\sigma^2} = 0 \leftrightarrow \hat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2 \tag{2}$$

Notice that while the MLE estimate of the mean corresponds to the sample mean, the MLE estimate of the variance differs from the sample variance estimator. Indeed, in the MLE estimate of the variance we divide by n, as opposed to $n-1$. This also means that the MLE produces a biased estimate of the variance. Obviously, this matters when n is small. If n is sufficiently large, then it matters much less whether we divide by n or $n-1$, and both estimators are asymptotically equivalent.

Example 2: Let $X_1, X_2, \ldots, X_n$ be a random sample from the uniform distribution on the interval $(0, \theta)$, where $\theta > 0$. We want to find the MLE of θ. Notice that the probability density function of the uniform is

$$f(x) = \begin{cases} 1/\theta & \text{if } 0 \le x \le \theta \\ 0 & \text{otherwise} \end{cases}.$$

Hence, the likelihood function is given by

$$L(\theta|x) = \begin{cases} 1/\theta^n & \text{if } 0 \le x_1, x_2, \ldots, x_n \le \theta \\ 0 & \text{otherwise} \end{cases}.$$

Let us sort the data from smallest to largest, i.e. let $x_{(1)} \le x_{(2)} \cdots \le x_{(n)}$ be the order statistics. Then, taking the derivative of the log-likelihood function with respect to θ (recall that maximizing the log-likelihood is equivalent to maximizing the likelihood function), we obtain

$$\partial_\theta \log L(\theta|x) = -\frac{n}{\theta} < 0.$$

Hence, the log-likelihood function (and thus the likelihood function) is a decreasing function of θ. Consequently, it is maximized at $\hat{\theta}_{MLE} = x_{(n)}$.

In general, the procedure to find MLE consists of the following steps:

- Define the likelihood function $L(\theta|x)$
- Take the natural logarithm of $L(\theta|x)$ (often easier to work with the log as it transforms a product into a sum, which is easier to differentiate)
- When possible, differentiate $\log L(\theta|x)$ and then equate the derivative to zero
- Solve for the parameter θ, and obtain $\hat{\theta}_{MLE}$
- Check that this is a global maximizer, by verifying that the log-likelihood function is concave in θ.

Things may be more difficult. For instance, a closed form expression for $\hat{\theta}_{MLE}$ may not be obtained by analytically differentiating the log-likelihood function, or there may exist multiple local optima, and it may be hard to compare them. In this case, one needs to resort to numerical methods to find the maximum likelihood estimates. These methods include Newton's method, quasi-Newton method, direct search method, etc... They can be implemented by the R functions optimize(), optim().

Pro and Cons of the MLE: The pros are the following. It uses information about the statistical distribution which generates the data. We will discuss next that when the sample size n is large enough, it is unbiased, consistent, normally distributed, and efficient. Efficiency means that it produces the minimum MSE among all methods, including the methods of moments. The cons are the following. MLE can be highly biased for small samples. In many cases, the MLE has no closed-form solution. In this case, numerical methods need to be employed. These methods are sensitive to the starting values in the optimization, and might not give a global optimum. This problem is especially common when θ is high dimensional.

Lecture 5: Fisher Information and Interval Estimators

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1 Fisher Information

Let $L(\theta|x)$ denote the likelihood function parameterized by θ. The *Fisher information* associated with the model is the matrix given by

$$I(\theta) := \mathbb{E}_\theta \left[\nabla_\theta \log L(\theta|X) \nabla_\theta \log L(\theta|X)^\top \right], \tag{1}$$

where $\nabla_\theta f(\theta) = (f_{\theta_1}(\theta), f_{\theta_2}(\theta), \ldots, f_{\theta_d}(\theta))$ is the (column vector) gradient of f with respect to θ. Moreover, $\mathbb{E}_\theta$ denotes the expectation taken with respect to the probability density function $L(\theta|x)$ (fixing θ and viewed as a function of x, the likelihood function is a probability density function). Intuitively, the Fisher information captures the variability of the gradient of the log-likelihood function (i.e., of $\nabla_\theta \log L(\theta|X)$). The function $l(\theta|x) := \nabla_\theta \log L(\theta|X)$ is also called the *score function*.

In a family of distributions for which the score function has high variability, we intuitively expect the estimation of the parameter θ to be easier. This is because different θ's will produce changes in value for $l(\theta|x)$.

Let us make a few useful observations. First, we notice that the score function has mean zero under $\mathbb{P}_\theta$: we have

$$\begin{aligned} \mathbb{E}_\theta \left[l(\theta|X) \right] &= \int L(\theta|x) l(\theta|x) dx = \int \frac{\nabla_\theta L(\theta|x)}{L(\theta|x)} L(\theta|x) dx \\ &= \int \nabla_\theta L(\theta|x) dx = \nabla_\theta \int L(\theta|x) dx = \nabla_\theta 1 = 0, \end{aligned}$$

where we have assumed that the integral and differential operator can be exchanged (dominated convergence theorem applies).

Using this result, we can provide an alternative representation of the Fisher information matrix. Indeed, we know that

$$0 = \mathbb{E}_\theta \left[l(\theta|X) \right] = \int l(\theta|x) L(\theta|x) dx \tag{2}$$

Differentiating with respect to θ on both sides of equation (2) gives (recall that θ is thought of as

a column vector)

$$\begin{aligned} 0 &= \int \frac{\partial^2 \log L(\theta|x)}{\partial\theta\partial\theta^T} L(\theta\big|x)dx + \int l(\theta|x)\left(\nabla_\theta L(\theta|x)\right)^\top dx \\ &= \mathbb{E}_\theta\left[\frac{\partial^2 \log L(\theta|X)}{\partial\theta\partial\theta^\top}\right] + \mathbb{E}_\theta\left[l(\theta|X)l(\theta|X)^\top\right], \end{aligned} \tag{3}$$

where the (j,k)-th entry of $\left[\frac{\partial^2 \log L(\theta|x)}{\partial\theta\partial\theta^T}\right]$ is $\frac{\partial^2 \log L(\theta|x)}{\partial\theta_j\partial\theta_k}$. The second equality above follows from the fact that

$$\begin{aligned} \mathbb{E}_\theta\left[l(\theta|X)l(\theta|X)^\top\right] &= \int l(\theta|x)\nabla_\theta \log L(\theta|x)^\top L(\theta|x)dx \\ &= \int l(\theta|x)\frac{1}{L(\theta|x)}\nabla_\theta L(\theta|x)^\top L(\theta|x)dx \\ &= \int l(\theta|x)\nabla_\theta L(\theta|x)^\top dx \end{aligned}$$

Recall that $\mathbb{E}_\theta\left[l(\theta|X)\right] = 0$. Hence,

$$Var_\theta\left(l(\theta|X)\right) = \mathbb{E}_\theta\left[l(\theta|X)l(\theta|X)^\top\right] - \mathbb{E}_\theta\left[l(\theta|X)\right]\mathbb{E}_\theta\left[l(\theta|X)^\top\right] = \mathbb{E}_\theta\left[l(\theta|X)l(\theta|X)^\top\right]$$

Consequently, recalling the definition of $I(\theta)$ in Eq. (1), it follows from Eq. (3) that

$$I(\theta) = -\mathbb{E}_\theta\left[\frac{\partial^2 \log L(\theta|X)}{\partial\theta\partial\theta^\top}\right] = Var_\theta(l(\theta|X))$$

Hence, the (j,k)-th entry is $\left[I(\theta)\right]_{j,k} = -\mathbb{E}\left[\frac{\partial^2 \log L(\theta|x)}{\partial\theta_j\partial\theta_k}\right]$.

Remark 1.1. *The alternative representation of the Fisher Information matrix given above relies heavily on the fact that the score function has mean zero, which in turn relies crucially on the exchanges of differentiation and expectation rigorous. For this to hold, we must have some conditions on the probability density functions we consider. One simple condition sufficient to make this work is provided by the Lebesgue's dominated convergence theorem. Let $f : X \times \Theta \to \mathbb{R}$ be a differentiable function. For a fixed base measure μ, assume there exists a function g such that $g(x) \geq ||\nabla_\theta f(x,\theta)||$ for all θ, where $\int g(x)d\mu(x) < \infty$. Then, in this case, we have $\nabla_\theta \int f(x,\theta)d\mu(x) = \int \nabla_\theta f(x,\theta)d\mu(x)$ by the mean-value theorem and the definition of a derivative.*

The above representation has the intuitive property that if we have n i.i.d observations coming from the model $L(\theta|x)$, then the information content grows linearly because $\log L(\theta\big|(X_1, X_2, \ldots, X_n)) = \sum_{i=1}^n \log L(\theta|X_i)$.

Example: We provide an example to show the close connection of Fisher information to the level of estimation accuracy. Suppose we want to estimate the mean of a Bernolli(p) random variable, where $p \in [0,1]$, from a sample of i.i.d. Bernoulli random variables $X_1, X_2, \ldots, X_n$. First, notice that $\mathbb{P}(X_i = x) = p^x(1-p)^{1-x}$, where $x \in \{0,1\}$. Then $\nabla_p \log(\mathbb{P}(X_i = x)) = \frac{x}{p} - \frac{1-x}{1-p}$. Therefore, the Fisher information associated with an individual observation X_i is given by

$$I(p) = \mathbb{E}_p\left[(\nabla_p \log L(p\big|X_i))^2\right] = \mathbb{E}_p\left[\left(\frac{X_i}{p} - \frac{1-X_i}{1-p}\right)^2\right] = \mathbb{E}_p\left[\frac{(X_i-p)^2}{(p(1-p))^2}\right] = \frac{1}{p(1-p)}.$$

The sample mean $\bar{p} = \frac{1}{n}\sum_{i=1}^{n} X_i$ is an unbiased estimator, whose mean squared error is given by

$$\mathbb{E}\left[(\bar{p}-p)^2\right] = \frac{1}{n^2}\sum_{i=1}^{n} Var(X_i) = \frac{1}{n}p(1-p) = \frac{1}{nI(p)},$$

where $I(p)$ is the Fisher information for a single Bernoulli(p) observation (the denominator $nI(p)$ is the Fisher information associated with the data set $x_1, x_2, \ldots, x_n$). It also states that the more informative an observation is (higher $I(p)$), and the more the variance of an unbiased estimator is reduced. This is an instance of a more general result: the Fisher information provides a lower bound on the mean squared error of any unbiased estimator, i.e., the variance of any unbiased estimator cannot be smaller than the inverse of the Fisher Information matrix.

2 Cramer-Rao Lower Bound

In this section and going forward, we will use $I(\theta)$ to denote the Fisher information of the entire data set of observations associated with the parameter θ.

Proposition 2.1 (**Cramer-Rao lower bound).** *Suppose θ is a d-dimensional vector, and let $T(X)$ be an unbiased estimator of θ, i.e., $\mathbb{E}[T(X)] = \theta$. Then*

$$Cov(T(X)) \geq I(\theta)^{-1},$$

where a matrix $A \geq B$ means that $A - B$ is positive semidefinite, i.e. the smallest eigenvalue of this matrix is nonnegative. If θ is a scalar, then it simplifies to $Var(T(X)) \geq \frac{1}{I(\theta)}$

Above, the (j,k)-th entry of the covariance matrix is $[Cov(T(X))]_{jk} = Cov((T(X))_j, (T(X))_k)$.

Example: Let us compute the variance-covariance matrix of the sample mean estimator based on a sequence of n i.i.d. Gaussian random variables $X_1, X_2, \ldots, X_n$. We want to verify whether it attains the Cramer-Rao lower bound. Recall that the sample mean $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ is an unbiased estimator of μ, where μ is the mean of the Gaussian random variable X_i. We also know that $Var(\bar{X}) = \frac{\sigma^2}{n}$. Recall that the log-likelihood function of a Gaussian is given by

$$\log L(\mu|x) = -n\log\left(\sqrt{2\pi}\right) - \frac{n}{2}\log\left(\sigma^2\right) - \frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2\sigma^2}$$

Hence,

$$I(\mu) = -\mathbb{E}_\mu\left[\frac{\partial^2 \log L(\mu|X)}{\partial^2 \mu}\right] = \frac{n}{\sigma^2},$$

where $X = (X_1, X_2, \ldots, X_n)$. Hence, $Var(\bar{X}) = \frac{1}{I(\mu)}$. This means that the sample mean is the best estimator of μ, i.e., among all those unbiased estimator of μ it is one that achieves the CRLB.

Lecture 6: Confidence Intervals

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1 Properties of MLE Estimator:

Let $\hat{\theta}_{MLE}$ be the maximum likelihood estimator (MLE) of θ. The following properties hold:

- **Consistency.** $\hat{\theta}_{MLE}$ converges to θ in probability.
- **Asymptotic Normality.** $I(\theta)^{1/2}\left(\hat{\theta}_{MLE}-\theta\right)$ converges in distribution to a standard Gaussian vector, or equivalently, $\hat{\theta}_{MLE}$ follows a Gaussian distribution with mean θ and covariance matrix $I(\theta)^{-1}$. (A matrix B is said to be the square root of A if $BB=A$). Here, $I(\theta)$ is the Fisher information matrix computed using the entire set of observations $X_1, X_2, \ldots, X_n$.
- **Asymptotic Efficiency.** An estimator is said to be efficient if its variance equals the Cramer-Rao lower bound $I(\theta)^{-1}$. Since the MLE estimator $\hat{\theta}_{MLE}$ achieves this lower bound asymptotically (i.e., as the number of observations goes to infinity), it is said asymptotically efficient.
- **Invariance.** If $\hat{\theta}_{MLE}$ is the MLE of θ and g is a function, then $g(\hat{\theta}_{MLE})$ is the MLE of $g(\theta)$. For the proof, see the book "Statistical Inference" by Casella and Berger.

2 Interval Estimation

We have so far discussed about point estimation. In this section, we discuss *interval estimation*, i.e., a range of values computed from data which is predicted to contain the population parameter with a specified probability.

An *interval estimator* of a parameter θ is a pair of estimators $L(X), U(X)$, such that $L(X) \leq U(X)$. Consequently, $[L(X), U(X)]$ is an interval. We are interested in the coverage probability of an interval estimator, i.e., in computing $\mathbb{P}_\theta\left(\theta \in [L(X), U(X)]\right)$, where $\mathbb{P}_\theta$ indicates that the samples $X=(X_1,\ldots,X_n)$ are being generated from the probability distribution parameterized by the true parameter θ.

The *confidence coefficient* is the infimum, over all probabilities, that a confidence interval will contain the true value of the population parameter θ wit the desired confidence (i.e.. with the specified probability) For example, if the confidence coefficient is 0.95, then it means that $\inf_{\theta\in\Theta}\mathbb{P}_\theta\left(\theta \in [L(X), U(X)]\right) = 0.95$, where Θ is the space of allowed parameters.

An interval estimator with a specific confidence coefficient is called a *confidence interval*.

Example: Consider i.i.d. samples $X_1, X_2, \ldots, X_n$ from a Gaussian distribution with *unknown mean* μ and known variance σ^2. Our objective is to construct an interval estimator of μ with a high confidence coefficient. We know that the sample mean $\bar{X}$ has a Gaussian distribution with mean μ and variance σ^2/n. Therefore, $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$ follows a Gaussian distribution with mean 0 and variance 1. Then we have that

$$\begin{aligned}\mathbb{P}_\mu\left(\frac{|\bar{X}-\mu|}{\sigma/\sqrt{n}} \leq c\right) &= \mathbb{P}_\mu\left(\bar{X} - c\sigma/\sqrt{n} \leq \mu \leq \bar{X} + c\sigma/\sqrt{n}\right) \\ &= 1 - 2\Phi(-c),\end{aligned}$$

where $\Phi(x)$ is the cumulative distribution function of a standard (zero mean unit variance) Gaussian random variable. Notice that the interval is independent of the unknown parameter μ. Let $z_{1-\alpha}$ be defined as the quantile of the distribution with tail probability α, which means that $\mathbb{P}(Z > z_{1-\alpha}) = \alpha$. In order to have $1-\alpha$ confidence coefficient (for instance, $\alpha = 5\%$), we want $1 - 2\Phi(-c) = 1 - \alpha$, i.e. $\Phi(-c) = \frac{\alpha}{2}$. This means that $-c = z_{\frac{\alpha}{2}}$, and, because the Gaussian distribution is symmetric, this implies $c = -z_{\frac{\alpha}{2}} = z_{1-\frac{\alpha}{2}}$. The method described above yields exact confidence intervals, where the probability of coverage is at least the claimed $(1-\alpha)$.

Remark: In the previous example we have constructed an exact confidence interval. Many times, this is not possible, and one settles for confidence intervals that are calculated by means of an approximation (for example, a normal approximation). Suppose that $\hat{\theta}_{MLE}$ is the MLE of a scalar parameter θ. Because of asymptotic normality, we know that $I(\theta)^{1/2}\left(\hat{\theta}_{MLE} - \theta\right)$ follows approximately a standard Gaussian random variable. An approximate confidence interval of θ with coefficient $1-\alpha$ is then $\left[\hat{\theta}_{MLE} + z_{\frac{\alpha}{2}} I(\theta)^{-1/2}, \hat{\theta}_{MLE} + z_{1-\frac{\alpha}{2}} I(\theta)^{-1/2}\right]$. Such an approximate confidence interval is computable if $I(\theta)$ is known.

Remark 2.1. *If we were dealing with a vector $\theta = (\theta_1, \ldots, \theta_d)$, then the confidence interval of θ with coefficient $1-\alpha$ would be a hyper-rectangle, and given by $\left[\hat{\theta}_{MLE} + z_{\frac{\alpha}{2}} diag(I(\theta)^{-1/2}), \hat{\theta}_{MLE} + z_{1-\frac{\alpha}{2}} diag(I(\theta)^{-1/2})\right]$. Here, $diag(A) = (A_{11}, A_{22}, \ldots, A_{dd})^\top$ is the column vector consisting of the diagonal entries of the matrix A. Again, such an approximate confidence interval is computable if $I(\theta)$ is known.*

2.1 Estimating $I(\theta)$

In the case of a Gaussian random variable, we know that the sample mean is the MLE of the population mean μ. We also know that the MLE is asymptotically efficient and hence $I(\mu) = \frac{n}{\sigma^2}$. Hence, it is computable and independent of the parameter μ which we are trying to estimate. In many practical circumstances, however, the Fisher information of a parameter θ, $I(\theta)$, is not known, or difficult to compute. Indeed, if $I(\theta)$ depends on θ, we cannot compute it because we do not know θ (otherwise, we would not be trying to estimate it). In this case, it is important to estimate $I(\theta)$. There are three approaches which can be used:

- **Method 1.** If we can compute $I(\theta)$ explicitly, then we know by the invariance property that $I(\hat{\theta}_{MLE})$ is a maximum likelihood estimator of $I(\theta)$. Moreover, by the properties of the maximum likelihood estimator, we know that $I(\hat{\theta}_{MLE})$ is asymptotically unbiased. We then just use $I(\hat{\theta}_{MLE})$ to construct the confidence interval.

- **Method 2.** If we cannot compute $I(\theta)$ explicitly, then we can plug $\hat{\theta}_{MLE}$ inside the defining expression and estimate $I(\theta)$ as $V_E(\hat{\theta}_{MLE})$, where $V_E(\theta) = -\mathbb{E}_\theta \left[\frac{\partial^2 \log L(\theta|X)}{\partial\theta\partial\theta^\top} \right]$. We can approximate the latter quantity numerically using Monte-Carlo methods, i.e., $V_E(\hat{\theta}_{MLE}) = -\sum_{i=1}^{n} \frac{1}{n} \frac{\partial^2 \log L(\theta|X_i)}{\partial\theta\partial\theta^\top}\Big|_{\theta=\hat{\theta}}$, where X_i is the realization of the data vector X in the i-th Monte-Carlo iteration.

- **Method 3** If we cannot compute $I(\theta)$ explicitly, we can use a sample mean estimation, given by $V_B(\hat{\theta}_{MLE})$, where

$$V_B(\theta) := \sum_{i=1}^{n} \frac{1}{n} \left(\frac{\partial \log L(\theta|X_i)}{\partial\theta} \right) \left(\frac{\partial \log L(\theta|X_i)}{\partial\theta} \right)^\top,$$

  where X_i is the realization of the data vector X in the i-th Monte-Carlo iteration.

Lecture 7: Delta Method and Hypothesis Testing

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1 The Delta Method

This part is not covered in John Rice's textbook. You can consult Chapter 3 of the book "Asymptotic Statistics' by A. W. van der Vaart, from the Vrije Universiteit, Amsterdam. If you use your Columbia account, this book should be available at `https://www.cambridge.org/core/books/asymptotic-statistics/A3C7DAD3F7E66A1FA60E9C8FE132EE1D`.

Motivation: Suppose we observe $X_1, \ldots, , X_n$ independent Bernoulli(p) random variables. We are interested in estimating the odds, i.e., in the ratio $\frac{p}{1-p}$. For example, if the success of a medical treatment has probability $p = 2/3$, the odds of getting better is $2:1$. Furthermore, if there is another treatment with success probability r, then we might be interested in the odds ratio $\frac{p}{1-p}/\frac{r}{1-r}$, which gives the relative odds of one treatment over another.

Question: Suppose we use $\frac{\hat{p}_{MLE}}{1-\hat{p}_{MLE}}$ to estimate the odd $\frac{p}{1-p}$. What is the variance of $\frac{\hat{p}_{MLE}}{1-\hat{p}_{MLE}}$? Can we approximate the distribution of this sampling estimator?

Let $T_1, \ldots, , T_d$ be unbiased estimators (each T_i can be a function of random variables) of the parameters $\theta_1, \ldots, \theta_d$, respectively. Define $T = (T_1, \ldots, T_d)$ and $\theta = (\theta_1, \ldots, \theta_d)$. Consider a differentiable function $g(\cdot)$, and define the partial derivatives as

$$g_i'(\theta) = \left.\frac{\partial g(t)}{\partial t_i}\right|_{t_1=\theta_1,\ldots,t_d=\theta_d}.$$

The first-order Taylor expansion of g around θ is

$$g(t) = g(\theta) + \sum_{i=1}^{d} g_i'(\theta)(t_i - \theta_i) + Remainder$$

Let us construct a statistical approximation for θ using the estimator vector T. Then

$$g(T) \approx g(\theta) + \sum_{i=1}^{d} g_i'(\theta)(T_i - \theta_i)$$

Taking expectations on both sides, and using the equations above, we get

$$\begin{aligned}
\mathbb{E}\left[g(T)\right] &\approx g(\theta) + \sum_{i=1}^{d} g_i'(\theta)\mathbb{E}\left[T_i - \theta_i\right] \\
&= g(\theta) \\
Var\left[g(T)\right] &\approx \mathbb{E}\left[(g(T) - g(\theta))^2\right] \\
&\approx \mathbb{E}\left[\left(\sum_{i=1}^{d} g_i'(\theta)(T_i - \theta_i)\right)^2\right] \\
&= \sum_{i=1}^{d} g_i'(\theta)^2 Var(T_i) + 2\sum_{i>j} g_i'(\theta) g_j'(\theta) Cov(T_i, T_j),
\end{aligned}$$

where the last equality follows by expanding the square. Back to our example of the odds ratio estimator. We take $g(p) = \frac{p}{1-p}$ so that $g'(p) = \frac{1}{(1-p)^2}$. In this case $d = 1$, thus there is only one derivative. The MLE of p is $T_1 = \hat{p}_{MLE} = \frac{1}{n}\sum_{i=1}^{n} X_i$. Thus

$$Var\left[g(T_1)\right] \approx (g'(p))^2 Var[\hat{p}_{MLE}] = \left(\frac{1}{(1-p)^2}\right)^2 \frac{p(1-p)}{n} = \frac{p}{n(1-p)^3} = \frac{(g'(p))^2}{I(p)},$$

where we recall from lecture 5 that $\frac{1}{p(1-p)}$ is the Fisher information of the parameter p for a single Bernoulli random variable, so that $I(p) = \frac{n}{p(1-p)}$ is the Fisher information of the entire data sample consisting of n independent samples. This gives an approximation for the variance of our estimator.

We can formulate a general result. Recall that the MLE estimator is asymptotically efficient and normal, i.e.

$$I(\theta)^{1/2}\left(\hat{\theta}_{MLE} - \theta\right) \xrightarrow{D} N(0, I),$$

where I is the identity matrix. $I(\theta)$ is the Fisher information associated with the whole sample. Then, we have that

Proposition 1.1.

$$g(\hat{\theta}_{MLE}) - g(\theta) \sim^D N\left(0, \nabla_\theta g(\theta)^\top I^{-1}(\theta)\nabla_\theta g(\theta)\right),$$

where $\nabla_\theta g(\theta) = (\frac{\partial g}{\partial \theta_1}, \frac{\partial g}{\partial \theta_2}, \ldots, \frac{\partial g}{\partial \theta_d})^\top$ *is the vector of partial derivatives. Above, the symbol* $\sim^D$ *means approximately equal in distribution as the number of observations grows large.*

2 Some Remarks on the Estimators

The material in this section will not be discussed in class, but you are strongly encouraged to read it.

A popular model for describing the dynamics of a stock price is the geometric Brownian motion

$$dS(t) = \mu S(t)dt + \sigma S(t)dW(t),$$

where $W(t)$ is a Brownian motion. Suppose that we can observe the stock price continuously. This is, of course, a stylized setup but frequently used in finance in the context of derivative pricing.

Also, the continuous assumption is a good approximation if we are dealing with high-frequency data. We would like to get the MLE estimates of the model parameters μ and σ. Using Ito's lemma, we can show that

$$d\log S(t) = \nu dt + \sigma dW(t),$$

where $\nu = \mu - \frac{\sigma^2}{2}$. Based on this relation, it is enough to find the MLE of ν and σ^2. Suppose we observe the stock price in equally spaced time periods $t_0 < t_1 < \cdots < t_n$, where $\delta := t_i - t_{i-1}$. Then $X_i := \log S(t_i) - \log(S_{t_{i-1}})$, $i = 1, \ldots, n$, are i.i.d. Gaussian samples of the Gaussian distribution with mean $\nu\delta$ and variance $\sigma^2\delta$. Recall the log-likelihood function of a Gaussian. In our case, it is given by

$$\log L(X\big|\nu, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\delta\sigma^2) - \frac{1}{2\sigma^2\delta}\sum_{i=1}^{n}(X_i - \nu\delta)^2,$$

where $X = (X_1, X_2, \ldots, X_n)$. Using the first order condition, that is differentiating and setting the derivative equals to zero, yields the MLE estimates

$$\widehat{\nu}_{MLE} = \frac{1}{\delta}\bar{X}, \qquad \widehat{\sigma^2}_{MLE} = \frac{1}{n\delta}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

We first fix $\delta > 0$ (this can be one day, for instance, i.e. we only get end-of-day price observations) and send $n \to \infty$. We know that, by the properties of the MLE estimator, that the MLE is a consistent estimator of ν and σ.

In recent years, financial data such as stock prices are available at a very high frequency, i.e., δ can be infinitesimally small (on the order of microseconds). Fix $T = \delta n = t_n - t_0$ and let δ go to zero. Notice that

$$\begin{aligned}\hat{\nu}_{MLE} &= \frac{1}{\delta n}\sum_{i=1}^{n} X_i = \frac{1}{\delta n}\sum_{i=1}^{n}\left(\log S(t_i) - \log(S_{t_{i-1}})\right) \\ &= \frac{1}{\delta n}\left(\log S_{t_n} - \log S_{t_0}\right) = \frac{1}{T}\left(\log S_{t_n} - \log S_{t_0}\right)\end{aligned}$$

Hence, $\hat{\nu}$ does not converge in probability to ν as $\delta \to 0$.

Consider, instead, the estimator for the squared-volatility

$$\begin{aligned}\widehat{\sigma^2}_{MLE} &= \frac{1}{\delta n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2 = \frac{1}{n\delta}\left[\sum_{i=1}^{n} X_i^2 - n(\bar{X})^2\right] \\ &= \frac{1}{\delta n}\left(\sum_{i=1}^{n}\left(\log S(t_i) - \log(S_{t_{i-1}})\right)^2\right) - \delta\hat{\nu}^2\end{aligned}$$

It is well known (from stochastic calculus) that, as $\delta \to 0$, $\left(\sum_{i=1}^{n}\left(\log S(t_i) - \log(S_{t_{i-1}})\right)^2\right)$ is the quadratic variation of $\log S$ in $[t_0, t_n]$, which is almost surely $\sigma^2(t_n - t_0) = \sigma^2 T$.[1] Hence, $\widehat{\sigma^2}_{MLE}$ converges almost surely to σ^2, as $\delta \to 0$ (notice that $\delta\hat{\nu}^2_{MLE}$ converge almost surely to zero as $\delta \to 0$ because the random variable $\hat{\nu}^2_{MLE}$ is almost surely bounded). This means that high-frequency data improve the accuracy of the estimate of the volatility σ, but not the accuracy of the estimate of the growth rate of the stock μ. Indeed, it is difficult to estimate expected stock returns.

[1] The proof that the quadratic variation of a Brownian motion $W(t)$ is almost surely t can be found, for instance, in Shreve's textbook "Stochastic Calculus for Finance II", see Theorem 3.4.3 therein.

Lecture 8: Likelihood Ratio Tests

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1 Hypothesis Testing

Consider independent and identically distributed samples $X_1, X_2, \ldots, X_n$ coming from a Gaussian distribution $N(\mu, \sigma^2)$ with unknown mean μ and unknown standard deviation σ. We want to know whether $\mu = \mu_0$ for some pre-specified mean μ_0. For instance, if X_i's are returns of a stock observed over a period of time, we may be interested in testing whether the returns of this stock have zero mean. In this case, we would take $\mu_0 = 0$. Let us consider the sample mean estimator $\bar{X} = \frac{1}{n}\sum_{i=1}^n X_i$. This is an unbiased estimator of μ. Would you reject the hypothesis when $\bar{X}$ is close to zero? Intuitively, we tend not to reject in this case. For instance, suppose that $|\bar{X} - \mu_0| \leq c\frac{\hat{\sigma}_s}{\sqrt{n}}$, where $\hat{\sigma}_s = \sqrt{\frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X})^2}$ is the sample standard deviation and c is a chosen constant. Then we would not reject the hypothesis, because $\bar{X}$ is sufficiently close to μ_0. We would reject, instead, if this inequality is not satisfied.

Question: What is the criterion used? What is the probability of *false negatives*, that is of wrongly not rejecting the hypothesis? What is the probability of *false positives*, that is of wrongly rejecting the hypothesis? The probability of wrongly rejecting the hypothesis is

$$\mathbb{P}_{\mu_0}\left(|\bar{X} - \mu_0| > c\frac{\hat{\sigma}_s}{\sqrt{n}}\right) = \mathbb{P}_{\mu_0}\left(\frac{|\bar{X} - \mu_0|}{\frac{\hat{\sigma}_s}{\sqrt{n}}} > c\right) = 2T_{n-1}(-c),$$

The ratio term inside $\mathbb{P}_{\mu_0}$ is a random variable, whose distribution is the student's t distribution with $n-1$ degrees of freedom (we lose one degree of freedom because we have replaced the standard deviation of the Gaussian with its sample estimator). We denote the CDF of a Student's t distribution with n degree of freedom by T_n.

The probability of wrongly not rejecting the hypothesis is the maximum, over all values $\mu \neq \mu_0$,

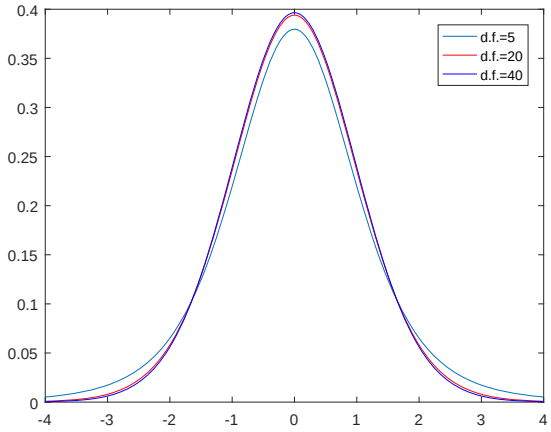


Figure 1: The probability density function of the t-distribution with 5, 20, and 40 degrees of freedom, respectively.

of the following probabilities:

$$\begin{aligned}
\mathbb{P}_\mu\left(|\bar{X}-\mu_0| < c\frac{\hat{\sigma}_s}{\sqrt{n}}\right) &= \mathbb{P}_\mu\left(-c\frac{\hat{\sigma}_s}{\sqrt{n}} \le \bar{X}-\mu_0 \le c\frac{\hat{\sigma}_s}{\sqrt{n}}\right) \\
&= \mathbb{P}_\mu\left(-c \le \frac{\bar{X}-\mu_0}{\frac{\hat{\sigma}_s}{\sqrt{n}}} \le c\right) \\
&= \mathbb{P}_\mu\left(-c-\frac{\mu-\mu_0}{\hat{\sigma}_s/\sqrt{n}} \le \frac{\bar{X}-\mu}{\frac{\hat{\sigma}_s}{\sqrt{n}}} \le c-\frac{\mu-\mu_0}{\hat{\sigma}_s/\sqrt{n}}\right) \\
&= T_{n-1}\left(c-\frac{\mu-\mu_0}{\hat{\sigma}_s/\sqrt{n}}\right) - T_{n-1}\left(-c-\frac{\mu-\mu_0}{\hat{\sigma}_s/\sqrt{n}}\right)
\end{aligned}$$

We would like both the probability of wrongly rejecting the hypothesis, and the probability of wrongly not rejecting the hypothesis, to be small.

Let us now try to formalize all this at a more general level. Assume that the parameter θ belongs to the space Θ. Consider the *null hypothesis* $H_0 : \theta \in \Theta_0$, versus the *alternative hypothesis* $H_a : \theta \notin \Theta_0$, that is $H_a : \theta \in \Theta \setminus \Theta_0$. Suppose we obtain samples $X_1, X_2, \ldots, X_n$. The rejection region will be a subset R of $\mathbb{R}^n$: we reject H_0 if $X = X_1, \ldots, X_n \in R$, and do not reject otherwise. We have two type of errors which can occur:

- **Type I error:** This is also known as a *false positive*: the error of rejecting a null hypothesis when it is actually true. In plain words, this occurs when we are observing a difference even if in truth there is no difference. Formally, this is $\sup_{\theta\in\Theta_0} \mathbb{P}_\theta(X \in R)$
- **Type II error:** This is also known as a *false negative*: the error of not rejecting a null hypothesis when the alternative hypothesis is the true state of nature. In plain words, this

occurs when we are failing to observe a difference even if in reality there is one. Formally, this is $\sup_{\theta\in\Theta\setminus\Theta_0}\mathbb{P}_\theta(X\notin R) = 1-\inf_{\theta\in\Theta\setminus\Theta_0}\mathbb{P}_\theta(X\in R)$.

We would like to minimize both Type I and Type II errors simultaneously. Usually, we control for the Type I error first. Notice that null and alternative hypothesis are not symmetric: not rejecting H_0 is different from accepting H_0.

Let α be the probability of committing the error to reject the null hypothesis H_0. α is called the *significance level* of the test. Mathematically, this is equal to $\sup_{\theta\in\Theta_0}\mathbb{P}_\theta(X\in R)$. Usually, when we perform a test, we specify a significance level. The rejection region is usually specified via a **test statistic**, which is a function of the sample X. Given a test statistic $W(X)$, we define the rejection region

$$R := \{x : W(x) > c\},$$

where c is a given threshold. Such threshold c will be chosen based on the significance level of the test. For this reason, it will be important to know the distribution $W(X)$ under the null hypothesis H_0.

2 Statistical Tests

There are two types of statistical tests: One-tailed tests and two-tailed tests. Both are ways of computing the statistical significance of a parameter inferred from a data set, in terms of a test statistics. A two-tailed test is used if deviations of the estimated parameter in either direction from some benchmark value are considered theoretically possible; in contrast, a one-tailed test is used if only deviations in one direction are considered possible.

2.1 One-Side Hypotheses

Let us first focus on one-side hypotheses. Consider i.i.d. samples $X_1, X_2, \ldots, X_n$, coming from a Gaussian distribution with unknown mean μ and variance σ^2.

Objective: Test the hypothesis $H_0 : \mu \leq \mu_0$ versus the alternative hypothesis $H_a : \mu > \mu_0$. Consider the test statistic

$$W(X) = \frac{\bar{X}-\mu_0}{\hat{\sigma}_s/\sqrt{n}},$$

where $\bar{X}$ is the sample mean and $\hat{\sigma}_s$ is the sample (unbiased) standard deviation estimator. We reject the null hypothesis if and only $W(X) > c$ for some constant c. Under the Gaussian probability distribution with mean μ and variance σ^2, the random variable $(\bar{X}-\mu)/(\hat{\sigma}_s/\sqrt{n})$ follows a Student's t distribution with $n-1$ degrees of freedom. Therefore,

$$\mathbb{P}_\mu\left(W(X) > c\right) = \mathbb{P}_\mu\left(\frac{\bar{X}-\mu}{\hat{\sigma}_s/\sqrt{n}} > c - \frac{\mu-\mu_0}{\hat{\sigma}_s/\sqrt{n}}\right) = 1 - T_{n-1}\left(c - \frac{\mu-\mu_0}{\hat{\sigma}_s/\sqrt{n}}\right).$$

We can then calculate the **type I error**. This is given by

$$\sup_{\mu\leq\mu_0}\mathbb{P}_\mu\left(W(X) > c\right) = 1 - T_{n-1}(c).$$

Suppose we want significance level α, e.g. 5%. Then we can set $c = T_{n-1,1-\alpha}$, where c is the $1-\alpha$-quantile of the distribution, i.e., $\mathbb{P}(V > c) = \alpha$ if the random variable V follows a Student's t distribution with $n-1$ degrees of freedom. Hence, we would reject the hypothesis H_0 if and only if

$$\bar{X} > \mu_0 + T_{n-1,\alpha}\hat{\sigma}_s/\sqrt{n}.$$

3 Two Simple Hypotheses

Consider two *simple* hypotheses, that is $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$. Define $\beta = \mathbb{P}(X \notin R \,|H_1)$, i.e., the probability of failing to reject the null hypothesis, given that the alternative hypothesis holds (here, R denotes the rejection region). The *power* of the test is $1-\beta$. Usually we fix the type I error probability $\alpha = \mathbb{P}(reject\ H_0|H_0)$ also known as the size of the test (typical values of α are $\alpha = 0.05, 0.01$, etc...) and we look for a test which minimizes the probability β of a type II error: this is called a *most powerful* test of *size* α. Intuitively, we reject H_0 in favor of H_1 if the likelihood of θ_1 is much larger than the likelihood of θ_0 given the data.

4 Neyman-Pearson Lemma

To test two simple hypotheses, we use likelihood ratio tests inspired by the following lemma.

Lemma 4.1 (**Neyman-Pearson Lemma**). *Let $X = (X_1, X_2, \ldots, X_n)$ be a random sample from a distribution with parameter $\theta \in \{\theta_0, \theta_1\}$, and let $L(\theta; x)$ be the likelihood function. Suppose there exists a test at significance level α such that the test rejects the null hypothesis if*

$$\frac{L(\theta_1; x)}{L(\theta_0; x)} > k,$$

where the constant k is defined so that $\mathbb{P}_{\theta_0}\left(\frac{L(\theta_1;X)}{L(\theta_0;X)} > k\right) = \alpha$. Then, this test is most powerful at significance level α for testing the null hypothesis $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$. This test is called the likelihood ratio test.

Example: Suppose that we have a random sample $X_1, X_2, \ldots, X_n$ following a Gaussian distribution with unknown mean μ and known variance σ^2. Let μ_0, μ_1 be given, and assume that $\mu_1 > \mu_0$. We would like to test $H_0 : \mu = \mu_0$ against $H_1 : \mu = \mu_1$. Using the Neyman-Pearson Lemma, the best test is a likelihood ratio test. To calculate the test statistics, let us use $L(\mu_1|x)$ to denote the Gaussian likelihood with mean μ_1 and variance σ^2. Let us also use $L(\mu_0|x)$ to denote the Gaussian likelihood with mean μ_0 and variance σ^2. Notice that

$$\frac{L(\mu_1|x)}{L(\mu_0|x)} \geq k$$

if and only if

$$LL(\mu_1|x) - LL(\mu_0|x) \geq \log(k),$$

whic holds if and only if

$$-\sum_i \left[(x_i - \mu_1)^2 - (x_i - \mu_0)^2\right] \geq 2\sigma^2 \log(k),$$

which in turn holds if and only if

$$(\mu_1 - \mu_0)\bar{x} \geq k',$$

which is true if and only if $\bar{x} \geq c$, where $c = \frac{k'}{\mu_1 - \mu_0}$ is a constant (i.e., independent of $x = (x_1, x_2, \ldots, x_n)$), and $\bar{x} = \frac{1}{n}\sum_i x_i$. Hence, for a size α test, we choose c so that $\mathbb{P}_{\mu_0}(\bar{X} \geq c) = \alpha$, or equivalently such that

$$\mathbb{P}_{\mu_0}\left(\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \geq \frac{c - \mu_0}{\sigma/\sqrt{n}}\right) = \alpha.$$

Hence, we want

$$\frac{c - \mu_0}{\sigma/\sqrt{n}} = z_{1-\alpha},$$

where $P(Z \geq z_{1-\alpha}) = P(Z \leq z_\alpha) = \Phi(z_\alpha) = \alpha$, and Φ is the standard Gaussian cumulative distribution function, i.e. $c = \mu_0 + \sigma z_{1-\alpha}/\sqrt{n}$. So, the most powerful test of H_0 versus H_1 at level α becomes "reject H_0 if and only if $\bar{X} \geq \mu_0 + \sigma\frac{z_{1-\alpha}}{\sqrt{n}}$".

Lecture 9: Likelihood Ratio Statistics and p-Values

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1 Likelihood Ratio Tests

The Neyman-Pearson Lemma suggests likelihood ratio tests of the following form. Consider a k-dimensional parameter space $\Theta_0 \subset \Theta$. We can denote by $\theta = (\theta_1, \ldots, \theta_k)$ an element of the parameter space Θ. Define the test statistics

$$\lambda(x) := \frac{\sup_{\theta \in \Theta_0} L(\theta; x)}{\sup_{\theta \in \Theta} L(\theta; x)}$$

For a realization x, determine its best chance of occurrence under $H_0 : \theta \in \Theta_0$, and also its best chance overall. The ratio of these two chances can never exceed unity. If small, it provides evidence for rejection of the null hypothesis. A likelihood ratio test for testing $H_0 : \theta \in \Theta_0$ versus $H_1 : \theta \notin \Theta_0$ is a test whose rejection region (sometimes also referred to as the critical region) is of the form

$$C_1 = \{x : \lambda(x) \leq k\},$$

where $0 \leq k \leq 1$. The test will be at the significance level α if k can be chosen to satisfy

$$\sup \{\mathbb{P}_{\theta \in \Theta_0}(\lambda(X) \leq k)\} = \alpha,$$

i.e. the probability of rejecting the null hypothesis H_0 when it is true is equal to α. If the null hypothesis is simple, that is $\Theta_0 = \{\theta_0\}$, we obtain the simpler form $\mathbb{P}_{\theta_0}(\lambda(X) \leq k) = \alpha$. To determine k, we need to characterize the cumulative distribution function of the r.v. $\lambda(X)$, when the random sample $X = (x_1, x_2, \ldots, x_n)$ has joint pdf $L(\theta_0; x)$.

Example: The null hypothesis is $H_0 : \theta = \theta_0$ for the mean of a normal distribution with unknown variance σ^2. Hence, $\Theta_0 = \left\{(\theta, \sigma^2) : \theta = \theta_0, \sigma^2 \in \mathbb{R}^+\right\}$, and $\Theta = \left\{(\theta, \sigma^2) : \theta \in \mathbb{R}, \sigma^2 \in \mathbb{R}^+\right\}$. The test statistic is of the form

$$\lambda(x) := \frac{\sup_{\sigma^2} L(\theta_0; x)}{\sup_{(\theta, \sigma^2)} L(\theta; x)}$$

Recall that the likelihood of a Gaussian associated with the sample $x = (x_1, \ldots, x_n)$ is

$$L(\theta_0, \sigma^2; x) = (2\pi\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^n (x_i - \theta_0)^2},$$

and the log-likelihood associated with the sample $x = (x_1, \ldots, x_n)$ is

$$LL(\theta_0, \sigma^2; x) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \theta_0)^2$$

Let us try to find the maximum of LL over σ^2. Taking the derivative with respect with σ^2 and setting it to zero, we find that

$$\frac{\partial LL}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i - \theta_0)^2 = 0$$

is satisfied if $\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \theta_0)^2$. Plugging this optimum into the likelihood function, we obtain

$$\sup_{\sigma^2}\left\{L(\theta_0, \sigma^2; x)\right\} = \left(\frac{2\pi}{n}\sum_{i=1}^{n}(x_i - \theta_0)^2\right)^{-n/2} e^{-n/2}$$

For the denominator of the test statistics, we already know that the maximum likelihood estimator of θ is $\bar{x} := \frac{1}{n}\sum_{i=1}^{n} x_i$. Hence,

$$\sup_{\theta, \sigma^2}\left\{L(\theta, \sigma^2; x)\right\} = \left(\frac{2\pi}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2\right)^{-n/2} e^{-n/2}.$$

Then we can rewrite

$$\lambda(x) = \left(\frac{\sum_{i=1}^{n}(x_i - \theta_0)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right)^{-n/2}.$$

Notice that

$$\begin{aligned}\sum_{i=1}^{n}(x_i - \theta_0)^2 &= \sum_{i=1}^{n}((x_i - \bar{x}) + (\bar{x} - \theta_0))^2 = \sum_{i=1}^{n}(x_i - \bar{x})^2 + \sum_{i=1}^{n}(\bar{x} - \theta_0)^2 + 2\sum_{i=1}^{n}(x_i - \bar{x})(\bar{x} - \theta_0) \\ &= \sum_{i=1}^{n}(x_i - \bar{x})^2 + n(\bar{x} - \theta_0)^2 + 2(\bar{x} - \theta_0)\sum_{i=1}^{n}(x_i - \bar{x}) = \sum_{i=1}^{n}(x_i - \bar{x})^2 + n(\bar{x} - \theta_0)^2\end{aligned}$$

so that $\lambda(x) = \left(1 + \frac{n(\bar{x}-\theta_0)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right)^{-n/2}$. Define

$$\frac{|\bar{x} - \theta_0|}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}}. \tag{1}$$

Recall that the critical region is $C_1 = \{x : \lambda(x) \leq k\}$. So, it follows that H_0 is rejected when the value of the expression in (1) is larger than some constant. We can equivalently write the rejection, also called critical, region as

$$C_1 = \left\{\frac{|\bar{x} - \theta_0|}{s/\sqrt{n}} \geq c\right\},$$

where $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$. Recall from previous lectures and homework problem that

$$\frac{\bar{X} - \theta_0}{S/\sqrt{n}} \sim t_{n-1},$$

where $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$. This is the form of *a two-sided t-test* (as we reject both if the value of the statistic is too low (negative), and if it is too large (positive)). We can then use the quantile of the t_{n-1} distribution and specify the constant c based on the desired significance level α.

2 Likelihood Ratio Statistics and Asymptotic Distribution

Since the function $-2\log\lambda(x)$ is a decreasing function of $\lambda(x)$, it follows that the critical region of the likelihood ratio test can also be written as

$$C_1 = \{x : -2\log\lambda(x) \geq c\}.$$

Why do we focus on the statistics $-2\log\lambda(x)$? In general, it is rather difficult to compute the probability distribution of the test statistics $\lambda(x)$. However, what is remarkable, is that we can asymptotically (when the number of observations is large) estimate the distribution of $-2\log\lambda(x)$. Consider the case when the null hypothesis is simple, i.e. $H_0 : \theta = \theta_0$. Let us first define $\Lambda(x) = -2\log\lambda(x) = -2\Big(LL(\theta_0; x) - LL(\hat{\theta}; x)\Big)$, where $\hat{\theta}$ is the maximum likelihood estimate of the parameter θ. Then the critical region may be written as

$$C_1 = \{\Lambda(x) \geq c\},$$

and $\Lambda(x)$ is usually referred to as the likelihood ratio statistic. Let us now study the asymptotic distribution of the test statistics $\Lambda(X)$. The idea is that values of θ close to the MLE estimate $\hat{\theta}$ should be well supported by the data. Thus, if θ_0 is a possible value of θ, then we can approximate the log-likelihood function at θ_0 using a Taylor expansion around $\hat{\theta}$. Indeed,

$$LL(\theta_0; x) \approx LL(\hat{\theta}; x) + \Big(\theta_0 - \hat{\theta}\Big)\left.\frac{\partial LL(\theta; x)}{\partial\theta}\right|_{\theta=\hat{\theta}} + \frac{1}{2}\Big(\theta_0 - \hat{\theta}\Big)^2\left.\frac{\partial LL(\theta; x)}{\partial^2\theta}\right|_{\theta=\hat{\theta}}.$$

By definition, $\hat{\theta}$ is the MLE and, assuming that the maximum of the likelihood function is an interior point, so that the local maximum also coincides with the global maximum, then $\left.\frac{\partial LL(\theta;x)}{\partial\theta}\right|_{\theta=\hat{\theta}} = 0$. Hence,

$$\begin{aligned}\Lambda &\propto -(\hat{\theta} - \theta_0)^2\left.\frac{\partial LL(\theta; x)}{\partial^2\theta}\right|_{\theta=\hat{\theta}} \\ &= (\hat{\theta} - \theta_0)^2 I(\theta_0)\frac{J(\hat{\theta})}{I(\theta_0)},\end{aligned}$$

where $I(\theta_0)$ is the Fisher information matrix associated with the parameter θ_0 and computed using the n observations at disposal (recall that this is given by $-\sum_{i=1}^{n}\mathbb{E}\left[\frac{\partial LL(\theta;X_i)}{\partial^2\theta}\big|_{\theta=\theta_0}\right]$), while $J(\hat{\theta}) = -\sum_{i=1}^{n}\frac{\partial}{\partial^2\theta}LL(\theta; x_i)\big|_{\theta=\hat{\theta}}$ is the observed Fisher information matrix. We know that

$$(\hat{\theta} - \theta_0)I(\theta_0)^{1/2} \xrightarrow{D} Gaussian(0, 1)$$

under the null hypothesis. Thus $(\hat{\theta} - \theta_0)^2 I(\theta_0) \xrightarrow{D} \chi_1^2$, because the square of a Gaussian random variable follows a χ-squared distribution with one degree of freedom. Moreover, it can be shown

that, under the null hypothesis, $\frac{J(\hat{\theta})}{I(\theta_0)} \xrightarrow{\mathbb{P}} 1$. Using Slutsky's theorem, we can conclude that the likelihood ratio statistic

$$\Lambda \xrightarrow{D} \chi_1^2$$

under the null hypothesis that $\theta = \theta_0$.

Goodness of Fit Test and QQ Plots

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1 p-Values

The p value is the probability that the test statistic takes the value we observed or more extreme values away from the null hypothesis, if the null hypothesis is true. Recall the example from last class, in which the rejection region is

$$C_1 = \left\{ \frac{|\bar{x} - \theta_0|}{s/\sqrt{n}} \geq c \right\},$$

where $\bar{x}$ is the sample mean and s is the sample standard deviation. Suppose $n = 130$, and we use a significance level $\alpha = 0.05$. Because the above is a two-sided test, we should reject the null hypothesis if our test statistic exceeds $t_{n-1,1-\alpha/2} = t_{129,0.975}$ in absolute value. Here, $t_{n-1,1-\alpha/2}$ is the $1-\alpha/2$ quantile of a t distribution with $n-1$ degrees of freedom. We have that $t_{129,0.975} \approx 1.98$. A problem with the critical value method is the following. Suppose the test statistic happens to be $t = 1.94$. Then the conclusion of the test would be to not reject the null hypothesis at $\alpha = 0.05$. However, a value of $t = 1.94$ is right on the border of the rejection region and the evidence is still fairly strong for rejecting the null hypothesis, but not quite at the $\alpha = 0.05$ level. If we test at $\alpha = 0.05$, all we can state is that we fail to reject the null hypothesis at $\alpha = 0.05$ without saying anything on the fact that the actual evidence against the null hypothesis is quite strong. Could we have rejected it using $\alpha = 0.075$ or $\alpha = 0.1$? *What is the smallest significance level at which we can still reject the null hypothesis?* The answer is the **p-value**. The p-value is the smallest significance level at which we can reject the null hypothesis. For this reason, the p-value is also called the *observed significance level*.

2 Goodness of Fit Tests

These tests are used to test the null hypothesis that the data comes from a specific distribution, and they are based on a comparison of the theoretical and the empirical distribution.

Example: In practice, it is very difficult to estimate the probability distribution of defaults (bankruptcy events). Moody's KMV has defined the concept of distance to default to capture the likelihood of a firm's default. This is defined as the distance of the asset value of a firm from its liabilities threshold. If the asset value of the firm falls below its liability threshold, then the firm is unable to cover its debt payments, and therefore will fail. Consider the following procedure. Create

Distance to defaults	DD (0-0.1M)	DD(0.1M-1M)	DD(1M-1.5M)	DD(1.5M,2M)	$DD > 2M$
frequency	9	9	10	14	6

Table 1: Distance to default buckets (M stands for million dollars). The table report number of firms whose distance to default falls in that bucket.

different buckets, each associated with a value for the *distance to default*. Then, for each firm i calculate the distance to default DD (this can be defined, for instance, as the difference between the asset value of firm i, and the firm i's liabilities.) Suppose that this procedure produces the output in Table 1.

The objective is to estimate the probability distribution of the distance to default. Let X_j be the random variable designating in which bucket the distance to default of the j-th firm falls. Assume that the X_j's are independent. To place this assessment in terms of a formal hypothesis testing, let $\pi = (\pi_1, \pi_2, \ldots, \pi_k)$ be postulated values of the probability

$$\mathbb{P}_\pi(\text{a firm is member of the i-th bucket}) = \pi_i.$$

Let the parameter space be $\Theta = \left\{p = (p_1, p_2, \ldots, p_k); p_i \geq 0, \sum_{i=1}^k p_i = 1\right\}$, where k is the number of buckets. This parameter space has $k-1$ free parameters. Once these are chosen, the remaining parameter is determined by the constraint that the sum of the p_i's must equal one. Hence, $dim(\Theta) = k-1$. The hypothesis test is: $H_0 : p_i = \pi_i$ for all i versus $H_1 : p_i \neq \pi_i$ for some i. We have that $dim(\Theta_0) = 0$ because this is a single point in the k-simplex.

Let us use $n_i = \#\{X_j : X_j = i\}$. Hence, the vector $n = (n_1, n_2, \ldots, n_k)$ is a vector that records the number of observations in each bucket. Let $N = \sum_{j=1}^k n_j$ be the total number of observations. The likelihood function $L(p|n) = \frac{N!}{n_1!n_2!\ldots n_k!} p_1^{n_1} p_2^{n_2} \cdots p_k^{n_k}$. The likelihood ratio is the ratio of the maximum value of the likelihood function under the null hypothesis, and the maximum likelihood for any parameter value. Hence, the test statistic is

$$\lambda(n) = \frac{L(\pi|n)}{L(\hat{p}|n)} = \frac{\pi_1^{n_1}\pi_2^{n_2}\cdots\pi_k^{n_k}}{\hat{p}_1^{n_1}\hat{p}_2^{n_2}\cdots\hat{p}_k^{n_k}} = \left(\frac{\pi_1}{\hat{p}_1}\right)^{n_1}\left(\frac{\pi_2}{\hat{p}_2}\right)^{n_2}\cdots\left(\frac{\pi_k}{\hat{p}_k}\right)^{n_k},$$

where $\hat{p}$ is the maximum likelihood estimator, i.e., the maximizer of the likelihood function $L(p|n)$.

To find the MLE, we need to maximize the log-likelihood, $LL(p|n) = \log\left(\frac{N!}{n_1!n_2!\ldots n_k!}\right) + \sum_{i=1}^k n_i \log p_i$, subject to the constraint that $\sum_{i=1}^k p_i = 1$. Using Lagrangian multipliers (verify yourself), it can be checked that $\hat{p_i} = \frac{n_i}{N}$. This is in line with what we would expect intuitively. Hence,

$$\lambda(n) = \left(\frac{N\pi_1}{n_1}\right)^{n_1}\left(\frac{N\pi_2}{n_2}\right)^{n_2}\cdots\left(\frac{N\pi_k}{n_k}\right)^{n_k},$$

We reject the null hypothesis if this ratio is too low, i.e, the likelihood under the null hypothesis is sufficiently smaller than the maximum likelihood under the alternative hypothesis. Using an argument similar to what we did for the likelihood ratio tests in the previous lecture, we can show that, under the null hypothesis H_0, $-2\log\lambda(n)$ has approximately a chi-squared distribution with a

number of degrees of freedom equal to $dim(\Theta) - dim(\Theta_0)$, that is a χ^2_{k-1}-distribution in our setting. We have that

$$-2\log\lambda(n) = -2\sum_{i=1}^{k} n_i \log\frac{N\pi_i}{n_i} = 2\sum_{i=1}^{k} n_i \log\frac{n_i}{N\pi_i}.$$

The above statistics is usually expressed (in textbooks or articles) using $O_i = n_i$ for the number of observed occurrences of i, and $E_i = N\pi_i$ for the number of expected occurrences of i under the null hypothesis H_0. Hence, we can rewrite it as

$$-2\log\lambda(O) = 2\sum_{i=1}^{k} O_i \log\frac{O_i}{E_i}.$$

The expression above is called the G test statistic. Because the rejection rule is to reject the null hypothesis when the test statistic $\lambda(n) \leq c$, this means that we would reject when $-2\log\lambda(n) \geq c_1$. Because $-2\log\lambda(n)$ follows a χ^2_{k-1} random variable, we have that the p-value is the probability that a χ^2_{k-1} distributed random variable takes a value larger or equal than $-2\log\lambda(O)$.

Another statistic used for goodness of fit tests is the Pearson's statistic. This is defined by

$$\Lambda'(O) = \sum_{j=1}^{k} \frac{(E_j - O_j)^2}{E_j}$$

Next, we show the relation between the Pearson's statistic and the G statistic. First, it can be verified that the Taylor expansion of the function $x\log(x/x_0)$ around x_0 is given by

$$x\log(x/x_0) \approx x - x_0 + \frac{1}{2}\frac{(x-x_0)^2}{x_0}.$$

Then

$$\begin{aligned} -2\log\lambda(O) &= 2\sum_{i=1}^{k} O_i \log\frac{O_i}{E_i} \approx 2\sum_{i=1}^{k}\left(O_i - E_i + \frac{1}{2}\frac{(O_i-E_i)^2}{E_i}\right) \\ &= \sum_{i=1}^{k}\frac{(O_i-E_i)^2}{E_i} := \Lambda'(O), \end{aligned}$$

where we have used that $\sum_{i=1}^{k} E_i = \sum_{i=1}^{k} O_i = N$.

The statistics $\Lambda'(O)$ is called the Pearson's statistic. It was introduced between 1895 and 1900 by Karl Pearson, and has been in use for longer than the concept of likelihood ratio tests.

The above argument shows that the Pearson's statistic and the G test statistic are asymptotically equivalent.

Example: Consider Table 1. Assume we want to fit a **Poisson model** to the distribution of the distances to default.[1] Our null hypothesis is $H_0 : p_i = \pi_i$, where $\pi_i(\lambda) = e^{-\lambda}\frac{\lambda^i}{i!}$ for $i = 0, 1, 2, 3,$

[1] We assume each bucket in our data corresponds to an incremental index of the Poisson counts (i.e bucket 0 corresponds to $X = 0$, ..., bucket k corresponds to $X = k$, where X is poisson(λ)).

and $\pi_4 = 1 - \sum_{i=0}^{3} \pi_i$.[2] We obtain $\hat{\lambda} = 1.979$ if we calculate the MLE estimate of the Poisson distribution using the data in Table 1. Next, compute the expected frequency $E_j = N\pi_j(1.979)$, where N is the total number of observations. We get $E_0 = 6.63$, $E_1 = 13.13$, $E_2 = 12.99$, $E_3 = 8.57$, $E_4 = 6.68$. Compute the test statistics $-2\log\lambda(O) = 5.92$, and $\Lambda'(O) = 6.34$. Note that $\Theta = \{p = (p_0, \ldots, p_4) : \sum p_j = 1\}$, and so $dim(\Theta) = 5 - 1$, and $\Theta_0 = \{\lambda : \lambda > 0\}$, so that $dim(\Theta_0) = 1$. Then, both $-2\log\lambda(O)$ and $\Lambda'(O)$ follow a χ^2 distribution with degrees of freedom equal to $5 - 1 - 1 = 3$. The p-values of these two tests can then be readily computed, and it is equal to 0.1155 if we use the statistic $-2\log\lambda(O)$, and 0.0962 if we use the statistic $\Lambda'(O)$. This provides moderate evidence against the null hypothesis that the distance to default follows a Poisson distribution with parameter $\hat{\lambda} = 1.979$.

[2]For practical purposes, in order to apply the G test, we have to account for the fact that buckets are finite, and therefore we choose to truncate the Poisson up to k, and assign the tail distribution to the last bucket. Note that this is not the only way to approach this problem, and it is only a heuristic to proceed with the goodness-of-fit test.

Lecture 11: Normal QQ Plots, Probability Distribution Tests

IEOR E4709: Statistic Analysis and Time Series
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1 Quantile to Quantile (QQ) Plot

A Q-Q plot ("Q" stands for quantile) is a *probability plot*, that is a graphical method for comparing two probability distributions, usually the sample distribution function and a population distribution function. The idea is to plot the quantiles of population and empirical distribution against each other. A point (x, y) on the plot corresponds to one of the quantiles of the second distribution (y-coordinate) plotted against the same quantile of the first distribution (x-coordinate).

Consider i.i.d. samples $X_1, X_2, \ldots, X_n$ coming from the distribution function F. For each sample of size n, $\{X_1, \ldots, X_n\}$, an empirical probability distribution function F_n is constructed by assigning mass $1/n$ to each observation X_i. Hence, F_n may be represented as $F_n(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{X_i \leq x}$, for $-\infty < x < \infty$. For each fixed value x, $F_n(x)$ is a random variable when it is viewed as a function of the sample of size n. In fact $F_n(x)$ is a random distribution function. The Glivenko-Cantelli Theorem ensures that the CDF converges uniformly almost surely to the true distribution function, i.e., $\mathbb{P}\left(\sup_x |F_n(x) - F(x)| \to 0\right\} = 1$.

Consider order statistics $X_{(1)} \leq X_{(2)} \leq \cdots, \leq X_{(n)}$. In general, we want to test the hypothesis H_0 that the samples are from the distribution F. Under H_0, $F(X_i)$'s are uniformly distributed. It is well known that the distribution of k-order statistics of the uniform distribution is a beta distribution $beta(k, n+1-k)$, where n is the number of random samples (see, for instance, the book by Gentle, James E. (2009), Computational Statistics, Springer, p. 63). Recall that the mean of a beta distribution $beta(k, n+1-k)$ is $\frac{k}{n+1}$. Define $Y_i = F(X_i)$ and observe that the order statistics $Y_{(1)}, \ldots, Y_{(n)}$ are the same as the order statistics $F(X_{(1)}), \ldots, F(X_{(n)})$ because $F(x)$ is an increasing function. Hence, it follows that $\mathbb{E}\left[F(X_{(k)})\right] = \frac{k}{n+1}$.

With a large sample size, $F(X_{(k)}) \approx \frac{k}{n+1}$, so that $X_{(k)} \approx F^{-1}(\frac{k}{n+1})$.[1] Hence, if H_0 is true, we expect to observe a line if we plot $X_{(1)}, X_{(2)}, \ldots, X_{(n)}$ versus $F^{-1}(\frac{1}{n+1}), F^{-1}(\frac{2}{n+1}), \ldots, F^{-1}(\frac{n}{n+1})$. This is a plot of theoretical (population) quantiles ($F^{-1}(\frac{k}{n+1})$'s) vs empirical quantiles ($X_{(k)}$'s), and is called quantile-to-quantile plot (QQ plot). If the plot is not linear, it may show nonlinearity at the upper or lower ends, which may be an indication of the presence of outliers. If the nonlinearity shows

[1] We know that $F(X_{(k)})$ follows a beta distribution $beta(k, n+1-k)$. Therefore, we have $\mathbb{E}\left[F(X_{(k)})\right] = \frac{k}{n+1}$ and $Var\left(F(X_{(k)})\right) = \frac{k(n+1-k)}{(n+1)^2(n+2)}$. Chebyshev's inequality states that if we have an integrable random variable X with finite expected value μ and finite non-zero variance σ^2, then for any real number $c > 0$, we have $\mathbb{P}(|X-\mu| \geq c\sigma) \leq \frac{1}{c^2}$. Hence, let $\epsilon = c\sigma$, we have $\mathbb{P}(|F(X_{(k)}) - \frac{k}{n+1}| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2} = \frac{k(n+1-k)}{(n+1)^2(n+2)\epsilon^2}$. As $n \to \infty$, $\mathbb{P}(|F(X_{(k)}) - \frac{k}{n+1}| \geq \epsilon) \leq \frac{k(n+1-k)}{(n+1)^2(n+2)\epsilon^2} \to 0$. And we conclude that for each k, $F(X_{(k)})$ converges to $\frac{k}{n+1}$ in probability as $n \to \infty$.

up at other points as well, one could question the validity of the assumption that the probability distribution function of the random variables X_i's is F.

1.1 Normal Quantile to Quantile (QQ) Plot

The *normal* Q-Q plot is obtained by plotting $y = X_{(k)}$ versus $x = F^{-1}\left(\frac{k}{n+1}; \mu, \sigma\right)$, where X_i are distributed according to a Gaussian distribution with mean μ and variance σ^2, that is $F(x; \mu, \sigma^2) = \Phi(\frac{x-\mu}{\sigma})$. Here, $\Phi(x)$ is the cumulative distribution function of a Gaussian evaluated at x. Let $z_k = \Phi^{-1}\left(\frac{k}{n+1}\right)$, $k = 1, \ldots, n$ be the theoretical quantiles of a standard Gaussian distribution. Then $F^{-1}(\frac{k}{n+1}; \mu, \sigma) = \mu + \sigma z_k$. So, if we plot $X_{(1)}, X_{(2)}, \ldots, X_{(n)}$ versus $z_1, z_2, \ldots, z_n$, and the $X_i's$ are sample from a normal distribution, then we expect a linear line with intercept μ and slope σ.

2 Statistical Distribution Tests

We discuss statistical tests designed to test the hypothesis that observed samples come from a specific distribution, including the normal distribution.

2.1 The Kolmogorov-Smirnov (KS) Test

Suppose we have i.i.d. observations $X_1, X_2, \ldots, X_n$, which come from a distribution F. The KS test is used to test the null hypothesis H_0 that the samples come from F versus the alternative hypothesis H_1 that the samples do not come from F. Here, we have $F(x) := \mathbb{P}(X_i \leq x)$. We define the empirical cdf by $F_n(x) = \mathbb{P}_n(X \leq x) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{X_i \leq x}$, where $X = (X_1, \ldots, X_n)$, that counts the proportion of sample points below x. For any fixed point x, the strong law of large numbers implies that $F_n(x) \to F(x)$, i.e., the empirical distribution converges to the true distribution F. Moreover, the Glivenko-Cantelli theorem strengthened this result by showing that this approximation holds uniformly, i.e. $\sup_x |F_n(x) - F(x)| \to 0$, i.e. the largest difference between F_n and F goes to zero in probability. The key observation in the KS test is that the distribution of the supremum $\sup_x |F_n(x) - F(x)|$ does not depend on F. Specifically, it can be shown that

$$\mathbb{P}\left(\sqrt{n}(\sup_x |F_n(x) - F(x)|) \leq t\right) \to H(t),$$

where $H(t) := 1 - 2\sum_{i=1}^{\infty}(-1)^{i-1}e^{-2i^2t^2}$ is the CDF of the *Kolmogorov-Smirnov distribution*. Consider the following statistics

$$D_n = \sqrt{n}(\sup_x |F_n(x) - F(x)|)$$

The null hypothesis is rejected if the value of the statistics is large. Let α be the significance level of the test. We have

$$\alpha = \mathbb{P}\left(D_n \geq c | H_0\right)$$

Since, under H_0, the cumulative probability distribution function of D_n is $H(t)$, and the quantiles of $H(t)$ can be tabulated, we can find the threshold $c = c_\alpha$ from the tables.

2.2 Normality tests

A distinguishing characteristic of the normal distribution, relative to other distributions, is its symmetry and lightness of the tails (i.e., extreme outcomes are very unlikely to occur for a normally distributed random variable).

Skewness measures the lack of symmetry in a distribution. A distribution is symmetric if it looks the same to the left and right of the center point. Formally, the skewness of X is defined as $\gamma_1(X) = \mathbb{E}\left[\frac{(X-\mu)^3}{\sigma^3}\right]$. Since any normal distribution is symmetric around its mean μ, its skewness is zero. The sample skewness is defined by $S := \frac{1}{n}\sum_{j=1}^{n}(X_j - \bar{X})^3/(s_X)^3$, where $\bar{X}$ is the sample mean, and $s_X = \left[\frac{1}{n}\sum_{j=1}^{n}(X_j - \bar{X})^2\right]^{1/2}$ is the sample standard deviation.

Kurtosis is a measure of whether the data are heavy-tailed or light-tailed relative to a normal distribution. That is, data sets with high kurtosis tend to have heavy tails. Data sets with low kurtosis tend to have light tails. The kurtosis is defined by $\gamma_2(X) = \frac{\mathbb{E}\left[(X-\mu)^4\right]}{\sigma^4} - 3$. Any normal distribution has zero kurtosis. The sample kurtosis is $K := -3 + \frac{1}{n}\sum_{j=1}^{n}\frac{(X_j-\bar{X})^4}{(s_X^4)}$.

Jarque-Bera Test: Suppose $X_1, X_2, \ldots, X_n$ are normally distributed i.i.d. random variables, and n is fairly large. Then, we expect S and K to be close to zero. A statistic for testing normality is the *Jarque–Bera statistic* defined as

$$JB = \frac{n}{6}\left(S^2 + \frac{1}{4}K^2\right).$$

As n becomes large, if normality holds, then the distribution of JB converges to a χ^2-distribution with two degrees of freedom. Normality is rejected if JB is sufficiently large.

Shapiro-Wilk Test: The test statistic is

$$W := \frac{\left(\sum_{i=1}^{n} a_i X_{(i)}\right)^2}{\sum_{i=1}^{n}(X_i - \bar{X})^2},$$

where $\bar{X}$ is the sample mean and $X_{(i)}$, $i = 1, \ldots, n$, is the i-th order statistic. Each number a_i is computed from $\mathbb{E}\left[X_{(i)}\right]$. Normality is rejected if W is small.

Lecture 12: Bayesian Statistics

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1 Bayesian Statistics

In our first class on probability, we all learned the Bayes formula. Let us briefly restate it:

$$\mathbb{P}\left(\mathcal{H}|\mathcal{D}\right) = \frac{\mathbb{P}\left(\mathcal{D}|\mathcal{H}\right)\mathbb{P}\left(\mathcal{H}\right)}{\mathbb{P}\left(\mathcal{D}\right)}$$

The prior $\mathbb{P}(H)$ is the probability that the hypothesis H is true before the data is considered. The posterior $\mathbb{P}\left(\mathcal{H}|\mathcal{D}\right)$ is the probability that H is true conditional on observing the data $\mathcal{D}$. The probability of observing the data $\mathcal{D}$ given that H is true is $\mathbb{P}\left(\mathcal{D}|\mathcal{H}\right)$. $\mathbb{P}\left(\mathcal{D}\right)$ is the total probability of the data taking into account all possible hypotheses. If the prior and likelihood are known for all hypotheses, then Bayes formula computes the posterior exactly.

In most experiments, the prior probabilities on hypotheses are not known. In this case, we either postulate a prior (Bayesian approach) or do our best using only the likelihood (frequentist approach).

Next, let us write the Bayes rule in terms of probability density functions. Let θ be unknown parameter, which we treat as a random variable and we put a probability distribution on it. This probability is then updated based on the received data. More specifically, let $f(\theta)$ be the *prior* probability density function on θ. We are also given the *conditional probability density* function, that is $f(x|\theta)$, which is the probability of observing the data x conditioning on the parameter θ. The *Bayesian posterior* belief of θ after observing the data x is the conditional probability density of θ given the sample, and can be computed by a direct application of Bayes' rule

$$f\left(\theta|x\right) = \frac{f(x|\theta)f(\theta)}{\int f(x|\theta')f(\theta')d\theta'}$$

Example: Let $X_1, X_2, \ldots, X_{90}$ stand for indicators of economic shocks in the past 90 trading days. An economic shock is an unexpected event that produces a significant change within an economy. Examples include shocks in the supply of commodities, rapid devaluation of a currency producing a shock to the import/export industry, or unpredictable political events (e.g. Trumps' election). Suppose no event occurs, i.e, $X_i = 0$, for $i = 1, \ldots, 90$. A frequentist statistician would estimate that the probability p of observing a shock is zero. Consider a Bayesian, who puts a beta distribution on the parameter p, given by $f(p) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}p^{\alpha-1}(1-p)^{\beta-1}$. (if $\alpha = \beta = 1$, the Beta distribution becomes a $(0,1)$ uniform distribution and $f(p) = 1$ for any $p \in [0,1]$). The conditional

probability density is given by

$$f(X|p) = \prod_{i=1}^{n} p^{X_i}(1-p)^{1-X_i} = p^{n\bar{X}}(1-p)^{n(1-\bar{X})}$$

We can apply Bayes rule to obtain the posterior distribution of p given the data X. Applying Bayes's rule, we obtain

$$f(p|X) \propto f(X|p)f(p) \propto p^{n\bar{X}+\alpha-1}(1-p)^{n(1-\bar{X})+\beta-1}.$$

Hence, the posterior distribution of the parameter p is $Beta(\alpha+n\bar{X}, \beta+n(1-\bar{X}))$. It is well known that the mean of a β distribution with (α, β) parameters is $\frac{1}{1+\frac{\beta}{\alpha}}$. Hence, the mean of our posterior distribution is

$$\frac{\alpha+n\bar{X}}{\alpha+\beta+n} = \frac{\alpha+\beta}{\alpha+\beta+n} \times \frac{\alpha}{\alpha+\beta} + \frac{n}{\alpha+\beta+n}\bar{X} \tag{1}$$

The above expression admits an intuitive interpretation. It is a convex combination of the prior mean $\frac{\alpha}{\alpha+\beta}$, and the sample mean $\bar{X}$. Assume that the Bayesian uses a totally uninformative prior, that is the uniform distribution $f(p) = 1$. Using Eq. (1), the mean of the posterior distribution equals $\frac{1+1}{1+1+90} \times \frac{1}{1+1} + \frac{90}{1+1+90} \times 0 = 0.0109$. Which one do you find more reasonable, the Bayesian or the frequentist estimate of the probability of no shocks to the economy?

1.1 Conjugate Prior

Bayesian statistics is often used in a sequential procedure. One starts from a prior, then combines it with the likelihood of the data, to obtain a posterior. This posterior would then be viewed as the new prior, combined with the likelihood of new data, to obtain another posterior. To make this sequential procedure computationally tractable, it is desirable that the posterior probability belongs to the same parametric class of probability distributions as the prior. This is achieved through the concept of conjugate prior, which we review next.

A family of distributions is **conjugate** with respect to a likelihood if, for any prior in this family, the posterior is in the same family. Conjugate prior and likelihood facilitate the computation of a posterior.

Example 1: Assume a Bernoulli random variable X with parameter p. Assume that the prior distribution on p is $f(p) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}p^{\alpha-1}(1-p)^{\beta-1}$. The likelihood function is $f(x|p) = p^x(1-p)^{1-x}$. Thus the posterior is $f(p|x) \propto f(x|p)f(p) \propto p^{\alpha+x-1}(1-p)^{\beta-x}$. We can thus conclude that the posterior distribution is $Beta_{(\alpha+x,\beta+1-x)}$. So, the beta distribution is conjugate with respect to the Bernoulli distribution.

Example 2: Consider a Gaussian distribution, where $X_1, X_2, \ldots, X_n$ are i.i.d. normal samples with unknown mean μ and known variance σ^2. Suppose the prior of μ follows a $Gaussian(a, b^2)$. The likelihood function is

$$f(x|\mu) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}},$$

where $x = (x_1, x_2, \ldots, x_n)$. The posterior distribution is

$$f(\mu|x) \propto f(x|\mu)f(\mu) \propto e^{-\frac{1}{2\sigma_n^2}(\mu-\mu_n)^2},$$

where $\sigma_n^2 := \frac{b^2\sigma^2}{\sigma^2+nb^2}$, $\mu_n := \frac{\sigma^2}{\sigma^2+nb^2} \times a + \frac{nb^2}{\sigma^2+nb^2} \times \bar{X}$. Hence the posterior is $Gaussian(\mu_n, \sigma_n^2)$.

Question: What is a 95% Bayesian confidence interval for the mean? It is $[\mu_n - z_{97.5\%}\sigma_n, \mu_n + z_{97.5\%}\sigma_n]$, where $z_{97.5\%}$ is the 97.5% quantile of the standard normal distribution.

Bayesian hypothesis testing. We use the posterior probabilities of the null and alternative hypothesis. More specifically, consider the null hypothesis $H_0 : \theta \in \Theta_0$ versus $H_1 : \theta \notin \Theta_0$. The Bayesian computes $\mathbb{P}(\theta \in \Theta_0|X)$, and $\mathbb{P}(\theta \notin \Theta_0|X)$, and rejects H_0 in favor of H_1 if the posterior $\mathbb{P}(\theta \notin \Theta_0|X) > \mathbb{P}(\theta \in \Theta_0|X)$.

1.2 How to choose priors?

One possibility is to choose a prior which conjugates to the likelihood. Another possibility is to choose subjective priors. A third possibility is to estimate priors from data, following an empirical Bayesian approach.

If you have no knowledge of the parameter a priori, a fourth possibility is to choose a non-informative prior. For instance, assume $X \sim Bin(n, p)$. A plausible noninformative prior on the parameter p would be $p \in U[0, 1]$, i.e., uniform on the $[0, 1]$ interval. However, by doing so, we would obtain that $\sqrt{p}$ has a higher density near one than near 0. This is because the square root of a number in $(0, 1)$ is larger than the number itself, thus we know intuitively that the distribution of $\sqrt{p}$ must concentrate its probability toward the right end of the interval $(0, 1)$. (Specifically, the density function would be $f_{\sqrt{p}}(x) = 2x$ for $0 < x < 1$). Thus *ignorance* about p seems to lead to *knowledge* about $\sqrt{p}$, which is paradoxical. We would like the prior to be invariant under reparametrization. We will achieve this by using the Fisher information.

Recall that the Fisher information equals $I(\theta) = \mathbb{E}_\theta \left[\left(\frac{\partial \log f(X|\theta)}{\partial \theta} \right)^2 \right]$. Let us define the *Jeffreys prior* as $\pi(\theta) \propto I(\theta)^{1/2}$. Consider a model $X \sim f(x|\theta)$, and its reparametrized version $X \sim g(x|\eta)$, where $\eta = h(\theta)$ and h is a differentiable monotone transformation of θ. Applying the chain rule, we would have that

$$\begin{aligned} I(\theta) &= \int \left(\frac{\partial}{\partial \theta} \log f(x|\theta) \right)^2 f(x|\theta) dx \\ &= \int \left(\frac{\partial}{\partial \theta} \log g(x|\eta) \right)^2 g(x|\eta) dx \\ &= \int \left(\frac{\partial}{\partial \eta} \log g(x|\eta) \Big|_{\eta=h(\theta)} h'(\theta) \right)^2 g(x|h(\theta)) dx \\ &= (h'(\theta))^2 I(h(\theta)). \end{aligned}$$

Next, recall that if X is a r.v. with pdf $f_X(x)$ that is continuous on its support, and $Y = g(X)$ where g is a monotone function of X, then the pdf f_Y of Y is given by

$$f_Y(y) = f_X(g^{-1}(y)) | \frac{\partial}{\partial y} g^{-1}(y) |$$

In our case $y = h(\theta)$, hence

$$\pi_Y(y) = \pi_x(h^{-1}(y)) \big| \frac{d}{dy} h^{-1}(y) \big|,$$

thus

$$\pi(h(\theta)) = \pi(\theta)\Big|\frac{d}{dh(\theta)}h^{-1}(h(\theta))\Big| = \pi(\theta)\left|\frac{d\theta}{dh(\theta)}\right| = \frac{\pi(\theta)}{|h'(\theta)|}.$$

Hence, for the Jeffreys prior $\pi(\theta)$, we have

$$\pi(h(\theta)) = \pi(\theta)\left|h'(\theta)\right|^{-1} \propto I(\theta)^{1/2}\left|h'(\theta)\right|^{-1} = I(h(\theta))^{1/2}$$

and thus the prior is invariant under reparametrization. The Jeffreys prior favors values of θ for which $I(\theta)$ is large. Hence, it minimizes the effect of the prior distribution relative to the information in the data.

Example: Let $X \sim Bin(n, p)$, where n is known, so that $f(x|p) = \binom{n}{x}p^x(1-p)^{n-x}$. Then

$$\frac{\partial^2 \log f(x|p)}{\partial^2 p} = -\frac{x}{p^2} - \frac{n-x}{(1-p)^2}.$$

Taking expectation and multiplying by -1 (we are using the equivalent representation $I(p) = -\mathbb{E}_p\left[\frac{\partial^2 \log f(X|p)}{\partial^2 p}\right]$, we obtain: $I(p) = \frac{n}{p} + \frac{n}{1-p} = \frac{n}{p(1-p)}$. Thus the Jeffreys prior is

$$\pi(p) \propto (p(1-p))^{-1/2},$$

which we recognize as the $Beta_{(1/2,1/2)}$ distribution.

Lecture 13: Linear Regression

IEOR E4709: Statistic Analysis and Time Series
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1 Annoucement

Midterm exam will be on Wednesday, March 12, 10:00-11:30am. Sample exam and midterm review sheet have been posted on Courseworks in a folder named "Midterm". Solutions to the sample exam will be posted on March 6 by 5pm. There will be a midterm review session conducted by the TA, Boxuan Li on Friday, March 7, 12-1:30pm, in 523 Mudd.

2 How to choose priors in Bayesian analysis?

One possibility is to choose a prior which conjugates to the likelihood. Another possibility is to choose subjective priors. A third possibility is to estimate priors from data, following an empirical Bayesian approach.

If you have no knowledge of the parameter a priori, a fourth possibility is to choose a non-informative prior. For instance, assume $X \sim Bin(n, p)$. A plausible noninformative prior on the parameter p would be $p \in U[0, 1]$, i.e., uniform on the $[0, 1]$ interval. However, by doing so, we would obtain that $\sqrt{p}$ has a higher density near one than near 0. This is because the square root of a number in $(0, 1)$ is larger than the number itself, thus we know intuitively that the distribution of $\sqrt{p}$ must concentrate its probability toward the right end of the interval $(0, 1)$. (Specifically, the density function would be $f_{\sqrt{p}}(x) = 2x$ for $0 < x < 1$). Thus *ignorance* about p seems to lead to *knowledge* about $\sqrt{p}$, which is paradoxical. We would like the prior to be invariant under reparametrization. We will achieve this by using the Fisher information.

Recall that the Fisher information equals $I(\theta) = \mathbb{E}_\theta \left[\left(\frac{\partial \log f(X|\theta)}{\partial \theta} \right)^2 \right]$. Let us define the *Jeffreys prior* as $\pi(\theta) \propto I(\theta)^{1/2}$. Consider a model $X \sim f(x|\theta)$, and its reparametrized version $X \sim g(x|\eta)$, where $\eta = h(\theta)$ and h is a differentiable monotone transformation of θ. Applying the chain rule,

we would have that

$$\begin{aligned} I(\theta) &= \int \left(\frac{\partial}{\partial \theta} \log f(x|\theta) \right)^2 f(x|\theta) dx \\ &= \int \left(\frac{\partial}{\partial \theta} \log g(x|\eta) \right)^2 g(x|\eta) dx \\ &= \int \left(\frac{\partial}{\partial \eta} \log g(x|\eta) \Big|_{\eta=h(\theta)} h'(\theta) \right)^2 g(x|h(\theta)) dx \\ &= (h'(\theta))^2 I(h(\theta)). \end{aligned}$$

Next, recall that if X is a r.v. with pdf $f_X(x)$ that is continuous on its support, and $Y = g(X)$ where g is a monotone function of X, then the pdf f_Y of Y is given by

$$f_Y(y) = f_X(g^{-1}(y)) | \frac{\partial}{\partial y} g^{-1}(y) |$$

In our case $y = h(\theta)$, hence

$$\pi_Y(y) = \pi_x(h^{-1}(y)) \big| \frac{d}{dy} h^{-1}(y) \big|,$$

thus

$$\pi(h(\theta)) = \pi(\theta) \big| \frac{d}{dh(\theta)} h^{-1}(h(\theta)) \big| = \pi(\theta) \left| \frac{d\theta}{dh(\theta)} \right| = \frac{\pi(\theta)}{|h'(\theta)|}.$$

Hence, for the Jeffreys prior $\pi(\theta)$, we have

$$\pi(h(\theta)) = \pi(\theta) \left| h'(\theta) \right|^{-1} \propto I(\theta)^{1/2} \left| h'(\theta) \right|^{-1} = I(h(\theta))^{1/2}$$

and thus the prior is invariant under reparametrization. The Jeffreys prior favors values of θ for which $I(\theta)$ is large. Hence, it minimizes the effect of the prior distribution relative to the information in the data.

Example: Let $X \sim Bin(n, p)$, where n is known, so that $f(x|p) = \binom{n}{x} p^x (1-p)^{n-x}$. Then

$$\frac{\partial^2 \log f(x|p)}{\partial^2 p} = -\frac{x}{p^2} - \frac{n-x}{(1-p)^2}.$$

Taking expectation and multiplying by -1 (we are using the equivalent representation $I(p) = -\mathbb{E}_p \left[\frac{\partial^2 \log f(X|p)}{\partial^2 p} \right]$, we obtain: $I(p) = \frac{n}{p} + \frac{n}{1-p} = \frac{n}{p(1-p)}$. Thus the Jeffreys prior is

$$\pi(p) \propto (p(1-p))^{-1/2},$$

which we recognize as the $Beta_{(1/2,1/2)}$ distribution.

3 Simple Linear Regression

Consider a linear model

$$y = \alpha + \beta x,$$

where x is the *explanatory variable*, and y is the *response variable/dependent variable*. Suppose we are confronted with the following question: given observed data $(x_1, y_1), \ldots, (x_n, y_n)$, estimate α and β. The most direct approach is to minimize the ordinary least square (OLS), i.e.,

$$\min_{\alpha,\beta} \sum_{i=1}^{n} (y_i - (\alpha + \beta x_i))^2.$$

By setting the first order derivative to zero, it is straightforward to see that we obtain

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}, \qquad \hat{\beta} = S_{xx}^{-1} S_{xy} = Cov(x, y)/Var(x), \tag{1}$$

where $\bar{y} := \frac{1}{n}\sum_{i=1}^{n} y_i$, $\bar{x} := \frac{1}{n}\sum_{i=1}^{n} x_i$ are the sample means of the response variables, and of the explanatory variables, respectively. The quantities $S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2$, and $S_{xy} = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$. We also use $Cov(x, y)$ to denote the sample covariance of x and y, and $Var(x)$ to denote the sample variance of x.

Next, consider the statistical model

$$y_i = \alpha + \beta x_i + \epsilon_i,$$

where ϵ_i is a zero-mean random variable. This model is called a *simple linear regression model*. It is a linear model because $\mathbb{E}[y_i|x_i] = \alpha + \beta x_i$, that is the *best linear* estimate for y_i given x_i, $\mathbb{E}[y_i|x_i]$, is linear in x_i.

Let us make the following assumptions: ϵ_i's have equal variance, ϵ_i's are independent, ϵ_i's have zero mean and variance σ^2. We define $\hat{y}_i := \hat{\alpha} + \hat{\beta} x_i$ as the *prediction* of the response given the explanatory variable x_i. The quantity $\hat{\epsilon}_i = y_i - \hat{y}_i$ is called the *residual*.

In linear regression models, the explanatory variable x is assumed to be observed (known) and not random. The linear model describes the relation between the mean of the response variable and the controlled explanatory variable. Consider the OLS estimates $\hat{\alpha}$ and $\hat{\beta}$ in Eq. (1). We can show that both of them are unbiased estimates of α and β in the simple linear regression model. Indeed

$$\begin{aligned}
\mathbb{E}\left[\hat{\beta}\right] &= \mathbb{E}\left[S_{xx}^{-1} S_{xy}\right] = S_{xx}^{-1}\mathbb{E}\left[S_{xy}\right] \\
&= S_{xx}^{-1} \sum_{i=1}^{n} (x_i - \bar{x})(\mathbb{E}[y_i] - \mathbb{E}\left[\bar{y}\right]) \\
&= S_{xx}^{-1} \sum_{i=1}^{n} (x_i - \bar{x})\left((\alpha + \beta x_i) - (\alpha + \beta\bar{x})\right) \\
&= S_{xx}^{-1} \beta S_{xx} \\
&= \beta.
\end{aligned}$$

Moreover,

$$\mathbb{E}\left[\hat{\alpha}\right] = \mathbb{E}\left[\bar{y}\right] - \mathbb{E}\left[\hat{\beta}\right]\bar{x} = \alpha + \beta\bar{x} - \beta\bar{x} = \alpha.$$

Assume further that ϵ_i's are Gaussian and independent. It can be shown that the estimators $\hat{\alpha}$ and $\hat{\beta}$ are jointly normal with variances and covariances given by

$$\begin{aligned}
Var(\hat{\beta}) &= S_{xx}^{-1}\sigma^2 \\
Cov(\hat{\alpha}, \hat{\beta}) &= -\bar{x} S_{xx}^{-1}\sigma^2 \\
Var(\hat{\alpha}) &= \frac{\sigma^2}{n} + \bar{x}^2 S_{xx}^{-1}\sigma^2.
\end{aligned} \tag{2}$$

Notice that the variances of $\hat{\alpha}$ and $\hat{\beta}$ and their covariance all depend on the unknown parameter σ^2. As σ is usually unknown, we would need to estimate σ^2 from the data. Define the sum of the squares of residuals

$$S_{rsdl} := \sum_{i=1}^{n} \hat{\epsilon}_i^2$$

It holds that that $\frac{S_{rsdl}}{\sigma^2}$ follows a χ^2 distribution with $n-2$ degrees of freedom. The number of degrees of freedom is equal to the number of observations minus the number of estimated regression coefficients. Notice that the expected value of a χ^2 distribution with k degrees of freedom is k. Hence, the estimator $\hat{\sigma}^2_{rsld} := \frac{S_{rsdl}}{n-2}$ is such that $\mathbb{E}\left[\hat{\sigma}^2_{rsld}\right] = \sigma^2$, i.e. it is an unbiased estimator of σ^2. Moreover, it can be proven that $\hat{\alpha}$ and $\hat{\beta}$ are independent of S_{rsdl} (next homework problem).

Replacing σ^2 with $\hat{\sigma}^2_{rsld}$ in the expression for the variances of Eq. (2), we obtain unbiased estimates of the variances of $\hat{\alpha}$ and $\hat{\beta}$ (This can be easily checked). Such an estimate of the standard deviation is called the *standard error* (s.e.) of the estimate. Thus, the standard errors of $\hat{\alpha}$ and $\hat{\beta}$ are

$$\begin{aligned} s.e.(\hat{\alpha}) &= \hat{\sigma}_{rsld}\sqrt{\left(\frac{1}{n} + \bar{x}^2 S_{xx}^{-1}\right)} \\ s.e.(\hat{\beta}) &= \hat{\sigma}_{rsld}\sqrt{S_{xx}^{-1}}, \end{aligned}$$

where $\hat{\sigma}_{rsld} = \sqrt{\hat{\sigma}^2_{rsld}}$. The standard errors of $\hat{\alpha}$ and $\hat{\beta}$ are a measure of how precisely the intercept and the slope have been estimated.

4 Hypothesis Testing and Confidence Intervals for Simple Linear Regression

Under the normality assumption of ϵ_i's, the statistic $t_1 := \frac{\hat{\beta}-\beta}{s.e(\hat{\beta})}$ follows a Student's t distribution with $n-2$ degrees of freedom (there are $n-2$ degrees of freedom because we are estimating two regression coefficients). Similarly, we have that $t_2 := \frac{\hat{\alpha}-\alpha}{s.e(\hat{\alpha})}$ follows a student's t distribution with $n-2$ degrees of freedom. This allows us to construct confidence intervals and do hypothesis testing.

Assume we want to test the null hypothesis $\beta = 0$ versus the alternative hypothesis $\beta \neq 0$. Under the null hypothesis, the test statistics $t_1 = \frac{\hat{\beta}}{s.e(\hat{\beta})}$ is distributed as a Student's t random variable with $n-2$ degrees of freedom. The test is carried out by comparing the observed value of the statistics t_1 with the critical value obtained from the t-table, which is $t_{n-2,\frac{\varrho}{2}}$, where ϱ is the specified significance level. (We are using ϱ as opposed to α which was used in previous lectures, because here α is being used to denote the intercept in the regression model). Here, we recall that $t_{n-2,\frac{\varrho}{2}}$ is the $\left(1-\frac{\varrho}{2}\right)$ percentile of a t distribution with $n-2$ degrees of freedom. Notice also that we divided by 2 because we are considering a two-sided alternative hypothesis. Consequently, the null hypothesis is rejected at the significance level ϱ if

$$|t_1| \geq t_{n-2,\frac{\varrho}{2}}$$

Similar criteria can be used to construct confidence intervals for the regression coefficients. More precisely, the $(1-\varrho)\times 100\%$ confidence interval for α is given by

$$\hat{\alpha} \pm t_{n-2,\frac{\varrho}{2}} \times s.e.(\hat{\alpha})$$

Similarly, the $(1-\varrho)\times 100\%$ confidence interval for β is given by

$$\hat{\beta} \pm t_{n-2,\frac{\varrho}{2}} \times s.e.(\hat{\beta})$$

Lecture 14: Simple and Multiple Linear Regression

IEOR E4709: Statistic Analysis and Time Series
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1 Simple Linear Regression

Last lecture, we have introduced simple linear regression and the corresponding estimators.

Note: The OLS estimator is a consistent estimator, that is, as $n \to \infty$ the estimators $\hat{\alpha}$ and $\hat{\beta}$ converge, in probability, to the true value of the parameters α and β, respectively. We will provide a proof in the most general case of multiple linear regression, which also covers the simple linear regression as a special case.

Remark 1.1. *The software package R can be used to compute statistics of linear regression models. The R command is lm. For instance, if returns denotes the time series of stock returns and index denotes the time series of index returns lm(returns* $\sim$ *index) regresses the stock returns versus the index returns. Define lm.r = lm(returns* $\sim$ *index). Then, type summary(lm.r) to obtain a summary of the statistics, including estimates of the regression coefficients, standard errors, values of the t statistics, etc...*

1.1 Mean Response versus Predicted Response

The mean response at a given value x_i is an estimate of the the mean of the y values in the population when $x = x_i$, that is

$$\hat{y}_i = \hat{\mathbb{E}}\left[y_i | x_i\right] = \hat{\alpha} + \hat{\beta} x_i.$$

The variance of the mean response is given by

$$Var\left(\hat{\alpha} + \hat{\beta} x_i\right) = Var\left(\hat{\alpha}\right) + x_i^2 Var\left(\hat{\beta}\right) + 2 x_i Cov\left(\hat{\alpha}, \hat{\beta}\right),$$

Assume that the ϵ_i's are i.i.d. Gaussian random variables. Using the expressions for variances and covariances of the least square estimates given above:

$$\begin{aligned} Var(\hat{\beta}) &= S_{xx}^{-1} \sigma^2 \\ Cov(\hat{\alpha}, \hat{\beta}) &= -\bar{x} S_{xx}^{-1} \sigma^2 \\ Var(\hat{\alpha}) &= \frac{\sigma^2}{n} + \bar{x}^2 S_{xx}^{-1} \sigma^2, \end{aligned}$$

along with the identity $\sum_j (x_j - \bar{x})^2 = \sum_j x_j^2 - \frac{1}{n}\left(\sum_j x_j\right)^2$, we can further simplify to

$$Var\left(\hat{\alpha} + \hat{\beta}x_i\right) = \sigma^2\left(\frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_j (x_j - \bar{x})^2}\right). \tag{1}$$

Suppose we observe a new explanatory variable x_{new}. We would like to predict the corresponding response y_{new}. The variability of this prediction has two sources: the variation in the estimates $\hat{\beta}_0$ and $\hat{\beta}_1$, and the variation due to the fact that y_{new} will not be equal to its expectation, since even if we knew the parameters exactly, the response value will generally not be equal to its expectation because of the error term ϵ_{new}. We have $\hat{y}_{new} = \hat{\alpha} + \hat{\beta}x_{new}$.

Next, compute the variance of the predicted distribution of residuals at the given point x_{new}. The distribution of the predicted response is distribution of the predicted residuals at the given point x_{new}. Hence, the variance is given by

$$Var\left(y_{new} - (\hat{\alpha} + \hat{\beta}x_{new})\right) = Var(y_{new}) + Var\left(\hat{\alpha} + \hat{\beta}x_{new}\right).$$

The second part of the above expression is given by (1). Since $Var(y_{new}) = \sigma^2$, we get that the variance of the predicted response is given by

$$Var\left(y_{new} - (\hat{\alpha} + \hat{\beta}x_{new})\right) = \sigma^2\left(1 + \frac{1}{n} + \frac{(x_{new} - \bar{x})^2}{\sum_j (x_j - \bar{x})^2}\right). \tag{2}$$

Notice that the variance of the mean response is smaller than the variance of the predicted response, which is intuitive. There is greater uncertainty in predicting one observation (the next observation) than in estimating the mean response given x_i. Notice that, as for mean response, the variance of the predicted response also depends on the unknown constant σ^2. We thus need to replace σ^2 with its estimate $\hat{\sigma}^2_{rsdl}$, and work with the standard error of the predicted response given by

$$\hat{\sigma}_{rsld}\sqrt{\left(1 + \frac{1}{n} + \frac{(x_{new} - \bar{x})^2}{\sum_j (x_j - \bar{x})^2}\right)},$$

and we use

$$s.e(\hat{y}_i) = \hat{\sigma}_{rsld}\sqrt{\left(\frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_j (x_j - \bar{x})^2}\right)}$$

for the standard error of the mean response.

It follows that the $(1 - \varrho) \times 100\%$ confidence interval for the mean response, i.e., the expected value of y_i given x_i, i.e. for $\mathbb{E}[y_i|x_i]$, is given by

$$\hat{y}_i \pm t_{n-2,\frac{\varrho}{2}} \times s.e.(\hat{y}_i).$$

Suppose we are interested in the predicted response, i.e., we want to predict a future value of y_{new} given a new observation x_{new}. The prediction interval for y_{new} given a new observation x_{new} is

$$\hat{y}_{new} \pm t_{n-2,\frac{\varrho}{2}} \times s.e.(y_{new} - \hat{y}_{new}).$$

The formula for the prediction interval is similar to that of a confidence interval, except that the variability is higher because there is the additional term "1" in the formula of the standard error in (2). This is expected, because there is more uncertainty around a specific predicted value than around an average.

2 Multiple Linear Regression

Consider the following *multiple linear regression* model:

$$y = \boldsymbol{x}^\top \boldsymbol{\beta} + \epsilon,$$

where $\boldsymbol{x}$ is a p dimensional (column) vector of explanatory variables. We will use bold lowercase symbols throughout to denote vectors, and bold uppercase symbols to denote matrices. For instance, if $\boldsymbol{x} = (1, x)^\top$ and $\boldsymbol{\beta} = (\alpha, \beta)^\top$, then we obtain the simple linear regression model. Let us denote by $\boldsymbol{X} := (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n)^\top$ the matrix of explanatory variables, by $\boldsymbol{y} := (y_1, \ldots, y_n)^\top$ the vector of response variables, and by $\boldsymbol{\epsilon} = (\epsilon_1, \ldots, \epsilon_n)^\top$ the vector of noise terms. The vector $\boldsymbol{x}_1 = (1, 1, \ldots, 1)^\top$. We may then write the multiple linear regression with the data set $\boldsymbol{X}$ as

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}.$$

To calculate the least squares estimates, we proceed as in the simple linear regression model. We calculate the first derivative of $S_{rsdl} = \sum_{i=1} \hat{\epsilon}_i^2 = \sum_{i=1}^n \left[y_i - \hat{\beta}_1 - \hat{\beta}_2 x_{i2} - \hat{\beta}_3 x_{i3} - \cdots - \hat{\beta}_p x_{ip} \right]^2$ with respect to each $\hat{\beta}_j$, $j = 0, 1, \ldots, p$, and set them to zero.

$$\begin{aligned}
\frac{\partial S_{rsdl}}{\partial \hat{\beta}_1} &:= 2 \sum_{i=1}^n \left[y_i - \hat{\beta}_1 - \hat{\beta}_2 x_{i2} - \hat{\beta}_3 x_{i3} - \ldots \hat{\beta}_p x_{ip} \right] \times (-1) = 0 \\
\frac{\partial S_{rsdl}}{\partial \hat{\beta}_2} &:= 2 \sum_{i=1}^n \left[y_i - \hat{\beta}_1 - \hat{\beta}_2 x_{i2} - \hat{\beta}_3 x_{i3} - \ldots \hat{\beta}_p x_{ip} \right] \times (-x_{i2}) = 0 \\
\frac{\partial S_{rsdl}}{\partial \hat{\beta}_3} &:= 2 \sum_{i=1}^n \left[y_i - \hat{\beta}_1 - \hat{\beta}_2 x_{i2} - \hat{\beta}_3 x_{i3} - \ldots \hat{\beta}_p x_{ip} \right] \times (-x_{i3}) = 0 \\
\ldots \quad \ldots \quad \ldots \quad \ldots & \\
\frac{\partial S_{rsdl}}{\partial \hat{\beta}_p} &:= 2 \sum_{i=1}^n \left[y_i - \hat{\beta}_1 - \hat{\beta}_2 x_{i2} - \hat{\beta}_3 x_{i3} - \ldots \hat{\beta}_p x_{ip} \right] \times (-x_{ip}) = 0.
\end{aligned}$$

Rewriting this in matrix notation, we obtain

$$\boldsymbol{X}^\top \boldsymbol{X} \hat{\boldsymbol{\beta}} = \boldsymbol{X}^\top \boldsymbol{y},$$

or equivalently

$$\hat{\boldsymbol{\beta}} = \left(\boldsymbol{X}^\top \boldsymbol{X} \right)^{-1} \boldsymbol{X}^\top \boldsymbol{y}.$$

Lecture 15: Multiple Linear Regression

IEOR E4709: Statistic Analysis and Time Series
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1 Least Square Estimators and Residual: A Projection View

Recall that the least square estimator in a multiple linear regression model is given by

$$\hat{\boldsymbol{\beta}} = \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top \boldsymbol{y}.$$

Define the matrix $P_{\boldsymbol{X}} := \boldsymbol{X}\left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top$ and I_n the n by n identity matrix. Both are symmetric matrices. The square matrix $P_{\boldsymbol{X}}$ is a projection matrix, i.e., $P_{\boldsymbol{X}}^2 = P_{\boldsymbol{X}}$. Moreover, $I_n - P_{\boldsymbol{X}}$ is also a projection matrix, i.e., $\left(I_n - P_{\boldsymbol{X}}\right)^2 = I_n - P_{\boldsymbol{X}}$, and $P_{\boldsymbol{X}}\left(I_n - P_{\boldsymbol{X}}\right) = 0$. Then

$$\hat{\boldsymbol{y}} = \boldsymbol{X}\hat{\boldsymbol{\beta}} = P_{\boldsymbol{X}}\boldsymbol{y} \tag{1}$$

and the residuals

$$\hat{\boldsymbol{\epsilon}} = \left(I_n - P_{\boldsymbol{X}}\right)\boldsymbol{y} = \left(I_n - P_{\boldsymbol{X}}\right)\boldsymbol{\epsilon}. \tag{2}$$

The last equality can also be geometrically understood from the fact that the projection of the vector $\boldsymbol{y}$ on the complement of $Span(\boldsymbol{X})$ (i.e., the span generated by the columns of the matrix $\boldsymbol{X}$) is the same as the projection of $\boldsymbol{\epsilon}$ on the complement of $Span(\boldsymbol{X})$. This is because the vector $\boldsymbol{X}\boldsymbol{\beta}$ belongs to the $Span(\boldsymbol{X})$ and hence its projection on the complement of the $Span(\boldsymbol{X})$ is zero. Formally,

$$\begin{aligned}
\left(I_n - P_{\boldsymbol{X}}\right)\boldsymbol{y} &= \left(I_n - \boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top\right)\boldsymbol{y} \\
&= \left(I_n - \boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top\right)(\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}) \\
&= \left(I_n - \boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top\right)\boldsymbol{\epsilon} \\
&= \left(I_n - P_{\boldsymbol{X}}\right)\boldsymbol{\epsilon}
\end{aligned}$$

It can be easily seen that $\hat{\boldsymbol{\beta}}$ is an unbiased estimator of $\boldsymbol{\beta}$. To see it, observe that $\mathbb{E}[\hat{\boldsymbol{\beta}}] = \left(\boldsymbol{X}^\top\boldsymbol{X}\right)^{-1}\boldsymbol{X}^\top\mathbb{E}[\boldsymbol{y}] = \left(\boldsymbol{X}^\top\boldsymbol{X}\right)^{-1}\boldsymbol{X}^T\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{\beta}$.

It follows immediately that $\hat{\boldsymbol{\epsilon}}$ has zero mean because ϵ_i's have zero mean. Moreover, if the ϵ_i's

are Gaussian, then $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\epsilon}}$ are jointly normal. They are independent because

$$\begin{aligned} cov\left(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\epsilon}}\right) &= cov\left(\left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top \boldsymbol{y}, (I_n - P_{\boldsymbol{X}})\boldsymbol{y}\right) \\ &= \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top cov(\boldsymbol{y})(I_n - P_{\boldsymbol{X}})^\top \\ &= \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top cov(\boldsymbol{\epsilon})(I_n - P_{\boldsymbol{X}})^\top \\ &= \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top \sigma^2 I_n (I_n - P_{\boldsymbol{X}})^\top \\ &= \sigma^2 \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \left[(I_n - P_{\boldsymbol{X}})\boldsymbol{X}\right]^\top \\ &= 0, \end{aligned}$$

where we have used that $P_{\boldsymbol{X}}\boldsymbol{X} = \boldsymbol{X}$ and that $cov(\boldsymbol{y}) = cov(\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}) = cov(\boldsymbol{\epsilon})$, because $\boldsymbol{X}\boldsymbol{\beta}$ is a deterministic quantity, and hence its covariance matrix is zero.

Next, let us compute the variance of the OLS estimator. We have that

$$\begin{aligned} cov\left(\hat{\boldsymbol{\beta}}\right) &= \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top cov\left(\boldsymbol{y}\right) \left(\left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top\right)^\top \\ &= \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top \sigma^2 I_n \left(\left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top\right)^\top \\ &= \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \sigma^2 \boldsymbol{X}^\top \boldsymbol{X} \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \\ &= \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \sigma^2. \end{aligned}$$

In the third equality above, we have used that $(\boldsymbol{A}\boldsymbol{B})^\top = \boldsymbol{B}^\top \boldsymbol{A}^\top$, and that the matrix $\boldsymbol{X}^\top \boldsymbol{X}$ is symmetric.

Remark 1.1. *The OLS estimator is not only an unbiased but also a consistent estimator of $\boldsymbol{\beta}$, that is, as $n \to \infty$, the estimators $\hat{\beta}$ converge, in probability, to β. The intuition is as follows. The entries of $\boldsymbol{X}^\top \boldsymbol{X}$ are unbounded (they go to infinity) as n gets very large. For example, the $(1,1)$-th entry is n and the (j,j)-th entry is $\sum_{i=1}^n x_{ij}^2$. Therefore,*

$$\lim_{n\to\infty} (\boldsymbol{X}^\top \boldsymbol{X})^{-1} = 0,$$

and the covariance of $\hat{\boldsymbol{\beta}}$ converges to zero. This means that the distribution collapses around its expected value, namely $\boldsymbol{\beta}$. So, the limit in the probability of $\hat{\boldsymbol{\beta}}$ is given by

$$plim_{n\to\infty} \hat{\boldsymbol{\beta}} = \boldsymbol{\beta},$$

and the OLS estimator is consistent.

Denote $S_{rsdl} := \sum_{i=1}^{n} \hat{\epsilon}_i^2 = \hat{\boldsymbol{\epsilon}}^\top \hat{\boldsymbol{\epsilon}}$. Then

$$\begin{aligned}
\mathbb{E}\left[S_{rsdl}\right] &= \mathbb{E}\left[\left(\left(I_n - P_{\boldsymbol{X}}\right)\boldsymbol{y}\right)^\top \left(\left(I_n - P_{\boldsymbol{X}}\right)\boldsymbol{y}\right)\right] \\
&= \mathbb{E}\left[\left(\left(I_n - P_{\boldsymbol{X}}\right)\boldsymbol{\epsilon}\right)^\top \left(\left(I_n - P_{\boldsymbol{X}}\right)\boldsymbol{\epsilon}\right)\right] \\
&= \mathbb{E}\left[tr\left(\left(\left(I_n - P_{\boldsymbol{X}}\right)\boldsymbol{\epsilon}\right)\left(\left(I_n - P_{\boldsymbol{X}}\right)\boldsymbol{\epsilon}\right)^\top\right)\right] \\
&= tr\left(\left(I_n - P_{\boldsymbol{X}}\right)\mathbb{E}\left[\boldsymbol{\epsilon}\boldsymbol{\epsilon}^\top\right]\left(I_n - P_{\boldsymbol{X}}\right)^\top\right) \\
&= tr\left(\left(I_n - P_{\boldsymbol{X}}\right)\sigma^2 I_n \left(I_n - P_{\boldsymbol{X}}\right)^\top\right) \\
&= \sigma^2\left(tr(I_n) - tr(P_{\boldsymbol{X}})\right) \\
&= (n-p)\sigma^2,
\end{aligned}$$

where in the first equality we have used the relation in (2). In the above chain of equalities, we have also used that the trace is a linear operator, $P_{\boldsymbol{X}}\left(I_n - P_{\boldsymbol{X}}\right) = 0$, and that $tr(P_{\boldsymbol{X}}) = tr(\boldsymbol{X}\left(\boldsymbol{X}^\top\boldsymbol{X}\right)^{-1}\boldsymbol{X}^\top) = tr(\boldsymbol{X}^\top\boldsymbol{X}\left(\boldsymbol{X}^\top\boldsymbol{X}\right)^{-1}) = tr(I_p) = p$ by the commutativity of the trace operator (i.e., if A is a $m \times n$ matrix and B is a $n \times m$ matrix, then $tr(AB) = tr(BA)$).

The above argument implies that an unbiased estimator of σ^2 is

$$\hat{\sigma}_{rsdl}^2 := \frac{1}{n-p} S_{rsdl}.$$

If the error terms ϵ_i's are Gaussian, the statistics $\frac{S_{rsdl}}{\sigma^2}$ follows a χ^2 distribution with $n-p$ degrees of freedom. We lose p degrees of freedom because we are estimating p parameters. We also have that, for $j = 1, \ldots, p$,

$$\frac{\hat{\beta}_j - \beta_j}{s.e(\hat{\beta}_j)}.$$

follows a Student's t-distribution with $n-p$ degrees of freedom. Above, $s.e(\hat{\beta}_j) = \sqrt{\left((\boldsymbol{X}^\top\boldsymbol{X})^{-1}\right)_{jj}}\hat{\sigma}_{rsdl}$. Using these statistics, we can then construct confidence intervals for each β_j. Moreover, we can use the above statistics to do hypothesis testing, paralleling to what we have done in the case of simple linear regression.

Lecture 16: BLUE Estimator, and R^2 Coefficient, and Model Building

1 Best Linear Unbiased Estimator (BLUE)

The OLS $\hat{\boldsymbol{\beta}}$ is the best linear unbiased estimator (BLUE) of $\boldsymbol{\beta}$. The Gauss-Markov theorem states that, among all linear unbiased estimators of $\boldsymbol{\beta}$, the estimator $\hat{\boldsymbol{\beta}}$ is the one with the smallest variance. More formally, consider any other linear unbiased estimator $\tilde{\boldsymbol{\beta}}$ of $\boldsymbol{\beta}$, i.e. of the form $\tilde{\boldsymbol{\beta}} = \boldsymbol{A}y$ where $\boldsymbol{A}$ is a $p \times n$ non-stochastic matrix such that $\mathbb{E}\left[\tilde{\boldsymbol{\beta}}\right] = \boldsymbol{\beta}$. Then $Var(\tilde{\boldsymbol{\beta}}) \geq Var(\hat{\boldsymbol{\beta}})$, i.e., the matrix $Var(\tilde{\boldsymbol{\beta}}) - Var(\hat{\boldsymbol{\beta}})$ is positive semidefinite. Notice that the theorem does not require the assumptions that the vector $\boldsymbol{\epsilon}$ of error terms is Gaussian, but just that ϵ_i's are i.i.d. with zero mean and finite variance σ^2.

We do not have time to see the proof of this result. But, the interested student is referred to any standard statistics textbook (e.g. Mathematical Statistics and Data Analysis by John Rice).

2 The R^2 Coefficient

Suppose we have p explanatory variables $\boldsymbol{x}$. Suppose we observe $(\boldsymbol{x}_1^\top, y_1), \ldots, (\boldsymbol{x}_n^\top, y_n)$. Let the linear regression model $y = \boldsymbol{x}^\top \boldsymbol{\beta} + \epsilon$, and refer to it as the *full* model. Consider a reduced model $y = \boldsymbol{x}_a^T \boldsymbol{\beta}'$, where $\boldsymbol{x}_a$ stands for the first q explanatory variables in $\boldsymbol{x}$, i.e. $\boldsymbol{x}^\top = (\boldsymbol{x}_a^\top, \boldsymbol{x}_b^\top)$. Denote $\hat{\boldsymbol{y}}_{full}$ and $\hat{\boldsymbol{y}}_{rdcd}$ as the in-sample prediction under the full and reduced models, respectively. Moreover, denote by $\hat{\boldsymbol{\epsilon}}_{full}$ and $\hat{\boldsymbol{\epsilon}}_{rdcd}$ the residuals under the full and reduced models, respectively, i.e.,

$$\hat{\boldsymbol{\epsilon}}_{full} = \boldsymbol{y} - \hat{\boldsymbol{y}}_{full}, \qquad \hat{\boldsymbol{\epsilon}}_{rdcd} = \boldsymbol{y} - \hat{\boldsymbol{y}}_{rdcd}.$$

Then

$$\sum_{i=1}^{n} \hat{\epsilon}_{rdcd,i}^2 = \sum_{i=1}^{n} \left(\hat{y}_{rdcd,i} - \hat{y}_{full,i} + \hat{y}_{full,i} - y_i\right)^2 = \sum_{i=1}^{n} \left(\hat{y}_{rdcd,i} - \hat{y}_{full,i}\right)^2 + \sum_{i=1}^{n} \hat{\epsilon}_{full,i}^2. \tag{1}$$

Above, we have used the fact that the residual $\hat{\boldsymbol{\epsilon}}_{full}$ is orthogonal to the column space of X, and $\hat{y}_{full} - \hat{y}_{rdcd}$ is in the column space of X. Adding more explanatory variables always reduces the residual sum of squares. When $\boldsymbol{x}_a^\top = 1$, i.e., the constant is the only explanatory variable in the reduced model, then we have $\hat{y}_{rdcd,i} = \bar{y}$. In this case, $\sum_{i=1}^n \hat{\epsilon}_{rdcd,i}^2 = \sum_{i=1}^n (y_i - \bar{y})^2 =: S_{ttl}$ becomes the *total sum of squares*, and $\sum_{i=1}^n \left(\hat{y}_{rdcd,i} - \hat{y}_{full,i}\right)^2 = \sum_{i=1}^n \left(\hat{y}_i - \bar{y}\right)^2 =: S_{rgsn}$. This is referred to as the *regression sum of squares* of the full model: it is the sum of squares of the differences

between the fitted values and the average response variable. In other words, it measures how far the regression line is from $\bar{y}$. Recall that $S_{rsdl} := \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$. Hence, we can rewrite Eq. (1) as

$$S_{ttl} = S_{rgsn} + S_{rsdl}.$$

The R **squared** of a linear regression model is defined by

$$R^2 := \frac{S_{rgsn}}{S_{ttl}},$$

or equivalently

$$R^2 = 1 - \frac{S_{rsdl}}{S_{ttl}}.$$

Hence, $R^2 \leq 1$.
Question: What does it mean a high R^2?

Answer: It means that the model fits the data well. Indeed, R^2 gives the proportion of variability in $\boldsymbol{y}$ explained by the predictors. However, a high R^2 does not necessarily mean that the model is correct. Indeed, some explanatory variables can still be missing, and the prediction can thus be very poor. One can always increase R^2 by adding more explanatory variables, but this may lead to over-fitting, that is, basically, you would be fitting the noise as opposed to fitting the data. A small R^2 does not necessarily mean that the model is wrong: it may still be the case that the response variable is purely random. Because financial data are usually very noisy, we do not expect a high R^2. It is more important that the model has a high prediction power rather than a perfect in-sample fitting: this means that out-of-sample testing is important.

3 Model Building

Example: Suppose we want to test if (H_0) the reduced model $y = \beta_1 + \beta_2 x_2 + \epsilon$ is true, versus the hypothesis (H_1) the full model $y = \beta_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \epsilon$ is true. If either β_3 or β_4 are statistically different from zero, it means that the full model accounts better for variation in the data. Notice that, even if the reduced model is correct, we still have that the sum of square residuals of the full model is lower or equal than the corresponding sum in the reduced model.

3.1 Comparison of Full and Reduced Models

A random variable $X = \frac{U_1/d_1}{U_2/d_2}$ follows a *F-distribution* with degrees of freedom d_1 and d_2 if U_1 and U_2 have chi-squared distributions with d_1 and d_2 degrees of freedom, respectively, and U_1 and U_2 are independent. Suppose it is desired to test the null hypothesis that $\beta_{q+1} = \beta_{q+2} = \ldots = \beta_p = 0$ versus the alternative hypothesis that at least one of the predictors is not zero. Consider the statistic

$$F := \frac{\frac{S_{rdcd,rsdl} - S_{full,rsdl}}{p-q}}{\frac{S_{full,rsdl}}{n-p}},$$

where $S_{rdcd,rsdl}$ and $S_{full,rsdl}$ are the residual sums of squares of the reduced and full models, respectively. Under the null hypothesis, and assuming that the error terms in the linear regression

model are Gaussian i.i.d. random variables, this statistics follows a F distribution with degrees of freedom $p - q$ and $n - p$.

The test rejects the null hypothesis when F is sufficiently large. The F statistics is also at the heart of sequential techniques to decide on the most suitable linear model. You first collect variables that have some plausible relationship with the response variable. After the data collection, you decide on a final model using one of these two popular procedures.

- **Forward Stepwise Selection:** start from a small set of explanatory variables. Consider all models obtained by adding one more predictor.

  1. For each variable not already included, calculate the *F-to-enter*, that is the value of the F statistic when this additional variable is included in the model. You then select the variable with the largest *F-to-enter*.
  2. If the largest F-to-enter is greater than some specified critical value F_{in}, then add the predictor to the model (typical choices are $2 \leq F_{in} \leq 4$).
  3. Repeat the two steps above until no predictors can be added.

An alternative procedure is

- **Backward Stepwise Elimination:** Start with the full model, that is inclusive of all collected predictor variables. Consider all models obtained by deleting one predictor.

  1. For each variable, calculate the *F-to-remove*, that is the value of the F statistics when this variable is removed. You then select the variable with the smallest *F-to-remove*.
  2. If the smallest F-to-remove is smaller than some specified critical value $2 \leq F_{out} \leq 4$, then remove the predictor variable from the model.
  3. Repeat the two steps above until no predictors can be removed.

A $F = t^2$

We show that, in a simple linear regression, we have that F statistics is equal to the square of the t statistics. To show this, we first prove that $R^2 = \rho^2$ in a simple linear regression.

We begin by defining some key quantities in simple linear regression. Let the observed values be (x_i, y_i) for $i = 1, \ldots, n$. Then:

- Total Sum of Squares (TSS):

$$\text{TSS} = \sum_{i=1}^{n} (y_i - \bar{y})^2 = \text{SS}_{yy}$$

- Residual Sum of Squares (RSS or SSR):

$$\text{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

We know that in simple linear regression:

$$\hat{y}_i = \hat{\alpha} + \hat{\beta} x_i$$

Using the least squares estimators, we have

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}, \quad \hat{\beta} = \frac{\mathrm{SS}_{xy}}{\mathrm{SS}_{xx}}$$

So we can write:

$$\hat{y}_i = \bar{y} + \hat{\beta}(x_i - \bar{x})$$

Then the residual sum of squares can be expressed as:

$$\mathrm{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} \Big[(y_i - \bar{y}) - \hat{\beta}(x_i - \bar{x}) \Big]^2$$

Expanding the square:

$$\mathrm{RSS} = \sum_{i=1}^{n} (y_i - \bar{y})^2 + \hat{\beta}^2 \sum_{i=1}^{n} (x_i - \bar{x})^2 - 2\hat{\beta} \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})$$

Using notation:

$$\mathrm{SS}_{yy} = \sum (y_i - \bar{y})^2, \quad \mathrm{SS}_{xx} = \sum (x_i - \bar{x})^2, \quad \mathrm{SS}_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y})$$

We get:

$$\mathrm{RSS} = \mathrm{SS}_{yy} + \hat{\beta}^2 \mathrm{SS}_{xx} - 2\hat{\beta}\mathrm{SS}_{xy}$$

Substitute $\hat{\beta} = \frac{\mathrm{SS}_{xy}}{\mathrm{SS}_{xx}}$:

$$\mathrm{RSS} = \mathrm{SS}_{yy} + \left(\frac{\mathrm{SS}_{xy}^2}{\mathrm{SS}_{xx}} \right) - 2 \left(\frac{\mathrm{SS}_{xy}^2}{\mathrm{SS}_{xx}} \right) = \mathrm{SS}_{yy} - \frac{\mathrm{SS}_{xy}^2}{\mathrm{SS}_{xx}}$$

Then we have:

$$R^2 = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}} = 1 - \frac{\mathrm{SS}_{yy} - \frac{\mathrm{SS}_{xy}^2}{\mathrm{SS}_{xx}}}{\mathrm{SS}_{yy}} = \frac{\mathrm{SS}_{xy}^2}{\mathrm{SS}_{xx} \cdot \mathrm{SS}_{yy}}$$

Now recall the definition of Pearson correlation coefficient:

$$\rho = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X) \cdot \mathrm{Var}(Y)}} = \frac{\mathrm{SS}_{xy}}{\sqrt{\mathrm{SS}_{xx} \cdot \mathrm{SS}_{yy}}} \Rightarrow \rho^2 = \frac{\mathrm{SS}_{xy}^2}{\mathrm{SS}_{xx} \cdot \mathrm{SS}_{yy}}$$

Hence, we conclude:

$$R^2 = \rho^2$$

Proof that $F = t^2$ in Simple Linear Regression

From the previous derivation, we know that:

$$\text{RSS} = \text{SS}_{yy} - \frac{\text{SS}_{xy}^2}{\text{SS}_{xx}} \qquad (\star)$$

We now prove that in simple linear regression, the F-statistic is equal to the square of the t-statistic when testing the null hypothesis $H_0 : \beta = 0$.

In the general linear model, the F-statistic is given by:

$$F = \frac{(\text{TSS} - \text{RSS})/(p - q)}{\text{RSS}/(n - p)}$$

where:

- p is the number of parameters in the full model ($p = 2$ in SLR),
- q is the number of parameters in the restricted model ($q = 1$ in SLR),
- n is the number of observations.

So in simple linear regression:

$$F = \frac{(\text{SS}_{yy} - \text{RSS})/1}{\text{RSS}/(n - 2)} = \frac{(n - 2)(\text{SS}_{yy} - \text{RSS})}{\text{RSS}}$$

The t-statistic for testing $H_0 : \beta = 0$ is given by:

$$t = \frac{\hat{\beta} - 0}{\text{SE}(\hat{\beta})}$$

$$t^2 = \frac{\hat{\beta}^2}{\text{Var}(\hat{\beta})}$$

We know:

$$\hat{\beta} = \frac{\text{SS}_{xy}}{\text{SS}_{xx}}, \quad \text{Var}(\hat{\beta}) = \frac{\hat{\sigma}^2}{\text{SS}_{xx}}, \quad \hat{\sigma}^2 = \frac{\text{RSS}}{n - 2}$$

So:

$$t^2 = \frac{\left(\frac{\text{SS}_{xy}}{\text{SS}_{xx}}\right)^2}{\frac{\text{RSS}}{(n-2)\text{SS}_{xx}}} = \frac{\text{SS}_{xy}^2}{\text{SS}_{xx}} \cdot \frac{(n - 2)}{\text{RSS}}$$

Now use Equation $\star$:

$$\frac{\mathrm{SS}_{xy}^2}{\mathrm{SS}_{xx}} = \mathrm{SS}_{yy} - \mathrm{RSS} \Rightarrow t^2 = \frac{(n-2)(\mathrm{SS}_{yy} - \mathrm{RSS})}{\mathrm{RSS}} = F$$

$$F = t^2$$

A.1 Relation between F-statistic and t-statistics in MLR:

The F-statistic tests the joint hypothesis

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0.$$

It is computed as:

$$F = \frac{(\mathrm{RSS}_{\mathrm{reduced}} - \mathrm{RSS}_{\mathrm{full}})/k}{\mathrm{RSS}_{\mathrm{full}}/(n-k-1)}.$$

For testing whether a single predictor $H_0 : \beta_j = 0$, the corresponding t-statistic is:

$$t_j = \frac{\hat{\beta}_j}{\mathrm{SE}(\hat{\beta}_j)}.$$

The F-statistic involves **all** k predictors simultaneously. The t-statistic tests **only one** predictor at a time, conditioning on all others being in the model. Thus, $F = t^2$ holds in MLR **only** if the F-test tests a single restriction (i.e., $H_0 : \beta_j = 0$ for some fixed j), in which case the test is equivalent to the corresponding two-sided t-test.

If $k > 1$ and the F-test involves multiple restrictions, the test evaluates a **joint null hypothesis** and aggregates evidence across predictors — hence, $F \neq t^2$ in general.

Lecture 17: Principal Component Analysis

IEOR E4709: Statistic Analysis and Time Series
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1 Principal Component Analysis

We begin with a quick digression about *exploratory* versus *confirmatory* analysis. The goal of exploratory analysis is to reveal structures of the data that would remain concealed if data were represented naively. The goal of confirmatory analysis is to test hypotheses about the data. Our next focus is on principal component analysis, that is part of exploratory analysis.[1] Consider a $n \times p$ data matrix, $X = (x_{ij})$, where each observation $(x_{i1}, x_{i2}, \ldots, x_{ip})$ consists of p variables (e.g. we can think of each row of this matrix as consisting of the vector of DJ30 stock prices at time t_i). Principal component analysis (PCA) seeks a compact summary of the variation of the data matrix X. That is, PCA seeks a few linear combinations of the variables that summarize as much of the variation in the data as possible.

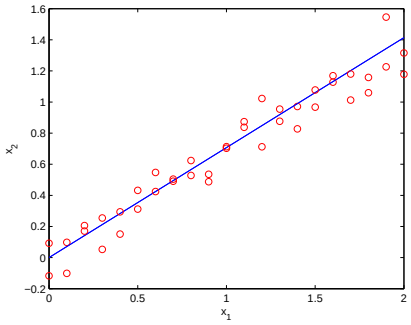


Figure 1: The new variable $\alpha = \frac{1}{\sqrt{2}}(x_1 + x_2)$ describes most of the variation in the data.

Denote by μ the mean vector, i.e., $\mu_j = \frac{1}{n}\sum_{i=1}^{n} x_{ij}$, for $j = 1, \ldots, p$. In general, we seek *principal components* $\alpha_i = (X - \mu)v_i$ such that, for $||v_i|| = 1$, the first principal component is the linear combination of the variables that has the largest variance. The second principal component is the linear combination of the variables that has the largest variance subject to being uncorrelated

[1] We will avoid using bold symbols for vector and matrices to avoid cluttered notation. It should be clear from the context if a quantity is a scalar, vector, or matrix.

with the first principal component, etc... Let us further assume that the mean μ of the columns of X is zero (if it not zero, then we subtract the mean from each column of the data matrix), hence $\alpha_i = Xv_i$. Recall that, given two zero-mean n-dimensional vectors x and y, their sample covariance is

$$cov(x, y) = \frac{1}{n-1} x^\top y.$$

In particular, the sample variance of the principal component α_i is

$$var(\alpha_i) = cov(\alpha_i, \alpha_i) = \frac{1}{n-1} \alpha_i^\top \alpha_i = \frac{1}{n-1} v_i^\top X^\top X v_i = v_i^\top \Sigma v_i,$$

where $\Sigma = \frac{1}{n-1} X^\top X$ is the sample covariance of the data. We want to choose v_1 so to maximize the variance of the first principal component subject to the constraint that $||v_1|| = 1$. This amounts to maximizing

$$\chi^2 := v_1^\top \Sigma v_1 - \lambda_1 (v_1^\top v_1 - 1),$$

where λ_1 is the Lagrangian multiplier. We then obtain that $\frac{\partial}{\partial v_1} \chi^2 = 0$ if and only if $2\Sigma v_1 - 2\lambda_1 v_1 = 0$, i.e., $\Sigma v_1 = \lambda_1 v_1$. The weight vector v_1 is called the principal component vector, and is the eigenvector of the covariance matrix Σ with the largest eigenvalue.[2] To see why v_1 must be the eigenvector associated with the largest eigenvalue, observe that $var(\alpha_1) = \frac{1}{n-1} \alpha_1^\top \alpha_1 = v_1^\top \Sigma v_1 = v_1^\top \lambda_1 v_1 = \lambda_1$,

The principal component $\alpha_2 = Xv_2$, i.e., the projection of the data matrix X onto the eigenvector associated with the second largest eigenvalue of Σ. It can be easily seen that α_2 is uncorrelated with α_1. This follows from the fact that the eigenvector v_2 is orthogonal to v_1 because Σ is a symmetric matrix and, for a real symmetric matrix, the eigenvectors corresponding to different eigenvalues are orthogonal. First, from the formula of the covariance matrix of two vectors, we obtain that

$$cov(\alpha_i, \alpha_j) = \frac{1}{n-1} \alpha_i^\top \alpha_j = \frac{1}{n-1} v_i^\top X^\top X v_j = v_i^\top \Sigma v_j.$$

Hence, $cov(\alpha_1, \alpha_2) = v_1^\top \Sigma v_2 = \lambda_2 v_1^\top v_2 = 0$. Hence, α_2 is the linear combination of the variables with the largest variance subject to being uncorrelated with the first principal component. Hence, the PCA yields uncorrelated principal components that are ordered according to decreasing variance λ_i's.

1.1 Definitions

More formally, the *principal component vectors* (aka principal component loadings or empirical orthogonal functions) v_i are the eigenvectors of the sample covariance matrix Σ, i.e., satisfying $\Sigma v_i = \lambda_i v_i$, where we use the convention that $\lambda_1 \geq \lambda_2 \ldots \geq \lambda_p \geq 0$. Notice that each eigenvector is of length p.[3] The *principal components* α_i (p component vectors of length n) are linear combinations of the columns of the data matrix X, whose weights in the linear combination are the components

[2] A real symmetric matrix Σ may have eigenvalues with algebraic multiplicity higher than 1, hence there may be multiple eigenvectors corresponding to the same eigenvalue. Then any linear combination of those vectors is also an eigenvector, with the same eigenvalue. For instance, if Σ is the identity matrix, then every vector is an eigenvector, and most PCA routines would return an essentially arbitrary collection of mutually perpendicular vectors. Typically, however, any arbitrarily small perturbation to Σ will break the degeneracy, i.e. it will guarantee that each eigenvalue is associated with only one eigenvector.

[3] The expection is if $n < p$, in which case there are only n distinct eigenvectors and eigenvalues.

of the eigenvector v_i: $\alpha_i = Xv_i$. The j-th element of the principal component α_i is called the *score* of the i-th principal component on the j-th observation.

1.2 Variance of Principal Components

The eigenvalue λ_i associated with the principal component vector v_i represents the variance of the i-th principal component, given by $var(\alpha_i) = \frac{1}{n-1}\alpha_i^\top \alpha_i = v_i^\top \Sigma v_i = v_i^\top \lambda_i v_i = \lambda_i$, using that $v_i^\top v_i = 1$. Divide the i-th eigenvalue λ_i by the sum of all eigenvalues

$$\frac{\lambda_i}{\sum_{i=1}^p \lambda_i} = \frac{\lambda_i}{tr(\Sigma)} = \frac{Var(i^{th} \text{ principal component})}{\text{Total variance}},$$

where $tr(\Sigma)$ is the trace of the matrix Σ, i.e. the sum of its diagonal entries that also coincide with the sum of its eigenvalues. This ratio is the proportion of total variation accounted for by the i-th principal component. Hence,

$$\frac{\sum_{i=1}^k \lambda_i}{\sum_{i=1}^p \lambda_i}$$

is the proportion of total variation accounted for by the first k principal components.

1.3 Property of maximum variance

No linear combination of the columns of X, Xu, with $||u|| = 1$, has a variance larger than λ_1, which is the variance of the first principal component α_1. If Xu is a linear combination of variables (with $||u|| = 1$) that is uncorrelated with the first k principal components $\alpha_1, \ldots, \alpha_k$, then the variance of Xu is maximized when Xu is the $(k+1)$-th principal component $\alpha_{k+1} = Xv_{k+1}$.

Lecture 18: PCA and Linear Time Series Analysis

IEOR E4709: Statistic Analysis and Time Series
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1 Principal Component Analysis

1.1 Property of minimum mean-squared error of representation

It can be shown that $X = \sum_{i=1}^{p} \alpha_i v_i^\top$. To see why this is true, let us first consider the variance-covariance matrix $\Sigma = \frac{1}{n-1} X^\top X$. This is a symmetric matrix and thus can be *diagonalized*, i.e.,

$$\Sigma = V \Lambda V^\top, \tag{1}$$

where Λ is a diagonal matrix whose entries are the corresponding eigenvalues ($\Lambda_{1,1}$ is the largest eigenvalue, $\Lambda_{2,2}$ is the second largest eigenvalue, etc...), V is a matrix whose columns are the eigenvectors of Σ (the first column of V is the eigenvector corresponding to the largest eigenvalue, the second column of V is the eigenvector corresponding to the second largest eigenvalue etc...).

We also know from linear algebra that the matrix X admits a *singular value decomposition*:

$$X = USV^\top, \tag{2}$$

where U is a unitary matrix, i.e., $U^\top U = UU^\top = I_n$, where I_n is the n-dimensional identity matrix. S is the matrix of singular values. Hence, using Eq. (2), we obtain

$$\Sigma = \frac{1}{n-1} X^\top X = \frac{1}{n-1} VSU^\top USV^\top = V \frac{S^2}{n-1} V^\top.$$

Hence, the eigenvalue λ_i of the covariance matrix is equal to $\frac{S_{i,i}^2}{n-1}$. Recall the matrix of principal components is given by $XV = USV^\top V = US$, where we have used that the eigenvectors of the symmetric matrix Σ are orthogonal. We know that the i-th column of the matrix XV is $(XV)^i = \alpha_i$. It then follows from Eq. (2) that X can be represented as $X = \sum_{i=1}^{p} \alpha_i v_i^\top$.

The so called "rank-r approximation" of the data matrix X in terms of the first r principal components is given by

$$X_r := \sum_{i=1}^{r} \alpha_i v_i^\top.$$

It is well known that X_r is the linear rank-r approximation of X that minimizes the Frobenius norm over all matrices of rank at most r (this is the Eckart–Young–Mirsky theorem whose proof can be

seen in linear algebra textbooks). We take this fact for granted, and compute the approximation error next as

$$\begin{aligned}
\frac{1}{n}||X - X_r||_F^2 &= \frac{1}{n}|| \sum_{i=r+1}^{p} \alpha_i v_i^\top ||_F^2 \\
&= \frac{1}{n} tr \left(\left(\sum_{i=r+1}^{p} \alpha_i v_i^\top \right) \left(\sum_{i=r+1}^{p} \alpha_i v_i^\top \right)^\top \right) \\
&= \frac{1}{n} tr \left(\sum_{i,j=r+1}^{p} \alpha_i v_i^\top v_j \alpha_j^\top \right) \\
&= \frac{1}{n} tr \left(\sum_{i=r+1}^{p} \alpha_i \alpha_i^\top \right) \\
&= \frac{1}{n} tr \left(\sum_{i=r+1}^{p} \alpha_i^\top \alpha_i \right) \\
&= \frac{1}{n} \sum_{i=r+1}^{p} ||\alpha_i||^2 = \frac{n-1}{n} \sum_{i=r+1}^{p} \lambda_i,
\end{aligned}$$

where $||A||_F^2 = \sum_{i,j} A_{ij}^2$ is the Frobenius norm of the matrix A, i.e. the sum of the absolute squares of its elements, while $||x||^2 = \sum_i x_i^2$ is the L_2 norm of the vector x. In the second equality of the derivation above, we have used the well known result that, for a matrix A, we have that $||A||_F = \sqrt{tr(AA^\top)}$. In the fourth equality, we have used that v_i and v_j are orthonormal. In the fifth equality, we have used that $tr(AB) = tr(BA)$. The above result has implications for data compression.

Remark 1.1. *The rank r approximation is not intended in the sense that the matrix X_r has rank r, but rather that the matrix has rank of at most r. X_r is the approximation of the data matrix X, which only makes use of the first r principal components, i.e. the principal components that have the r largest variances.*

1.2 Lack of Scale Invariance

Consider a transformation of the data, i.e., $\tilde{X} = XQ$. For example, if Q is a diagonal matrix, this means that we are re-scaling the units of the observation variables. Under a transformation Q, the transformed covariance matrix becomes $\tilde{\Sigma} = \frac{Q^\top X^\top XQ}{n-1} = Q^\top \Sigma Q$. Next, we want to ask the question: when do we have invariance of the PCA, i.e. under which transformation of the data, we have that $\lambda_i = \tilde{\lambda}_i$ and $\tilde{v}_i = Q^\top v_i$, i.e. the principal component vectors transform the same way as the data and the captured variances are the same? We are using the notation $\tilde{\Sigma}\tilde{v}_i = \tilde{\lambda}_i \tilde{v}_i$. For the scale invariance property to hold, we must guarantee that

$$\tilde{\lambda}_i = \tilde{v}_i^\top \tilde{\Sigma} \tilde{v}_i = v_i^\top \Sigma v_i = \lambda_i,$$

where $\tilde{\Sigma} = Q^\top \Sigma Q$ and $\tilde{v}_i = Q^\top v_i$. Plugging this expression in the equation above, we obtain that $v_i^\top QQ^\top \Sigma QQ^\top v_i = v_i^\top \Sigma v_i$, which holds true if $QQ^\top = I$, so the scale invariance property is maintained only under orthogonal transformations of the data.

Remark 1.2. *It is usually a good idea to standardize the data, i.e. demand each column of the data matrix, and divide each column by the standard deviation. The reason is that if some variables have a large variance and some others a small variance, the PCA will place higher loading on the columns with large variances. For example, if you change one variable from kilometers to meters (thus increasing its variance), it may go from having small impact to dominating the first principal component. To make the PCA independent of such rescaling, standardizing the variables is a good approach to follow. If we standardize the data, the covariance matrix of the data becomes the correlation matrix.*

2 Principal Component Regression

Recall that the variance of the least square estimator $\hat{\beta}$ in linear regression is $\sigma^2(X^\top X)^{-1}$. If X is column-rank deficient, meaning that one or more of its columns is equal to a linear combination of the others, then it means that one predictor variable is redundant. We refer to this problem as *collinearity*. In that case, the matrix $(X^\top X)$ would be non-invertible. Even if there is not an exact linear relationship among the predictors, but one predictor is approximately a linear combination of the others, then we would have that the inverse matrix $(X^\top X)^{-1}$ would be huge, and thus the variance of the least square estimator $\hat{\beta}$ would be very large. This happens, for instance, if we have many predictor variables compared to the number of observations (i.e., $p >> n$). In such case, one possibility is to reduce the dimensionality of the data matrix X first, which is at the heart of principal component regression.

The key idea is that often a small number of principal components is sufficient to explain most of the variability of the data, as well as the relationship with the response. Hence, we modify the multiple linear regression model from $y = X\beta + \epsilon$ to $y = X'\gamma + \epsilon$. where the matrix $X' = (\alpha_1, \alpha_2, \ldots, \alpha_r)$ is the matrix whose first column is the principal component α_1, the second column is the principal component α_2, ..., and the r-th column is the principal component α_r.

Remark 2.1. *Let us call the least square estimate of the principal component regression $\hat{\gamma}$. We can recover the coefficient β associated with the regression using the original data matrix X, i.e., the linear regression model $y = X\beta + \epsilon$, from the coefficient vector γ. To do so, we use the following relation*

$$\beta = V\gamma, \tag{3}$$

where $V = (v_1, ..., v_r)$ and v_i is the i-th eigenvector. To see why the relation (3) *holds, observe the following: $y = X'\gamma + \epsilon = \sum_{i=1}^{r} \gamma_i \alpha_i + \epsilon = \sum_{i=1}^{r} X v_i \gamma_i + \epsilon = XV\gamma + \epsilon = X\beta + \epsilon$, where the last equality holds true if $\beta = V\gamma$. If the standard PCA and PCR packages of R are used, one gets the principal components α_i and the least square estimates $\hat{\gamma}$.*

Lecture 19: Autoregressive Linear Models

IEOR E4709: Statistic Analysis and Time Series
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1 Announcement

No in-person class on Wednesday, April 9, 2024. The video-recorded lecture will be broadcast and made available from the video library on Courseworks at 10:10am (usual class time).

2 Linear Time Series Analysis

Most of the material from this part of the class on Time Series is taken from the textbook "Analysis of financial Time Series" by Ruey Tsay.

The objective of financial time series analysis is to set up a family of probability models to represent data over time. Our goal is to estimate parameters of the model, and check the model for goodness of fit. Treating an asset return (e.g. the return r_t of a stock) as a collection of random variables over time, we have a time series $\{r_t\}_{t\geq 0}$. We will focus on linear time series analysis, where the data point at time t depends linearly or previous data points of interest. Linear time series analysis provides a framework for studying the dynamic structure of such a series.

The theories of linear time series usually discussed include stationarity, dynamic dependence, autocorrelation function, modeling and forecasting. Correlation plays an important role in understanding these models. In particular, correlations between the variable of interest and its past values are the focus of linear time series analysis. These correlations are referred to as *serial correlations* or *autocorrelations*. For instance, to test whether stock returns in different periods are uncorrelated, we want to verify that the auto-correlation is equal to zero. Next, we discuss how this can be achieved, but first we introduce some preliminary concepts and definitions.

A time series is *strictly stationary* if the joint distribution of $(r_{t_1}, r_{t_2}, \ldots, r_{t_k})$ is the same as the distribution of any other portion of the series $(r_{t_1+\tau}, r_{t_2+\tau}, \ldots, r_{t_k+\tau})$, where τ can be any integer. This is a very difficult condition to verify empirically. We will work with a weak version of stationarity, and assume that the return series of a stock $\{r_t\}$ is *weakly stationary*, i.e., $\mathbb{E}[r_t]$ and $cov(r_t, r_{t-l})$ exist and are time invariant, i.e., independent of t for any l.

We will use $\mu := \mathbb{E}[r_t]$ to denote the *common mean*, and $\gamma_l = cov(r_t, r_{t-l})$ to denote the lag-l auto-covariance of $\{r_t\}$.

Remark 2.1. *In finance, it is common to assume that an asset return series is weakly stationary. This assumption can be verified empirically if we have a sufficient number of historical returns available. For instance, we can divide the data into subsamples and then see if the mean and lag-l autocovariances are the same across subsamples.*

Consider the volatility of a stock. It is reasonable to expect (and empirically observed in data) that the volatility tomorrow will be correlated with the volatility today. However, will the volatility five months from now depend in any way on today's volatility? In order to answer questions of this sort, we need to define the autocorrelation function. The auto-covariance function is defined as $\mathbb{E}\left[(r_{t_1} - \mu_{t_1})(r_{t_2} - \mu_{t_2})\right]$. The autocorrelation function can be computed from the autocovariance function dividing by the standard deviations of r_{t_1} and r_{t_2} which corresponds to our usual definition of correlation. In other words, the autocorrelation function is given by $\mathbb{E}\left[(r_{t_1} - \mu_{t_1})(r_{t_2} - \mu_{t_2})\right]/(\sigma_{t_1}\sigma_{t_2})$. In the case of a stationary time series, the *lag l auto-correlation* $\rho_l = corr(r_t, r_{t+l})$ is given by $\frac{\gamma_l}{\gamma_0}$, when $\gamma_0 \neq 0$.

Clearly, we have that $\gamma_l = \gamma_{-l}$, and γ_0 is the common variance of $\{r_t\}_{t\geq 0}$. Hence, let us use $\Gamma_n := [\gamma_{i-j}]_{i,j=1}^n$ to denote the covariance matrix of $r_{t+1}, \ldots, r_{t+n}$. The matrix Γ_n is positive semidefinite (its smallest eigenvalue is nonnegative).

In practice, the auto-covariance and auto-correlation functions are unknown and have to be estimated. Consider observations $\{r_1, r_2, \ldots, r_T\}$ of a weakly stationary time series. The *sample auto-covariance* of the observations is

$$\hat{\gamma}_l = \frac{1}{T}\sum_{t=l+1}^{T}(r_t - \bar{r})(r_{t-l} - \bar{r}), \qquad 0 \leq l < T, \tag{1}$$

where $\bar{r} := \frac{1}{T}\sum_{t=1}^{T} r_t$ is the sample mean. The *sample covariance* matrix of $r_{t+1}, \ldots, r_{t+n}$ is $\hat{\Gamma}_n := [\hat{\gamma}_{i-j}]_{i,j=1}^n$. It holds that if $\hat{\gamma}_0 > 0$, then $\hat{\Gamma}_n$ is invertible. The sample auto-correlation is given by $\hat{\rho}_l = \frac{\hat{\gamma}_l}{\hat{\gamma}_0}$. Alternatively, one could use

$$\hat{\gamma}_l = \frac{1}{T-l}\sum_{t=l+1}^{T}(r_t - \bar{r})(r_{t-l} - \bar{r}), \qquad 0 \leq l < T, \tag{2}$$

as an estimate of the autocovariance function. Both estimator in (1) and (2) are consistent. The estimator (1) produces positive definite autocovariance matrices in any sample. By contrast, the estimator (2) does not (necessarily) produce positive definite sequences. One can use either, and there is little difference when T is large. I will use the estimator (1) in this class, which is typically the most commonly used.

Remark 2.2. *It is worth observing that both estimators* (1) *and* (2) *are biased. If we divide by* $T - l - 1$ *instead* $T - l$*, i.e.,*

$$\hat{\gamma}_l = \frac{1}{T-l-1}\sum_{t=l+1}^{T}(r_t - \bar{r})(r_{t-l} - \bar{r}),$$

one can then show that $\hat{\gamma}_l$ *is an unbiased estimator of the lag* l *autocovariance* γ_l*.*

Example 1. *Consider a time series of length* 2*, where* $x_t = 0$ *and* $x_{t+1} = 1$*. It is then simple to see that the matrix* $\hat{\Gamma}$*, whose entries are computed dividing by* $T - l = 2 - 1 = 1$ *in this case, is given by* $\hat{\gamma}(1,1) := \hat{\gamma}_{1-1} = 1/4$*,* $\hat{\gamma}(1,2) := \hat{\gamma}_{1-2} = -1/4$*,* $\hat{\gamma}(2,1) := \hat{\gamma}_{2-1} = -1/4$*,* $\hat{\gamma}(2,2) := \hat{\gamma}_{2-2} = 1/4$*, which is singular. To compute these entries, we have used the formula*

$$\frac{1}{T-h}\sum_{t=1}^{T-h}(x_{t+h} - \bar{x})(x_t - \bar{x})$$

Instead, the matrix $\hat{\Gamma}$, whose entries are computed dividing by $T = 2$ is given by $\hat{\gamma}(1,1) = 1/4$, $\hat{\gamma}(1,2) = -1/8$, $\hat{\gamma}(2,1) = -1/8$, $\hat{\gamma}(2,2) = 1/4$, which has eigenvalues 3/8 and 1/8, and thus it is positive-definite. To compute these entries, we have used the formula

$$\frac{1}{T}\sum_{t=1}^{T-h}(x_{t+h} - \bar{x})(x_t - \bar{x})$$

3 Randomness of a Time Series

With many time series, a trend or seasonal pattern may be evident from the time plot indicating that the series is not random. The lack of randomness may be evident by looking at the short term autocorrelation. For instance, if the time series alternates up and down around the mean in successive values, the lag-1 autocorrelation will be negative. On the other hand, if an observation above the mean tends to be followed by several other observations that are above the mean, then the autocorrelations for lags k will be positive (or, vice-versa, if an observation below the mean tends to be followed by successive observations that are also below the mean is an indication of positive lag k autocorrelations). If a time series exhibits these types of correlations, then the series is not completely random. One of the goals of time series analysis is to model this sort of autocorrelation. However, before pursuing this more complicated modeling, it would be useful to know if the series is completely random or not.

Consider the lag l-autocorrelation, which we recall is defined as

$$\hat{\rho}_l = \frac{\sum_{t=l+1}^{T}(r_t - \bar{r})(r_{t-l} - \bar{r})}{\sum_{t=1}^{T}(r_t - \bar{r})^2}.$$

If $\{r_t\}$ is an i.i.d. sequence such that $\mathbb{E}\left[r_t^2\right] < \infty$, then it can be shown that $\hat{\rho}_l$ is asymptotically normal with mean zero and variance $1/T$ for any fixed positive integer l; see the book by Brockwell and Davis 1991, Theorem 7.2.2, for a proof of this result. This result can be used to test, for instance, the null hypothesis H_0 that $\rho_1 = 0$ versus the alternative hypothesis H_a that $\rho_1 \neq 0$. Under the null hypothesis, $\sqrt{T}\hat{\rho}_1$ follows asymptotically the normal distribution.[1]

Ljung and Box (1978) introduced the statistics

$$Q(m) = T(T+2)\sum_{l=1}^{m}\frac{\hat{\rho}_l^2}{T-l},$$

where T is the sample size, as a test statistic for the null hypothesis $H_0 : \rho_1 = \rho_2 = \cdots = \rho_m = 0$ against the alternative hypothesis that $H_a : \rho_i \neq 0$ for some $i = 1, \ldots, m$. If $\{r_t\}_{t\geq 0}$ is an i.i.d. sequence satisfying certain moment conditions, we have that the asymptotic distribution of $Q(m)$ under the null hypothesis H_0 is a χ^2 distribution with m degrees of freedom. Hence, the test rejects if $Q(m) > \chi^2_{m,1-\alpha}$, where $\chi^2_{m,1-\alpha}$ is the $1-\alpha$ quantile of a χ^2 distribution with m degrees of freedom.[2]

[1] More generally, if the time series $\{r_t\}$ is weakly stationary satisfying $r_t = \mu + \sum_{i=0}^{p}\phi_i a_{t-i}$, where $\phi_0 = 1$ and $\{a_j\}$ is a sequence of i.i.d. zero-mean random variables, then $\hat{\rho}_l$ is asymptotically normal with mean zero and variance $(1 + 2\sum_{i=1}^{p}\rho_i^2)/T$ for $l > p$, where ρ_i is the actual lag i autocorrelation coefficient.

[2] See the paper by Ljung and Box (1978): "On a Measure of a Lack of Fit in Time Series Models", Biometrika, Vol. 65, No. 2, pp. 297-303 for the details.

4 AR(1) Model

Autoregressive (AR) models provide a representation for random time-varying processes arising in finance and economics. The autoregressive model specifies that the response (output) variable depends linearly on its own previous values and on a stochastic term (an unpredictable term). The model takes the form of a stochastic difference equation.

$$r_t = \phi_0 + \phi_1 r_{t-1} + a_t, \tag{3}$$

where $\{a_t\}$ is a white noise sequence, i.e., an i.i.d. sequence with zero mean and variance σ_a^2. The terms ϕ_0 and ϕ_1 are constants. Next, let us identify the conditions for the weak stationarity of the time series. Assuming that the time series $\{r_t\}$ is weakly stationary, we have that $\mathbb{E}[r_t] = \mu$ and $cov(r_t, r_{t-l}) = \gamma_l$. Let $\mathcal{F}_t$ be the sigma algebra generated by the return observations $r_1, r_2, \ldots, r_t$. We can compute the conditional mean given by

$$\mathbb{E}\left[r_t \middle| \mathcal{F}_{t-1}\right] = \phi_0 + \phi_1 r_{t-1}, \qquad Var\left[r_t \middle| \mathcal{F}_{t-1}\right] = \sigma_a^2$$

The unconditional mean is given by

$$\mathbb{E}[r_t] = \mathbb{E}\left[\mathbb{E}\left[r_t \middle| \mathcal{F}_{t-1}\right]\right] = \phi_0 + \phi_1 \mathbb{E}[r_{t-1}]. \tag{4}$$

As $\{r_t\}$ is weakly stationary, we have that $\mathbb{E}[r_t] = \mu$ for any t. Using Eq. (4), we deduce $\mu = \phi_0 + \phi_1 \mu$, or equivalently

$$\mu = \frac{\phi_0}{1 - \phi_1}. \tag{5}$$

We can also compute the unconditional variance, given by

$$Var(r_t) = \mathbb{E}\left[Var\left(r_t \middle| \mathcal{F}_{t-1}\right)\right] + Var\left(\mathbb{E}\left[r_t \middle| \mathcal{F}_{t-1}\right]\right) = \sigma_a^2 + \phi_1^2 Var(r_{t-1}). \tag{6}$$

Because r_t is weakly stationary, i.e., $Var(r_t) = \gamma_0$, we can use Eq. (6) to deduce $\gamma_0 = \sigma_a^2 + \phi_1^2 \gamma_0$, hence

$$\gamma_0 = \frac{\sigma_a^2}{1 - \phi_1^2}, \tag{7}$$

provided that $\phi_1^2 < 1$. The requirement that $\phi_1^2 < 1$ results from the fact that the variance of a random variable is bounded and nonnegative. Consequently, the weak stationarity of an $AR(1)$ model implies that $-1 < \phi_1 < 1$, i.e., $|\phi_1| < 1$. It can be shown that this condition is also sufficient for weak stationarity, hence the *necessary and sufficient condition for the AR(1) model* (3) *to be weakly stationary is* $|\phi_1| < 1$. *See Section 2.4.1 of Ruey Tsay's textbook, Box and Jenkins, 1976, Ch. 3 or Pagano, 1973, for the formal proof.*

Using Eq. (5), we can rewrite the AR(1) model as

$$r_t - \mu = \phi_1 (r_{t-1} - \mu) + a_t,$$

or, equivalently, as

$$r_t = (1 - \phi_1)\mu + \phi_1 r_{t-1} + a_t.$$

This model is often used in finance, and ϕ_1 measures the persistence of an AR(1) time series, where high persistence means that past shocks have a strong impact on the current value of the variable.

We can also compute the auto-correlation function of the AR(1) time series. Using that r_t is a weakly stationary time series, we have

$$\gamma_l = cov(r_t, r_{t-l}) = \phi_1 cov(r_{t-1}, r_{t-l}) + cov(a_t, r_{t-l}) = \phi_1 \gamma_{l-1},$$

which admits the solution

$$\gamma_l = \phi_1^l \gamma_0, \qquad l \geq 1.$$

Because $\rho_0 = 1$, the lag-l auto-correlation function (ACF) is then given by

$$\rho_l = \phi_1^l, \tag{8}$$

where $l \geq 1$. The ACF of the AR(1) times series decays exponentially to zero, as the lag l goes to infinity.

4.1 Estimation of AR(1) Model

Suppose r_t, $t = 1, \ldots, T$, are observed. In order to estimate the parameters of the AR(1) model, we can proceed as follows:

- Estimate ϕ_1 by matching the population and the sample lag-1 autocorrelation, i.e. set $\hat{\phi}_1 := \hat{\rho}_1$, where $\hat{\rho}_1$ is the sample lag-1 autocorrelation.
- Using Eq. (5), i.e., $\mu = \frac{\phi_0}{1-\phi_1}$, we can estimate ϕ_0. This yields $\hat{\phi}_0 = \hat{\mu}(1 - \hat{\phi}_1)$, where $\hat{\mu}$ is the sample mean.
- Estimate σ_a^2 using Eq. (7), by matching the population and sample variance. This yields $\hat{\sigma}_a^2 = \hat{\gamma}_0(1 - \hat{\phi}_1^2)$, where $\hat{\gamma}_0$ is the sample variance.

An alternative estimation method is the maximum likelihood estimation, which can be used if we have knowledge of the returns distribution. Assume that, given r_j, $j = 0, \ldots, t-1$, then r_t is normally distributed with mean $\phi_0 + \phi_1 r_{t-1}$ and variance σ_a^2. We can then use maximum likelihood estimation to estimate the parameters. The likelihood of r_t, $t = 1, \ldots, T$, given r_0 is:

$$\log L(r_{1:T} | \phi_0, \phi_1, \sigma_a, r_0) = -\frac{T}{2}\log(2\pi) - \frac{T}{2}\log\sigma_a^2 - \frac{1}{2\sigma_a^2}\sum_{t=1}^{T}\left(r_t - (\phi_0 + \phi_1 r_{t-1})\right)^2.$$

We can then find the estimates by maximizing the conditional log-likelihood. It can be shown that the estimator is the same as the OLS estimator in the linear regression model where r_{t-1}, $t = 1, \ldots, T$ are the explanatory variables, and r_t, $t = 1, \ldots, T$ are the response variables. When T is large, the choice of r_0 does not have a big impact on the parameter estimates.

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Lecture 20: AR(p) Models and Identification of AR Models

IEOR E4709: Statistic Analysis and Time Series
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1 AR(p) Model

The auto-regressive model of order p, denoted as $AR(p)$, is

$$r_t = \phi_0 + \sum_{i=1}^{p} \phi_i r_{t-i} + a_t, \tag{1}$$

where $\{a_t\}$ is a white noise sequence, i.e., an i.i.d. sequence with mean zero and variance σ_a^2. The quantities ϕ_i's are constants. Moreover, it is assumed that $\Phi(z) := 1 - \sum_{i=1}^{p} \phi_i z^i$, $z \in \mathbb{C}$, where $\mathbb{C}$ denotes the set of complex numbers, has no zero within the unit circle, i.e. $\Phi(z) \neq 0$ for any z such that $|z| \leq 1$. This condition is necessary to ensure that $\{r_t\}$ is weakly stationary.

The conditional mean of r_t is given by

$$\mathbb{E}\left[r_t \middle| \mathcal{F}_{t-1}\right] = \phi_0 + \sum_{i=1}^{p} \phi_i r_{t-i}.$$

The unconditional mean is given by

$$\mu := \mathbb{E}\left[\mathbb{E}\left[r_t \middle| \mathcal{F}_{t-1}\right]\right] = \phi_0 + \sum_{i=1}^{p} \phi_i \mathbb{E}\left[r_{t-i}\right] = \phi_0 + \sum_{i=1}^{p} \phi_i \mu. \tag{2}$$

This leads to $\mu = \frac{\phi_0}{1-\sum_{i=1}^{p}\phi_i} = \frac{\phi_0}{\Phi(1)}$. Using this equation to replace ϕ_0 in Eq. (1), we can rewrite the $AR(p)$ model as

$$r_t - \mu = \sum_{i=1}^{p} \phi_i (r_{t-i} - \mu) + a_t.$$

For each $l \geq 0$, we have

$$\begin{aligned} \gamma_l &= cov(r_t, r_{t-l}) = cov(a_t, r_{t-l}) + \sum_{i=1}^{p} \phi_i cov(r_{t-i}, r_{t-l}) \\ &= cov(a_t, r_{t-l}) + \sum_{i=1}^{p} \phi_i \gamma_{l-i}. \end{aligned}$$

When $l = 0$, we have that $cov(a_t, r_{t-l}) = cov(a_t, r_t) = \sigma_a^2$, and consequently

$$\gamma_0 = \sigma_a^2 + \sum_{i=1}^{p} \phi_i \gamma_{-i} = \sigma_a^2 + \sum_{i=1}^{p} \phi_i \gamma_i = \sigma_a^2 + \gamma_0 \sum_{i=1}^{p} \phi_i \rho_i,$$

where we have used that the lag l autocorrelation is $\rho_l = \frac{\gamma_l}{\gamma_0}$. Therefore, the unconditional variance is given by

$$\gamma_0 = \frac{\sigma_a^2}{1 - \sum_{i=1}^{p} \phi_i \rho_i}. \tag{3}$$

For $l \geq 1$, we have that $cov(a_t, r_{t-l}) = 0$, and consequently

$$\gamma_l = \sum_{i=1}^{p} \phi_i \gamma_{l-i} = \sum_{i=1}^{p} \phi_i \gamma_{|l-i|}.$$

Therefore, we obtain

$$\rho_l = \sum_{i=1}^{p} \phi_i \rho_{|l-i|}$$

The ρ_l's, $l = 1, \ldots, p$, satisfy the **Yule-Walker equation:**

$$\underbrace{\begin{pmatrix} 1 & \rho_1 & \cdots & \rho_{p-1} \\ \rho_1 & 1 & \cdots & \rho_{p-2} \\ \vdots & \vdots & \vdots & \vdots \\ \rho_{p-1} & \rho_{p-2} & \cdots & 1 \end{pmatrix}}_{correlation\ matrix\ of\ (r_1, r_2, \ldots, r_p)} \begin{pmatrix} \phi_1 \\ \phi_2 \\ \cdots \\ \phi_p \end{pmatrix} = \underbrace{\begin{pmatrix} \rho_1 \\ \rho_2 \\ \cdots \\ \rho_p \end{pmatrix}}_{ACF} \tag{4}$$

The estimation of the $AR(p)$ model proceeds similarly to the estimation of the AR(1) model. Suppose r_t, $t = 1, \ldots, T$ are observed. We can use the method of moments as follows:

- Use the Yule-Walker equation and the sample correlation to estimate ϕ_i, $i = 1, \ldots, p$.
- We use Eq. (2) for the unconditional mean and the sample mean estimate for μ, to estimate ϕ_0.
- We use the sample variance to estimate γ_0. We then use Eq. (3) along with the sample correlations and the coefficients ϕ_i estimated in the previous step to estimate σ_a^2.

Alternatively, assuming that r_j, $j = 0, \ldots, t-1$ are such that r_t conditional on $\mathcal{F}_{t-1}$ (where we recall that $\mathcal{F}_{t-1}$ is the filtration generated by the returns $r_1, r_2, \ldots, r_{t-1}$) is normally distributed with mean $\phi_0 + \sum_{i=1}^{p} \phi_i r_{t-i}$ and variance σ_a^2, we can use the MLE method. The objective is to maximize the conditional likelihood of r_t, $t = 1, \ldots, T$, given $r_0, r_{-1}, r_{-2}, \ldots, r_{-(p-1)}$.

2 Identification of AR Models

How do we specify the order of an AR model? One approach is to use the partial autocorrelation function (PACF). What is the PACF? The PACF at lag k is defined as the correlation between r_t and r_{t-k} after removing the effect of the variables in between: $r_{t-1}, \ldots, r_{t-k+1}$.

For example, the formula for the partial correlation between r_t and r_{t+2}, i.e., the correlation of r_t and r_{t+2} conditional on r_{t+1} is given by

$$\rho_{r_t,r_{t+2}|r_{t+1}} = \frac{\rho_{r_t,r_{t+2}} - \rho_{r_t,r_{t+1}}\rho_{r_{t+1},r_{t+2}}}{\sqrt{1-\rho^2_{r_t,r_{t+1}}}\sqrt{1-\rho^2_{r_{t+2},r_{t+1}}}}.$$

For an AR(1) model, $\rho_{r_t,r_{t+2}} = \phi_1^2$ and $\rho_{r_t,r_{t+1}} = \rho_{r_{t+1},r_{t+2}} = \phi_1$. Hence, $\rho_{r_t,r_{t+2}|r_{t+1}} = 0$.

Remark 2.1. *Observe that the PACF at lag 1 is the same as the lag-1 autocorrelation function. However, for lags $k > 1$, the PACF at lag k and the ACF at lag k can be quite different.*

PACF is at the heart of the order identification procedure for an AR model. Consider a sequence of AR models:

$$\begin{aligned}
r_t &= \phi_{0,1} + \phi_{1,1}r_{t-1} + e_{1t} \\
r_t &= \phi_{0,2} + \phi_{1,2}r_{t-1} + \phi_{2,2}r_{t-2} + e_{2t} \\
r_t &= \phi_{0,3} + \phi_{1,3}r_{t-1} + \phi_{2,3}r_{t-2} + \phi_{3,3}r_{t-3} + e_{3t} \\
r_t &= \phi_{0,4} + \phi_{1,4}r_{t-1} + \phi_{2,4}r_{t-2} + \phi_{3,4}r_{t-3} + \phi_{4,4}r_{t-4} + e_{4t},
\end{aligned}$$

where $\phi_{0,j}, \phi_{i,j}$, and $\{e_{jt}\}$ are, respectively, the constant term, the coefficient of r_{t-i}, and the error term of an $AR(j)$ model. These models are in the form of a multiple linear regression and can be estimated using the least squares method. Hence, we can use the idea of F tests in multiple linear regression analysis.

The estimate $\hat{\phi}_{1,1}$ of the first equation is called the lag-1 sample PACF of r_t. The estimate $\hat{\phi}_{2,2}$ of the second equation is the lag-2 sample PACF of r_t. The estimate $\hat{\phi}_{3,3}$ is the lag-3 sample PACF of r_t and so on. From the definition, we see that the lag-2 PACF $\hat{\phi}_{2,2}$ is the added contribution of r_{t-2} to r_t over the AR(1) model $r_t = \phi_0 + \phi_1 r_{t-1} + e_{1t}$. The lag-3 PACF is the added contribution of r_{t-3} to r_t over an AR(2) model, and so on. Hence, for an $AR(p)$ model, the lag-p sample PACF should not be zero, but $\hat{\phi}_{j,j}$ should be close to zero for all $j > p$.

For a stationary Gaussian $AR(p)$ model, we can show that the sample PACF ha the property that $\hat{\phi}_{p,p}$ converges to ϕ_p, and $\hat{\phi}_{l,l}$ converges to 0 for all $l > p$, as the sample size T goes to infinity. Moreover, the asymptotic variance of $\hat{\phi}_{l,l}$ is $1/T$ for $l > p$. All these results state that, for an $AR(p)$ series, the sample PACF *cuts off* at lag p.

Lecture 21: Root of Characteristic Polynomial of AR Models and Moving Average Models

1 Interpreting Roots of Characteristic Polynomial

Let us consider an AR(2) model. Recall the Yule-Walker equation $\gamma_l = \sum_{i=1}^{p} \phi_i \gamma_{l-i}$. The moment equation of a stationary AR(2) model is given by

$$\rho_l = \phi_1 \rho_{l-1} + \phi_2 \rho_{l-2}, \qquad l \geq 1.$$

For a stationary AR(2) series, one gets $\rho_1 = \frac{\phi_1}{1-\phi_2}$, and $\rho_l = \phi_1 \rho_{l-1} + \phi_2 \rho_{l-2}$ for $l \geq 2$. Hence, the autocorrelation function (ACF) of a stationary AR(2) series satisfies the second order differential equation

$$(1 - \phi_1 B - \phi_2 B^2)\rho_l = 0, \tag{1}$$

where B is called the *back-shift* operator, and defined as $B\rho_l = \rho_{l-1}$. This difference equation determines the properties of the ACF. Consider the second-order polynomial equation above, i.e., $1 - \phi_1 x - \phi_2 x^2 = 0$. The solutions of this equation are

$$x = \frac{\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{-2\phi_2}.$$

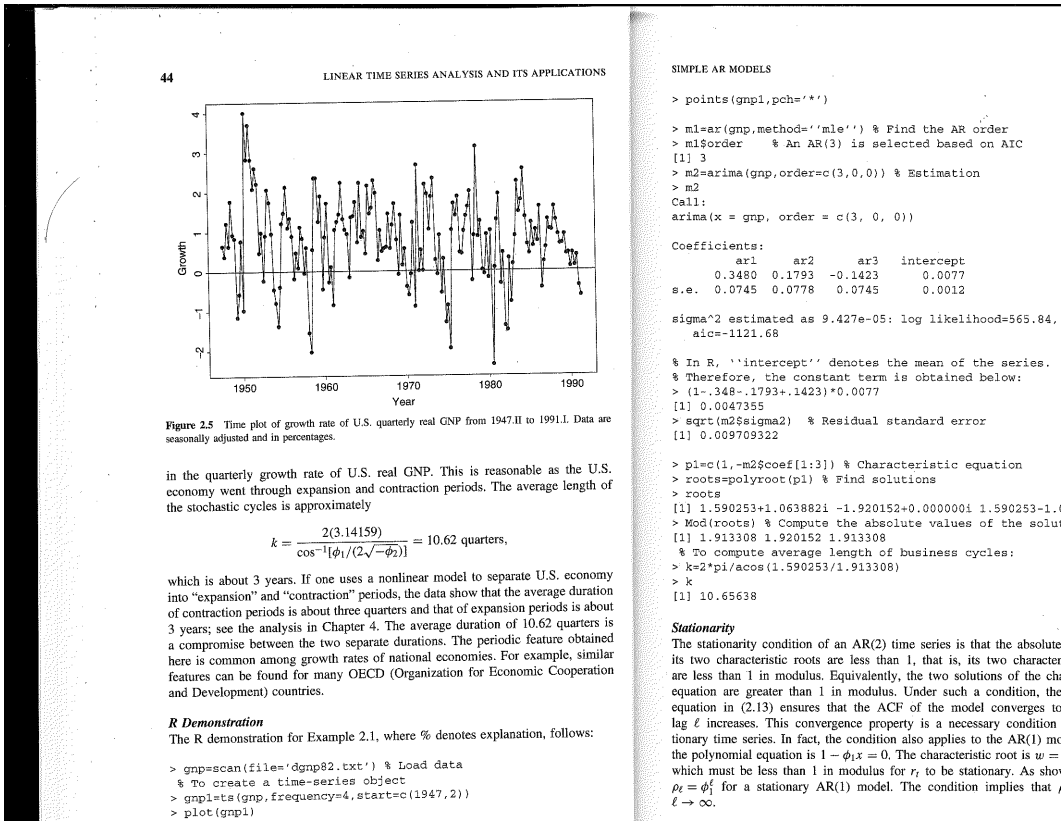
In the literature on time series, the reciprocals of the two solutions above are called the characteristic roots of the AR(2) model. Let ω_1 and ω_2 be the two characteristic roots (notice that $\omega_1 = \omega_2$ if $\phi_1^2 + 4\phi_2 = 0$). If they are both real valued, then the second-order difference equation of the model (1) can be factored as $(1 - \omega_1 B)(1 - \omega_2 B)$, where ω_1 and ω_2 are real. Then, this can be regarded as an AR(1) model that operates on top of another AR(1) model.

If $\phi_1^2 + 4\phi_2 < 0$, then we would have that ω_1 and ω_2 are complex numbers (a complex conjugate pair). The plot of the ACF of the return series r_t would then show a picture of damping sine and cosine waves. In financial applications, complex roots are important. They indicate business cycles. By writing the complex solutions $x = a \pm bi$, where $i = \sqrt{-1}$ is the imaginary unit, we can characterize the average length of the business cycle: this is given by the value k such that $2\pi/k = \theta$, where θ is the angle between $a + bi$ and the x-axis, so that in k steps, the angle goes back to 2π, i.e., we get a cycle. In other words, θ is the angle between (a, b) and $(1, 0)$. Notice that the cosine of this angle is given by the inner product of the corresponding unit norm vectors, i.e., $\langle \frac{1}{\sqrt{a^2+b^2}}(a, b), (1, 0) \rangle = \frac{a}{\sqrt{a^2+b^2}}$. Hence, $\theta = \cos^{-1}\left(a/(\sqrt{a^2 + b^2})\right)$, and as a result, $k = \frac{2\pi}{\cos^{-1}\left(a/(\sqrt{a^2+b^2})\right)}$, where the $\cos^{-1}$ is stated in radians. In our specific example where $x =$

$\frac{\phi_1 \pm \sqrt{\phi_1^2+4\phi_2}}{-2\phi_2}$, we have that[1]

$$k = \frac{2\pi}{\cos^{-1}\left(\phi_1/(2\sqrt{-\phi_2})\right)}. \tag{2}$$

Example: Consider the quarterly growth rate of the U.S. gross national product (GNP), seasonally adjusted[2], from the second quarter of 1947 to the first quarter of 1991. We report the time plot in Figure 1.



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Figure 2.5 Time plot of growth rate of U.S. quarterly real GNP from 1947.II to 1991.I. Data are seasonally adjusted and in percentages.

in the quarterly growth rate of U.S. real GNP. This is reasonable as the U.S. economy went through expansion and contraction periods. The average length of the stochastic cycles is approximately

$$k = \frac{2(3.14159)}{\cos^{-1}[\phi_1/(2\sqrt{-\phi_2})]} = 10.62 \text{ quarters},$$

which is about 3 years. If one uses a nonlinear model to separate U.S. economy into "expansion" and "contraction" periods, the data show that the average duration of contraction periods is about three quarters and that of expansion periods is about 3 years; see the analysis in Chapter 4. The average duration of 10.62 quarters is a compromise between the two separate durations. The periodic feature obtained here is common among growth rates of national economies. For example, similar features can be found for many OECD (Organization for Economic Cooperation and Development) countries.

R Demonstration

The R demonstration for Example 2.1, where % denotes explanation, follows:

```
> gnp=scan(file='dgnp82.txt') % Load data
 % To create a time-series object
> gnp1=ts(gnp,frequency=4,start=c(1947,2))
> plot(gnp1)
```

SIMPLE AR MODELS

```
> points(gnp1,pch='*')

> m1=ar(gnp,method=''mle'') % Find the AR order
> m1$order    % An AR(3) is selected based on AIC
[1] 3
> m2=arima(gnp,order=c(3,0,0)) % Estimation
> m2
Call:
arima(x = gnp, order = c(3, 0, 0))

Coefficients:
         ar1     ar2      ar3   intercept
      0.3480  0.1793  -0.1423      0.0077
s.e.  0.0745  0.0778   0.0745      0.0012

sigma^2 estimated as 9.427e-05: log likelihood=565.84,
   aic=-1121.68

% In R, ''intercept'' denotes the mean of the series.
% Therefore, the constant term is obtained below:
> (1-.348-.1793+.1423)*0.0077
[1] 0.0047355
> sqrt(m2$sigma2)  % Residual standard error
[1] 0.009709322

> p1=c(1,-m2$coef[1:3]) % Characteristic equation
> roots=polyroot(p1) % Find solutions
> roots
[1] 1.590253+1.063882i -1.920152+0.000000i 1.590253-1.(
> Mod(roots) % Compute the absolute values of the solut
[1] 1.913308 1.920152 1.913308
 % To compute average length of business cycles:
> k=2*pi/acos(1.590253/1.913308)
> k
[1] 10.65638
```

Stationarity

The stationarity condition of an AR(2) time series is that the absolute its two characteristic roots are less than 1, that is, its two character are less than 1 in modulus. Equivalently, the two solutions of the cha equation are greater than 1 in modulus. Under such a condition, the equation in (2.13) ensures that the ACF of the model converges to lag ℓ increases. This convergence property is a necessary condition tionary time series. In fact, the condition also applies to the AR(1) mo the polynomial equation is $1 - \phi_1 x = 0$, The characteristic root is $w =$ which must be less than 1 in modulus for r_t to be stationary. As sho $\rho_\ell = \phi_1^\ell$ for a stationary AR(1) model. The condition implies that $\ell \to \infty$.

Next, let us fit an AR(3) model to the data. Suppose we have already estimated the model, and found that the fitted model is given by

$$r_t = 0.0047 + 0.348 r_{t-1} + 0.179 r_{t-2} - 0.142 r_{t-3} + a_t,$$

and $\hat{\sigma}_a = 0.0097$. The ACF of this AR(3) satisfies the third-order difference equation

$$1 - 0.348B - 0.179B^2 + 0.142B^3 = 0,$$

[1]Observe that $\phi_2 < 0$, thus $\sqrt{\phi_2^2} = -\phi_2$.

[2]Seasonally adjusted refers to a process of removing the influence of regular, periodic fluctuations (like those related to seasons) from a time series data.

which can be factored as

$$(1 + 0.521B)(1 - 0.869B + 0.274B^2) = 0.$$

This representation captures the important features of the time series. The first factor shows the exponentially decaying feature of the GNP. Next, let us focus on the second-order factor $1-0.869B-(-0.274)B^2$. We deduce that $\phi_1^2 + 4\phi_2 = 0.869^2 + 4(-0.274) = -0.341 < 0$. Hence, the roots of the polynomial are complex, which confirms the existence of business cycles in the quarterly growth rate of U.S. GNP. This is in line with economic intuition, considering that the U.S. economy went through contraction and expansion periods. We can also compute the average length of the cycle k using Eq. (2). This leads to $k = \frac{2(3.14159)}{\cos^{-1}\left(\phi_1/(2\sqrt{-\phi_2})\right)} = 10.62$ quarters, which is roughly three years.

2 Moving Average Models

Moving average (MA) models is another class of tractable models that are useful for modeling return series in finance. They are useful to predict future values of a signal using the weighted moving average of past prediction errors, similar to the regression. We will introduce MA models as an infinite AR model, with some parameter constraints. Consider an AR model with infinite order as

$$r_t = \phi_0 + \phi_1 r_{t-1} + \phi_2 r_{t-2} + \ldots + a_t.$$

Such an AR model is not realistic because it has infinitely many parameters. To reduce the dimensionality and make the model practical, we can assume that the coefficients ϕ_i's satisfy some constraints so that they are determined by a finite number of parameters. For example,

$$r_t = \phi_0 - \theta_1 r_{t-1} - \theta_1^2 r_{t-2} - \theta_1^3 r_{t-3} + \ldots a_t, \tag{3}$$

where we have effectively imposed that $\phi_i = -\theta_1^i$. We require $|\theta_1| < 1$, otherwise the series will explode and the time series cannot be stationary. This, in turn, implies that $\theta_1^i \to 0$ as $i \to \infty$. Hence, the influence of r_{t-i} on r_t decays exponentially as i increases. We may rewrite Eq. (3) as

$$r_t + \theta_1 r_{t-1} + \theta_1^2 r_{t-2} + \theta_1^3 r_{t-3} + \cdots = \phi_0 + a_t. \tag{4}$$

This means that the model for r_{t-1} is

$$r_{t-1} + \theta_1 r_{t-2} + \theta_1^2 r_{t-3} + \cdots = \phi_0 + a_{t-1}. \tag{5}$$

Multiplying Eq. (5) by θ_1, and then subtracting it from Eq. (4), we obtain

$$r_t = \phi_0(1 - \theta_1) + a_t - \theta_1 a_{t-1}. \tag{6}$$

This means that, apart from the constant term $\phi_0(1 - \theta_1)$, r_t is a weighted average of the shocks a_t and a_{t-1}. For this reason, the model is called a MA model of order 1, or MA(1) for short.

Lecture 22: MA Models and Forecasting

1 Moving Average Models

The general form of a $MA(q)$ model is

$$r_t = c_0 + a_t - \sum_{i=1}^{q} \theta_i a_{t-i} \tag{1}$$

It can be seen easily that the unconditional mean is $\mathbb{E}[r_t] = c_0$, and the unconditional variance is

$$\gamma_0 := Var(r_t) = \left(1 + \sum_{i=1}^{q} \theta_i^2\right) \sigma_a^2.$$

Moving-average models are always weakly stationary, because they are finite linear combinations of a white noise sequence, for which the first two moments are time invariant.

Let us next compute the auto-covariance function of the model. For $l > q$, $\gamma_l := cov(r_t, r_{t-l}) = 0$. For $1 \leq l \leq q$,

$$\begin{aligned}
\gamma_l &= cov(r_t, r_{t-l}) = -\sum_{i=l}^{q} \theta_i cov(a_{t-i}, r_{t-l}) \\
&= -\sum_{i=l}^{q} \theta_i cov\left(a_{t-i}, a_{t-l} - \sum_{j=1}^{q} \theta_j a_{t-l-j}\right) \\
&= \left[-\theta_l + \sum_{i=l+1}^{q} \theta_i \theta_{i-l}\right] \sigma_a^2
\end{aligned}$$

Recall that the lag-l autocorrelation is given by $\rho_l = \frac{\gamma_l}{\gamma_0}$. This yields

$$\rho_l = \frac{\gamma_l}{\gamma_0} = \frac{-\theta_l + \sum_{i=l+1}^{q} \theta_i \theta_{i-l}}{1 + \sum_{i=1}^{q} \theta_i^2}$$

Forecasting: Assume the time series follows $MA(q)$. The l-step ahead forecast is given by

$$\hat{r}_T(l) := \mathbb{E}\left[r_{T+l} \middle| \mathcal{F}_T\right] = c_0 - \sum_{i=l}^{q} \theta_i a_{T+l-i}.$$

The forecasting error is given by

$$\hat{e}_T(l) := r_{T+l} - \hat{r}_T(l) = a_{T+l} - \sum_{i=1}^{\min(l-1,q)} \theta_i a_{T+l-i}.$$

The variance of the forecasting error is then given by

$$\mathbb{E}\left[\left(\hat{e}_T(l)\right)^2\right] = Var\left(\hat{e}_T(l)\right) = \sigma_a^2 \left(1 + \sum_{i=1}^{\min(l-1,q)} \theta_i^2\right).$$

2 Estimation of MA(q) models

The maximum likelihood method is typically used to estimate a $MA(q)$ model. Suppose r_t, $t = 1, \ldots T$, are observed. We consider the conditional likelihood function of r_t, $t = 1, \ldots, T$, given $a_0, a_{-1}, \ldots, a_{-q+1}$. For each $t = 1, \ldots, T$, we have

$$a_t = r_t + \sum_{i=1}^{q} \theta_i a_{t-i} - c_0.$$

Given $a_{t-1}, \ldots, a_{t-q}$, we have that r_t follows a Gaussian with mean $c_0 - \sum_{i=1}^{q} \theta_i a_{t-i}$ and variance σ_a^2. The conditional log-likelihood of $r_1, \ldots, r_T$ is given by

$$\begin{aligned} \log L(r_{1:T} \big| c_0, \theta_{1:q}, a_{-q+1:0}) &= -\frac{T}{2}\log(2\pi) - \frac{T}{2}\log\sigma_a^2 \\ &\quad -\frac{1}{2\sigma_a^2}\sum_{i=1}^{T}\left(r_t - \left(c_0 - \sum_{i=1}^{q}\theta_i a_{t-i}\right)\right)^2 \end{aligned}$$

2.1 Example

We present an application to estimation of bid-ask spreads in financial time series. The example is based on the paper R. Roll (1984), A Simple Implicit Measure of the Effective Bid-Ask Spread in an Efficient Market, Journal of Finance. Suppose that P_t^* is the unobserved fundamental price of the asset without the bid-ask spread. Moreover, assume that $\eta_t := P_t^\star - P_{t-1}^\star$ are i.i.d. with zero mean and variance σ_η^2. Let $s > 0$ be the bid-ask spread. Then the observed asset price is given by

$$P_t = P_t^\star + \frac{s}{2} I_t,$$

where $I_t = 1$ if it is a buyer initiated transaction (the buyer would have to buy the asset at the ask price given by $P_t = P_t^\star + s/2$). Instead, $I_t = -1$ if it is a seller initiated transaction (the seller would have to see the asset at the bid price given by $P_t = P_t^\star - s/2$). Assume that I_t's are i.i.d. and have zero mean, i.e., $P(I_t = 1) = P(I_t = -1) = \frac{1}{2}$. This implies $Var(I_t) = 1$ for each t.

Consider the observed stock price change given by

$$y_t := P_t - P_{t-1} = \eta_t + \frac{s}{2}\left(I_t - I_{t-1}\right).$$

The variance is $Var(y_t) = \sigma_\eta^2 + \frac{s^2}{2}$. The auto-covariance is given by

$$\gamma_1 := cov(y_t, y_{t-1}) = -\frac{s^2}{4}.$$

Moreover, $\gamma_l = 0$ for $l \geq 2$. It then follows directly the the lag-1 autocorrelation is

$$\rho_1 = -\frac{s^2/4}{\sigma_\eta^2 + s^2/2},$$

while $\rho_l = 0$ for $l \geq 2$. Hence, the observed price change y_t follows a time-series resembling a MA(1) model with negative autocorrelation, although the fundamental price change is i.i.d.

Lecture 23: ARMA Models

1 ARMA Model

The generalized version of the ARMA model with parameters p and q is

$$r_t = \phi_0 + \sum_{i=1}^{p} \phi_i r_{t-i} + a_t - \sum_{i=1}^{q} \theta_i a_{t-i}. \tag{1}$$

For any time series x_t, introduce the characteristic functions

$$\phi(z) := 1 - \sum_{i=1}^{p} \phi_i z^i, \qquad \theta(z) := 1 - \sum_{j=1}^{q} \theta_j z^j,$$

for any $z \in \mathcal{C}$. Recall that back-shift operator $Bx_t := x_{t-1}$. Then, we can rewrite the $ARMA(p,q)$ model as

$$\phi(B) r_t = \phi_0 + \theta(B) a_t,$$

or, in an extended form, as

$$\left(1 - \phi_1 B - \ldots \phi_p B^p\right) r_t = \phi_0 + \left(1 - \theta_1 B - \ldots - \theta_q B^q\right) a_t,$$

We require that $\phi(\cdot)$ and $\theta(\cdot)$ have no common factors, otherwise the order (p,q) can be reduced. In other words, we require that the polynomial $\phi(\cdot)$ and $\theta(\cdot)$ are co-prime.

Remark 1.1. *Some of the material presented in these notes is based on fundamental concepts from Complex Analysis. We refer to the online book, Taylor, Michael. Introduction to Complex Analysis, for an in-depth coverage.*

We will use the fact that a holomorphic function, i.e., one that is complex differentiable in a neighborhood of each point in its domain admits a power series representation within its domain.

Stationarity: The $ARMA(p,q)$ model defines a stationary, linear process if and only if all the roots of the AR characteristic equation $\phi(z) = 0$ are not on the unit circle in the complex plane.

Causality: The requirement that ϕ has no complex zeros inside or on the unit circle guarantees *causality*, i.e., that the return at time t depends on present and past innovation (shock) terms a_s, $s \leq t$, but not on future innovations. We also say that the $AR(p)$ process is causal.

Invertibility. We require that θ has no complex zeros inside or on the unit circle guarantees that the ARMA model is *invertible*, i.e., we can recover the innovation terms a_t using present and past observations of the returns r_u, $u \leq t$.

2 MA Representation of ARMA(p,q) models

Given two polynomials $\phi(B) = 1 - \sum_{i=1}^{p} \phi_i B^i$ and $\theta(B) = 1 - \sum_{i=1}^{q} \theta_i B^i$, we can obtain, by long division, that

$$\frac{\theta(B)}{\phi(B)} = 1 + \psi_1 B + \psi_2 B^2 + \ldots =: \psi(B)$$

and

$$\frac{\phi(B)}{\theta(B)} = 1 - \pi_1 B - \pi_2 B^2 - \ldots =: \pi(B).$$

For instance, if $\phi(B) = 1 - \phi_1 B$ and $\theta(B) = 1 - \theta_1 B$, then

$$\psi(B) = \frac{1 - \theta_1 B}{1 - \phi_1 B} = 1 + (\phi_1 - \theta_1) B + \phi_1(\phi_1 - \theta_1) B^2 + \phi_1^2 (\phi_1 - \theta_1) B^3 + \ldots,$$

and

$$\pi(B) = \frac{1 - \phi_1 B}{1 - \theta_1 B} = 1 - (\phi_1 - \theta_1) B - \theta_1(\phi_1 - \theta_1) B^2 - \theta_1^2 (\phi_1 - \theta_1) B^3 + \ldots.$$

It follows directly from the definitions that $\psi(B)\pi(B) = 1$. Recall the representation of an $ARMA(p, q)$ model as $\phi(B) r_t = \phi_0 + \theta(B) a_t$. The MA representation of an $ARMA(p, q)$ model is given by

$$r_t = \phi(B)^{-1}\phi_0 + \phi(B)^{-1}\theta(B) a_t = c_0 + \psi(B) a_t = c_0 + a_t + \sum_{i=1}^{\infty} \psi_i a_{t-i}, \tag{2}$$

where we have set $c_0 := \phi(B)^{-1}\phi_0$, and we can easily see that $\phi(B)^{-1}\phi_0 = \frac{\phi_0}{1-\phi_1-\cdots-\phi_p}$ using the following argument: $\phi(B)^{-1}\phi_0 = \frac{\phi_0}{1-\phi_1-\cdots-\phi_p}$ if and only if $\phi_0 = \phi(B)\frac{\phi_0}{1-\phi_1-\cdots-\phi_p}$ where we have applied the operator $\phi(B)$ on the left and right hand side of the previous equality. Observe that, for a constant c, Bc is a constant, which further implies that $B^n c = c$ for any integer n. Setting $c := \frac{\phi_0}{1-\phi_1-\cdots-\phi_p}$ we would obtain that $\phi(B)c = c - \phi_1 c - \phi_2 c - \ldots - \phi_p c = c(1 - \phi_1 - \ldots - \phi_p) = \phi_0$, hence proving the result.

Remark 2.1. *In writing the representation* (2)*, we are using the fact that ϕ has no complex root within the unit circle. If ϕ had complex roots within or at the boundary of the unit circle, then the series r_t would take the form*

$$r_t = \phi(B)^{-1}\phi_0 + \phi(B)^{-1}\theta(B) a_t = c_0 + \psi(B) a_t = c_0 + \sum_{i=-\infty}^{\infty} \psi_i a_{t-i}, , \tag{3}$$

where $\psi_i \neq 0$ for some $i < 0$, so that the return at time t would also depend on future innovations a_s for $s > t$.

The representation (2) shows explicitly the impact of the past shock $a_{t-i} (i > 0)$ on the current return r_t. The coefficients ψ_i's are referred to as the *impulse response function* of the ARMA model. For a weakly stationary series, the ψ_i coefficients decay exponentially as i increases, consistently with intuition: the effect of shock a_{t-i} on the return r_t should decrease over time. Thus, the shock a_{t-i} does not have a permanent impact on the series.

Remark 2.2. *Suppose we want to recover the innovations as past observations of r_t. Then we look for a AR representation of the $ARMA(p,q)$ model, i.e., starting from*

$$\phi(B)r_t = \phi_0 + \theta(B)a_t, \tag{4}$$

where we recall that

$$\phi(z) := 1 - \sum_{i=1}^{p} \phi_i z^i, \qquad \theta(z) := 1 - \sum_{j=1}^{q} \theta_j z^j.$$

If we multiply LHS and RHS of (4) by $\theta(B)^{-1}$, we obtain

$$a_t = \theta(B)^{-1}\phi(B)r_t - \theta(B)^{-1}\phi_0.$$

Then, we can write

$$a_t = d_0 + \pi(B)r_t = d_0 + r_t - \sum_{i=1}^{\infty} \pi_i r_{t-i}.$$

where $d_0 = -\theta(B)^{-1}\phi_0$. The above representation holds only if the ARMA model is invertible, i.e., $\theta(B)$ does not have any complex root within or on the unit circle. If $\theta(B)$ has complex roots within or on the unit circle, then the representation would take the form $a_t = d_0 + \pi(B)r_t = d_0 - \sum_{i=-\infty}^{\infty} \pi_i r_{t-i}$, where $\pi_i \neq 0$ for some $i < 0$. Hence, the shock (innovation) at time t would depend on future returns r_s, $s > t$.

Lecture 24: Trend and Seasonality of Time Series Models and ARCH Models

1 Estimation and Forecasting with ARMA Models

Consider an ARMA model with parameters p and q, given by

$$r_t = \phi_0 + \sum_{i=1}^{p} \phi_i r_{t-i} + a_t - \sum_{i=1}^{q} \theta_i a_{t-i}. \tag{1}$$

ARMA models can be estimated using maximum likelihood. We do not have time to cover it, but you are referred to the book by Brockwell and Davis (2006), "Time Series: Theory and Methods" for the details.

Let us discuss forecasting using an ARMA model. The 1-step ahead forecast r_{T+1} can be easily obtained from the model and is given by

$$\hat{r}_T(1) := \mathbb{E}\left[r_{T+1}\middle|\mathcal{F}_T\right] = \phi_0 + \sum_{i=1}^{p} \phi_i r_{T+1-i} - \sum_{i=1}^{q} \theta_i a_{T+1-i},$$

and the corresponding forecast error is $e_T(1) = r_{T+1} - \hat{r}_T(1) = a_{T+1}$. The variance is $Var\left(e_T(1)\right) = \sigma_a^2$. For the l-steps ahead forecast, we have

$$\hat{r}_T(l) = \mathbb{E}\left[r_{T+l}\middle|\mathcal{F}_T\right] = \phi_0 + \sum_{i=1}^{p} \phi_i \hat{r}_T(l-i) - \sum_{i=1}^{q} \theta_i a_T(l-i),$$

where $\hat{r}_T(l-i) = r_{T+l-i}$ if $l-i \leq 0$, and $a_T(l-i) = 0$ if $l-i > 0$ and $a_T(l-i) = a_{T+l-i}$ if $l-i \leq 0$. Equivalently, we can use the MA representation of an $ARMA(p,q)$ given by

$$r_t = \phi(B)^{-1}\phi_0 + \phi(B)^{-1}\theta(B)a_t = c_0 + \psi(B)a_t = c_0 + a_t + \sum_{i=1}^{\infty} \psi_i a_{t-i}, \tag{2}$$

where $c_0 = \frac{\phi_0}{1-\sum_{i=1}^{p}\phi_i}$ is the common mean μ of the time series. This yields

$$\hat{r}_T(l) = c_0 + \sum_{i=l}^{\infty} \psi_i a_{T+l-i}. \tag{3}$$

We then obtain the forecast error

$$e_T(l) := r_{T+l} - \hat{r}_T(l) = a_{T+l} + \sum_{i=1}^{l-1} \psi_i a_{T+l-i}.$$

Then,

$$Var(e_T(l)) = \mathbb{E}\left[(e_T(l))^2\right] = \sigma_a^2 \left(1 + \sum_{i=1}^{l-1} \psi_i^2\right), \tag{4}$$

which, as expected, is a non-decreasing function of the forecast horizon l. The MA representation of the ARMA model gives a simple proof of mean reversion of a stationary time series. The stationarity implies that ψ_i approaches zero as $i \to \infty$. Therefore, by Eq. (3), we have that $\hat{r}_T(l) \to c_0$ as $l \to \infty$. Because $\hat{r}_T(l)$ is the conditional expectation of r_{T+l} at the forecast origin T, this result says that in the long term the return series is expected to approach its mean c_0, that is, the series is mean reverting. The speed by which $\hat{r}_T(l)$ approaches c_0 determines the speed of mean reversion. Furthermore, using the MA representation given in (2), we have $Var(r_t) = (1 + \sum_{i=1}^{\infty} \psi_i^2)\sigma_a^2$, if the series ψ_i is summable, i.e., if $\sum_{i=1}^{\infty} \psi_i^2 < \infty$. Consequently, comparing equations (2) and (4), we obtain that $Var(e_T(l)) \to Var(r_t)$ as $l \to \infty$.

2 Unit Root Test and Stationarity

We have so far focused on time series which are stationary. This may be a reasonable assumption for returns of securities (e.g. stock returns or bond returns). However, this may not be the case if we consider interest rates, foreign exchange rates, or the price series (as opposed to the return series) of an asset. If we consider a price series, the non-stationarity is typically due to the fact that there is no "fixed level" for the price. Assume, for instance, that the price of the stock follows a geometric Brownian motion. The log-price of the stock follows an arithmetic Brownian motion

$$p_t = \mu + p_{t-1} + a_t,$$

where $\{a_t\}$ is a white noise series. The log-price process p_t is nonstationary. We can rewrite the above equation as $\phi(B)p_t = \mu + a_t$, where the characteristic polynomial $\phi(z) := 1 - z$. This polynomial has a zero within the circle, and in particular it has a *unit root*. This time series fails to satisfy the weak stationarity condition of an AR(1) model.

Using a MA representation, we can write the above series as $p_t = p_0 + \mu t + \sum_{i=1}^{t} a_i$. The term μt is the time trend. Thus, the impact of any past shock a_{t-i} on p_t does not decay over time. Consequently, the series has a strong memory as it remembers all the past shocks. In economics, the shocks are said to have a *permanent effect* on the time series. The strong memory of a unit-root time series can be seen from the sample ACF of the observed series. If the sample ACF does not decay exponentially, it is likely that the time series is non-stationary due to a unit root.

Remark 2.1. *Many people use "nonstationary" interchangeably with "has a unit root". This is one form of nonstationarity, but there are lots of others. For example, if the time series has a structural shift occurring at one point in time, then it would be nonstationary.*

A classical test used for unit roots is the Dickey-Fuller test, which tests the null hypothesis that a unit root is present in an autoregressive model. For example, in an AR(1) model with characteristic polynomial $\phi(z) = 1 - \phi_1 z$, the null hypothesis would be $\phi_1 = 1$. The details of the test are reported in the paper by Dickey and Fuller (1979), and a sketch of the test is given in Section 2.7.5 of the textbook "Linear time series analysis and its applications" by Tsay.

Remark 2.2. *In general, unit root tests have very low power. That is, unit root tests cannot distinguish highly persistent stationary processes from nonstationary processes very well. Also, the*

power of unit root tests diminish as deterministic terms are added to the test regressions. That is, tests that include a constant and trend in the test regression have less power than tests that only include a constant in the test regression. Most recent tests proposed by Elliot, Rothenberg and Stock (1996) and Ng and Perron (2001) have been designed to alleviate these problems.

Lecture 25: ARCH Models

1 Dealing with nonstationarity of a time series

A conventional approach to handle unit-root nonstationarity is to use *differencing*. Although p_t is nonstationary, define the time series

$$r_t := p_t - p_{t-1} = \mu + a_t,$$

which is *stationary*. The time series $\{r_t\}_{t\geq 0}$ is the *first differenced* series of $\{p_t\}_{t\geq 0}$, i.e. $r_t := \Delta p_t$, where $\Delta := 1 - B$ is the difference operator. For a time series $\{p_t\}_{t\geq 0}$, it is possible that its first difference series Δp_t still has a unit root. This typically happens if the original time series p_t contain multiple unit roots. One may then consider its second, or d-th in general, differenced series $\Delta^d p_t$, until we obtain a weakly stationary time series.

An $ARMA(p, q)$ time series for $\Delta^d r_t$ is called an *autoregressive integrated moving average*, $ARIMA(p, d, q)$, for the time series r_t.

2 Trend and Seasonality

In general, a time series $\{r_t\}$ can be decomposed into "trend + seasonality + stationary time series". Indeed some financial time series, such as the quarterly earnings per share of a company, exhibits cyclical behavior. Both trends and seasonality can be removed by taking differences. For instance, suppose

$$r_t = f_t + y_t,$$

where $\{y_t\}_{t\geq 0}$, is a stationary process, and $f_t = at$ is a linear function representing a linear trend. Then, we have

$$\Delta r_t := r_t - r_{t-1} = \Delta f_t + \Delta y_t = a + \Delta y_t,$$

which is a stationary time series. Hence, the trend can be removed by taking differences. Next, suppose

$$r_t = g_t + y_t,$$

where $\{y_t\}_{t\geq 0}$ is a stationary process, and g_t is a period function of time t with periodicity s. Consider $\Delta_s g_t := (1 - B^s)g_t = 0$, where we recall that B is the back-shift operator. The operation $\Delta_s = (1 - B^s)$ is called seasonal differencing. Then

$$\Delta_s r_t = \Delta_s g_t + \Delta_s y_t = \Delta_s y_t$$

Hence, $\{\Delta_s r_t\}$ is stationary. Most economic data published by the US government are seasonally adjusted (e.g. the growth rate of gross domestic product, and the unemployment rate). Economic

growth is affected by different seasonal factors such as the weather and the holidays. Economists get a better picture of how an economy is moving when they adjust their analyses based on these factors. For example, roughly two-thirds of U.S. gross domestic product (GDP) is made up of consumer spending—which is a seasonal measure. The more consumers spend, the more the economy grows.

Remark 2.1. *How can we determine the ARMA signatures, i.e., whether we are dealing with an AR(p) model, or with a MA(q) model, or with an ARMA(p,q). We first plot the ACF and the PACF. Now, we know that for an AR(p) model, the ACF decays geometrically and PACF has p significant lags. If that is the case, then we have an ARMA(p,0) model, i.e., the signature is an AR model. Next, recall that a MA(q) model can be reproduced using an AR(∞) model. Hence, the ACF of a MA(q) model would have q significant lags (i.e., cut off at lag q), while the PACF would never cut off but decay geometrically. If our ARMA time series exhibit these characteristics, then we have a MA(q) model. If instead we have that both the ACF and PACF decay geometrically, then we have an ARMA model.*

3 ARCH Models and Stochastic Volatility

ARCH/GARCH models are suitable to model characteristics of volatility. Volatility is not directly observable. Consider, for example, the daily log returns of the IBM stock. The daily volatility is not directly observable from the return data because there is only one observation in a trading day. If intraday data such as 10-minute returns are available, then one could estimate the daily volatility even though the accuracy of such an estimate remains questionable.

Despite not being directly observable, the volatility presents characteristics that are commonly observed in asset returns. First, there exists volatility clusters, i.e., periods where volatility is high are followed by other periods with high volatility. Second, volatility evolves continuously over time, that is, volatility jumps are rare. Third, volatility does not diverge to infinity but varies within some fixed range. Statistically, this means that volatility is often stationary. Fourth, volatility reacts differently to a price increase or a price drop, also referred to as the *leverage* effect. Specifically, the observed tendency is for an asset's volatility to be negatively correlated with the asset's returns. Typically, rising asset prices are accompanied by declining volatility, and vice versa. The term "leverage" refers to one possible economic interpretation of this phenomenon, developed in Black (1976) and Christie (1982): as asset prices decline, companies become mechanically more leveraged because the relative value of their debt rises relative to that of their equity. As a result, it is natural to expect that their stock becomes riskier, hence more volatile. It has also been documented that the effect is generally asymmetric: other things equal, declines in stock prices are accompanied by larger increases in volatility than the decline in volatility that accompanies rising stock markets (see, e.g., Nelson, 1991; and Engle and Ng, 1993).

Lecture 26: GARCH Models

1 Announcements

Sample final exam and final review sheet have been posted on Courseworks. The final review sheet contains all details about the exam. There will be a review session conducted by the TA Jose Sidaoui Gali on May 9, 2-3:30pm in 1127 MUDD. All details are also available in the final review sheet.

2 ARCH Models

How do we know if ARCH is necessary for the time series we are considering? Construct a residual plot, in which we plot $\epsilon_t := r_t - \hat{r}_{t-1}(1)$, where r_t follows an ARMA model. First, we visually check if the residual plot displays any cluster of volatility. Second, we can display the plot of squared residuals of the time series. If the volatility is constant, the variance of the error terms should be close to constant through time. If we observe that the variances of data are different over different periods of time, then it means that a better model is needed to capture the stochastic behavior of the volatility.

Consider the monthly log stock returns of Intel Corporation from January 1973 to December 2008 shown in Figure 1. It suggests no significant serial correlation (except for a minor one at lag 7). Moreover, it shows that the sample ACF of the absolute log returns (i.e., $|r_t|$) and the squared log returns r_t^2 are more significant. Combining these plots, we deduce that the log returns are *serially uncorrelated but dependent.* Volatility models attempt to capture such dependence in the return series.

The first model that provides a systematic framework for volatility modeling is the *Auto-Regressive conditional heteroscedastic model* (ARCH) model of Engle (1982). The basic idea of an $ARCH(1)$ model is that the shocks $\{a_t\}$ of asset returns are serially uncorrelated, but dependent, and the dependence of a_t can be described by a quadratic function of its lagged values. The model is specified by

$$a_t = \sigma_t \epsilon_t, \qquad \sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2,$$

where $\{\epsilon_t\}$ is a white noise sequence with zero mean and variance 1, $\alpha_0 > 0$ and $\alpha_1 \geq 0$ are constants.

In practice, ϵ_t is assumed to follow the standard normal or the Student's t distribution. Note that the variance at time t is connected to the value of the series at time $t-1$. A large past squared shock a_{t-1}^2 implies a larger conditional variance σ_t^2 for the innovation a_t. Consequently a_t tends to assume a large value in absolute value. This implies that, under the ARCH framework, a large shock tends to be followed by another large shock. *Conditional heteroskedasticity* is said to occur when the variance of the shock a_t, conditioned on the past shocks $a_1, \ldots, a_{t-1}$ changes dynamically

with t. In other words, σ_t^2 is not constant but dynamically changes over time as new shocks are realized. It is called *conditional heteroscedasticity* to emphasize the fact that it is the sequence of conditional variances that changes and not the unconditional variance.

Notice that $\{a_t\}$ is a serially uncorrelated zero-mean time series. Indeed,

$$\begin{aligned}
\mathbb{E}\left[a_t\right] &= \mathbb{E}\left[\mathbb{E}\left[a_t\middle|\mathcal{F}_{t-1}\right]\right] = \mathbb{E}\left[\sigma_t \mathbb{E}\left[\epsilon_t\middle|\mathcal{F}_{t-1}\right]\right] = 0 \\
Cov(a_t, a_{t-l}) &= \mathbb{E}[a_t a_{t-l}] = \mathbb{E}\left[\mathbb{E}[a_t a_{t-l}|\mathcal{F}_{t-1}]\right] \\
&= \mathbb{E}\left[\mathbb{E}[\sigma_t \epsilon_t \sigma_{t-l}\epsilon_{t-l}|\mathcal{F}_{t-1}]\right] = \mathbb{E}\left[\sigma_t\sigma_{t-l}\epsilon_{t-l}\mathbb{E}[\epsilon_t|\mathcal{F}_{t-1}]\right] = 0,
\end{aligned}$$

for any $l \geq 1$. We have used that ϵ_t is a white noise sequence, and that $\sigma_t, \sigma_{t-l}, \epsilon_{t-l}$ are all known at time $t-1$. Notice that

$$\begin{aligned}
Var(a_t) &:= \mathbb{E}\left[a_t^2\right] = \mathbb{E}\left[\left(\alpha_0 + \alpha_1 a_{t-1}^2\right)\epsilon_t^2\right] \\
&= \mathbb{E}\left[\left(\alpha_0 + \alpha_1 a_{t-1}^2\right)\mathbb{E}\left[\epsilon_t^2\middle|\mathcal{F}_{t-1}\right]\right] \\
&= \mathbb{E}\left[\alpha_0 + \alpha_1 a_{t-1}^2\right]
\end{aligned}$$

using that $Var(\epsilon_t) = 1$. If the time series a_t is weakly stationary, it follows immediately that

$$Var(a_t) = \frac{\alpha_0}{1-\alpha_1}.$$

Since the variance of a_t must be positive, we require $0 \leq \alpha_1 < 1$.

3 Leptokurtic Feature of ARCH Models

Consider

$$a_t = \sigma_t \epsilon_t, \qquad \sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2,$$

where $\{\epsilon_t\}$ is a white noise sequence with zero mean and variance 1. Let us analyze the *leptokurtic* feature of the ARCH time series. The leptokurtic feature of a frequency distribution is the extent to which the distribution has greater kurtosis than the normal distribution, i.e, it has fatter tails than the normal distribution. The more extreme points along the x-axis of the distribution are also called outliers.

Suppose that $\{a_t\}$ has stationary fourth order moment. Under the normality assumption of ϵ_t, and recalling that the fourth moment of a zero-mean and unit variance normal distribution is 3, we have that the conditional 4-th order moment is

$$\mathbb{E}\left[a_t^4\middle|\mathcal{F}_{t-1}\right] = \left(\alpha_0 + \alpha_1 a_{t-1}^2\right)^2 \mathbb{E}\left[\epsilon_t^4\middle|\mathcal{F}_{t-1}\right] = 3\left(\alpha_0 + \alpha_1 a_{t-1}^2\right)^2.$$

The unconditional 4-th order moment is

$$\begin{aligned}
\mathbb{E}\left[a_t^4\right] &= 3\left(\alpha_0^2 + 2\alpha_0\alpha_1\mathbb{E}\left[a_{t-1}^2\right] + \alpha_1^2\mathbb{E}\left[a_{t-1}^4\right]\right) \\
&= 3\left(\frac{\alpha_0^2(1+\alpha_1)}{1-\alpha_1} + \alpha_1^2\mathbb{E}\left[a_{t-1}^4\right]\right)
\end{aligned}$$

Because of stationarity of the 4-th order moment, i.e., $\mathbb{E}[a_t^4] = \mathbb{E}\left[a_{t-1}^4\right]$, we obtain

$$\mathbb{E}\left[a_t^4\right] = \frac{3\alpha_0^2(1+\alpha_1)}{(1-3\alpha_1^2)(1-\alpha_1)}.$$

This result has two important implications. First, since the fourth moment of a_t is positive, we have that α_1 must satisfy the condition $1 - 3\alpha_1^2 > 0$, that is, $0 \leq \alpha_1^2 < 1/3$. Second, the unconditional kurtosis is

$$\frac{\mathbb{E}[a_t^4]}{Var(a_t)^2} = \frac{3(1-\alpha_1^2)}{1-3\alpha_1^2} > 3$$

Hence, the kurtosis increases with α_1. Notice that the kurtosis of a standard univariate Gaussian is 3. Hence, the above inequality implies that the tail distribution of a_t is heavier than that of a normal distribution. In other words, the shock a_t of a Gaussian ARCH(1) model is more likely than a Gaussian white noise series to produce "outliers". This is in line with the empirical finding that outliers appear more often in asset returns that what implied by a Gaussian i.i.d sequence. These properties continue to hold for higher order ARCH models, but the formulas become more complicated.

3.1 ARCH(m) model

The direct generalization of the ARCH(1) model is the ARCH(m) model given by

$$a_t = \sigma_t \epsilon_t, \qquad \sigma_t^2 = \alpha_0 + \sum_{i=1}^{m} \alpha_i a_{t-i}^2,$$

where $\{\epsilon_t\}$ is a white noise with variance 1. α_0 and α_i's are nonnegative constants. Under the assumption that $\{a_t\}$ is stationary, we have that

$$Var(a_t) = \frac{\alpha_0}{1 - \sum_{i=1}^{m} \alpha_i}.$$

For the variance to be positive, we require that $\sum_{i=1}^{m} \alpha_i < 1$.

3.2 Test for an ARCH Effect

A popular method to test for an ARCH effect is the Lagrange multiplier test proposed by Engle (1982). Let us assume that we have an $ARMA(p,q)$ model for the return series r_t. We build the *mean equation* $a_t = r_t - \mathbb{E}\left[r_t \middle| \mathcal{F}_{t-1}\right]$. Since ARCH suggests that a_t^2 is dependent of lags of itself, we can test the hypothesis of no ARCH by regressing a_t^2 on lags of itself, and test whether the parameters of this regression are statistically and economically significant. The test is equivalent to the usual F statistic for testing $\alpha_i = 0$, $i = 1, \ldots, m$, in the linear regression

$$a_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \alpha_2 a_{t-2}^2 + \ldots + \alpha_m a_{t-m}^2 + e_t, \qquad t = m+1, \ldots, T$$

where e_t denotes the error term, m is a prespecified positive integer, and $T - m$ is the sample size.[1] Specifically, the null hypothesis is $H_0 := \alpha_1 = \alpha_2 = \cdots = \alpha_m = 0$. Let $SSR_0 = \sum_{t=m+1}^{T} (a_t^2 - \bar{\omega})^2$,

[1] One way to choose m is to compare loglikelihood values for different choices of m.

where $\bar{\omega} = \frac{1}{T-m}\sum_{t=m+1}^{T} a_t^2$ is the sample mean of a_t^2, and $SSR_1 = \sum_{t=m+1}^{T} \hat{e}_t^2$, where $\hat{e}_t = a_t^2 - \widehat{a_t^2}$ is the least square residual of the previous linear regression. Then we have

$$F = \frac{(SSR_0 - SSR_1)/m}{SSR_1/(T-2m-1)},$$

follows a F distribution with degrees of freedom m and $T-2m-1$ under the null hypothesis (we have $T-m$ observations and we are estimating $m+1$ parameters so that we lose $m+1$ degrees of freedom) Morover, mF is asymptotically distributed as a chi-squared distribution with m degrees of freedom under the null hypothesis. The decision rule is to reject the null hypothesis if $mF > \chi^2_{m,1-\alpha}$, where $\chi^2_{m,1-\alpha}$ is the $100(1-\alpha)$th percentile of χ^2_m.

3.3 Forecast of ARCH model

At the forecast origin T, the 1-step ahead forecast of σ^2_{T+1} is

$$\hat{\sigma}^2_T(1) := \mathbb{E}\left[\sigma^2_{T+1}\middle|\mathcal{F}_T\right] = \alpha_0 + \sum_{i=1}^{m} \alpha_i a^2_{T+1-i}.$$

The 2-step ahead forecast is

$$\begin{aligned}
\hat{\sigma}^2_T(2) &:= \mathbb{E}\left[\sigma^2_{T+2}\middle|\mathcal{F}_T\right] \\
&= \alpha_0 + \alpha_1 \mathbb{E}\left[\epsilon^2_{T+1}\sigma^2_{T+1}|\mathcal{F}_T\right] + \alpha_2 \mathbb{E}\left[a^2_T|\mathcal{F}_T\right] + \ldots \alpha_m a^2_{T+2-m} \\
&= \alpha_0 + \alpha_1 \mathbb{E}\left[\epsilon^2_{T+1}\sigma^2_{T+1}|\mathcal{F}_T\right] + \alpha_2 a^2_T + \ldots \alpha_m a^2_{T+2-m} \\
&= \alpha_0 + \alpha_1 \mathbb{E}\left[\epsilon^2_{T+1}\right]\mathbb{E}\left[\sigma^2_{T+1}|\mathcal{F}_T\right] + \alpha_2 a^2_T + \ldots \alpha_m a^2_{T+2-m} \\
&= \alpha_0 + \alpha_1 \hat{\sigma}^2_T(1) + \alpha_2 a^2_T + \ldots \alpha_m a^2_{T+2-m}
\end{aligned}$$

The l-step ahead forecast is

$$\hat{\sigma}^2_T(l) := \mathbb{E}\left[\sigma^2_{T+l}\middle|\mathcal{F}_T\right] = \alpha_0 + \sum_{i=1}^{m} \alpha_i \hat{\sigma}^2_T(l-i),$$

where $\hat{\sigma}^2_T(j) := a^2_{T+j}$ for $j \leq 0$.

3.4 How to estimate the ARCH(m) model?

Several likelihood functions are commonly used in ARCH estimation, depending on the distributional assumptions of ϵ_t. Suppose $a_1, a_2, \ldots, a_T$ are observed. Given $a_{t-1}, a_{t-2}, \ldots, a_{t-m}$, we have that σ_t is known. Under the assumption that $\{\epsilon_t\}_{t\geq 0}$ is Gaussian distributed, the conditional likelihood is given by

$$L\left(a_{m+1:T}\middle|\alpha_0, \alpha_{1:m}, a_{1:m}\right) = \prod_{t=m+1}^{T} \frac{1}{\sqrt{2\pi\sigma^2_t}} e^{-\frac{a^2_t}{2\sigma^2_t}}$$

Hence, the conditional log-likelihood is given by

$$\begin{aligned}\log L\left(a_{m+1:T}\middle|\alpha_0,\alpha_{1:m},a_{1:m}\right) &= -\frac{T-m}{2}\log\left(2\pi\right)\\ &\quad -\frac{1}{2}\sum_{t=m+1}^{T}\log(\sigma_t^2)-\sum_{t=m+1}^{T}\frac{1}{2\sigma_t^2}a_t^2.\end{aligned}$$

We can then use a nonlinear optimization algorithm to find the best parameter estimates.

4 Pro and Cons of ARCH model

We have already discussed the pros of the ARCH model, which are the modeling of conditional heteroscedasticity, i.e. time varying variance that is able to capture clustering effects. However, the following are important cons of ARCH models.

- The model assumes that positive and negative shocks have the same effects on volatility, because the conditional variance depends on the square of the previous shocks. It practice, it is widely known that the price of a financial asset reacts differently to positive and negative shocks.
- The model is restrictive. To have finite fourth-order moment, we require that $\alpha_1^2 < 1/3$ in a ARCH(1). This constraint becomes complicated in higher order ARCH models. In practice, it limits the ability to capture excess kurtosis.
- It provides a mechanical way to describe the behavior of the conditional variance. But, it is silent on what causes such behavior to occur.

5 GARCH Model

The ARCH model is simple, but often requires many parameters to adequately describe the volatility process of an asset return. Bollerslev (1986) proposes a useful extension known as the generalized ARCH model, also referred to as the GARCH model. Let $a_t = r_t - \mathbb{E}\left[r_t\middle|\mathcal{F}_{t-1}\right]$, where r_t is a log-return series. Then the innovation a_t at time t follows a $GARCH(m,s)$ model if

$$a_t = \sigma_t\epsilon_t, \qquad \sigma_t^2 = \alpha_0 + \sum_{i=1}^{m}\alpha_i a_{t-i}^2 + \sum_{j=1}^{s}\beta_j\sigma_{t-j}^2, \tag{1}$$

where $\{\epsilon_t\}_{t\geq 0}$ is a sequence of i.i.d. random variables with zero mean and variance 1, $\alpha_0 > 0$, $\alpha_i \geq 0$, and $\beta_j \geq 0$. As in the case of the ARCH model, it is often assumed that ϵ_t follows a standard normal or a generalized Student's t-distribution. If $s = 0$, the model reduces to a pure $ARCH(m)$ model. The α_j and β_j coefficients are referred to as ARCH and GARCH parameters, respectively. To understand properties of GARCH models, it is informative to use the following representation. Let $\eta_t = a_t^2 - \sigma_t^2$ so that $\sigma_t^2 = a_t^2 - \eta_t$. By plugging $\sigma_{t-j}^2 = a_{t-j}^2 - \eta_{t-j}$, $j = 1,\ldots,s$ into Eq. (1), we may rewrite the GARCH model as

$$a_t^2 - \eta_t = \alpha_0 + \sum_{i=1}^{m}\alpha_i a_{t-i}^2 + \sum_{j=1}^{s}\beta_j a_{t-j}^2 - \sum_{j=1}^{s}\beta_j\eta_{t-j},$$

or, equivalently, as

$$a_t^2 = \alpha_0 + \sum_{i=1}^{\max(m,s)} (\alpha_i + \beta_i) a_{t-i}^2 + \eta_t - \sum_{j=1}^{s} \beta_j \eta_{t-j},$$

where it is understood that $\alpha_i = 0$ for $i > m$ and $\beta_j = 0$ for $j > s$. It can be easily checked that $\{\eta_t\}_{t\geq 0}$ has zero mean and it is a serially uncorrelated time series, i.e., $cov(\eta_t, \eta_{t-j}) = 0$ for $j \geq 1$ (Notice, however, that $\{\eta_t\}$ is not a i.i.d. sequence in general). In light of this, a GARCH model may be viewed as an application of the ARMA model to the squared series a_t^2. Using the expression for the unconditional mean of an ARMA model, we obtain that

$$\mathbb{E}\left[a_t^2\right] = \frac{\alpha_0}{1 - \sum_{i=1}^{\max(m,s)} (\alpha_i + \beta_i)},$$

provided that the denominator is positive. Hence, we need to impose the restriction $\sum_{i=1}^{\max(m,s)} (\alpha_i + \beta_i) < 1$ for a $GARCH(m, s)$ model to be well defined.

The strengths and weaknesses of GARCH models can be seen easily by focusing on the simplest GARCH(1,1) model where

$$\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \qquad 0 \leq \alpha_1, \beta_1 \leq 1, (\alpha_1 + \beta_1) < 1. \tag{2}$$

First, a large a_{t-1}^2 or σ_{t-1}^2 gives rise to a large σ_t^2. This means that a large a_{t-1}^2 tends to be followed by another large a_t^2, generating again the well known behavior of volatility clustering in financial time series. Second, it can be shown (as in the case of the ARCH model) that the kurtosis of the distribution of a_t is larger than 3, i.e.,

$$\frac{\mathbb{E}\left[a_t^4\right]}{\left(\mathbb{E}\left[a_t^2\right]\right)^2} = \frac{3\left(1 - (\alpha_1 + \beta_1)^2\right)}{1 - (\alpha_1 + \beta_1)^2 - 2\alpha_1^2} > 3.$$

Consequently, the tail distribution of a GARCH(1,1) process is heavier than that of a normal distribution. Third, the model provides a simple parametric function to describe the volatility evolution.

Forecasts of a GARCH model can be obtained using methods similar to those of an ARMA model. Consider the GARCH(1,1) model and assume the forecast origin is T. Then

$$\hat{\sigma}_T^2(1) = \alpha_0 + \alpha_1 a_T^2 + \beta_1 \sigma_T^2$$

For multistep-ahead forecasts, we use $a_t^2 = \sigma_t^2 \epsilon_t^2$, and rewrite the volatility equation in (2) as

$$\sigma_{t+1}^2 = \alpha_0 + (\alpha_1 + \beta_1)\sigma_t^2 + \alpha_1 \sigma_t^2 (\epsilon_t^2 - 1).$$

When $t = T + 1$, the equation becomes

$$\sigma_{T+2}^2 = \alpha_0 + (\alpha_1 + \beta_1)\sigma_{T+1}^2 + \alpha_1 \sigma_{T+1}^2 (\epsilon_{T+1}^2 - 1).$$

This leads to

$$\hat{\sigma}_T^2(2) = \alpha_0 + (\alpha_1 + \beta_1)\hat{\sigma}_T^2(1).$$

In general, we have

$$\hat{\sigma}_T^2(l) = \alpha_0 + (\alpha_1 + \beta_1)\hat{\sigma}_T^2(l-1), \qquad l > 1. \tag{3}$$

This result is exactly the same as that of an ARMA(1,1) model with AR polynomial $1-(\alpha_1+\beta_1)B$, where B is the back-shift operator. By repeated substitutions in Eq. (3), we obtain that the l-step ahead forecast is

$$\hat{\sigma}_T^2(l) = \frac{\alpha_0\left[1-(\alpha_1+\beta_1)^{l-1}\right]}{1-\alpha_1-\beta_1} + (\alpha_1+\beta_1)^{l-1}\hat{\sigma}_T^2(1).$$

As $l \to \infty$, $\hat{\sigma}_T^2(l) \to \frac{\alpha_0}{1-\alpha_1-\beta_1}$. Hence, the multi-steps ahead volatility forecast of a GARCH(1,1) model converges to the unconditional variance of a_t as the forecast horizon increases to infinity, provided that $Var(a_t)$ exists.

The conditional maximum likelihood method continues to apply to estimate the model. There is an enormous literature on GARCH models: see, for instance, Bollerslev, Chou and Kroner (1992), Bollerslev, Engle and Nelson (1994).

The model suffers from the same weaknesses as ARCH. For instance, it responds equally to positive and negative shocks. Studies of high-frequency financial time series suggest that the tail behavior of GARCH models remains too short even if we use standardized Student t's distribution for the innovation terms.

Remark 5.1. *It can be shown that a $GARCH(m,s) = ARCH(\infty)$. This means that the conditional variance depends on an infinite number of past squared innovations. Hence, the GARCH is more parsimonious than an ARCH model, because it uses just a few parameters to achieve what the ARCH model would do with an infinite number of parameters. In other words, the conditional variance may have a long memory with low values of m and s. This argument is very similar to how an ARMA model is more parsimonious than an AR or MA model, because we would need a $MA(\infty)$ or $AR(\infty)$ representation to achieve what an ARMA model does with a few parameters. In general, with low values of m and s, one can generate complex features, whereas you would need a higher value of m to generate the same features using an ARCH model. This means that you would be able to generate, say, an ACF of squared returns for the GARCH model with a richer pattern than if using an ARCH model with a similar number of parameters. Lower order GARCH models are used in most applications, say GARCH(1,1), GARCH(2,1) and GARCH(1,2) models.*

6 Relation to Stochastic Volatility Models

A popular stochastic volatility model is

$$a_t = \sigma_t \epsilon_t, \qquad \log(\sigma_t^2) = \alpha_0 + \sum_{i=1}^{m} \alpha_i \log(\sigma_{t-i}^2) + \nu_t,$$

where $\{\epsilon_t\}$ is a white noise sequence, α_0, α_i's are non-negative constants, $\{\nu_t\}$ is a white noise sequence with variance σ_ν^2, $\{\epsilon_t\}$ and $\{\nu_t\}$ can be correlated. Indeed, stock asset returns and volatility are negatively correlated. Such a negative correlation goes under the name of leverage effect, and we have discussed this in the context of ARCH models. We refer to Melino and Turnbull (1990), Harvey, Ruiz, and Shephard (1994) and Jacquier, Polson, and Rossi (1994), for a study of these models.

There is an important difference between ARCH and stochastic volatility models. In ARCH, the conditional variance σ_t^2 is a deterministic function of past a_t^2's. In stochastic volatility models, the conditional variance involves a new innovation term ν_t.

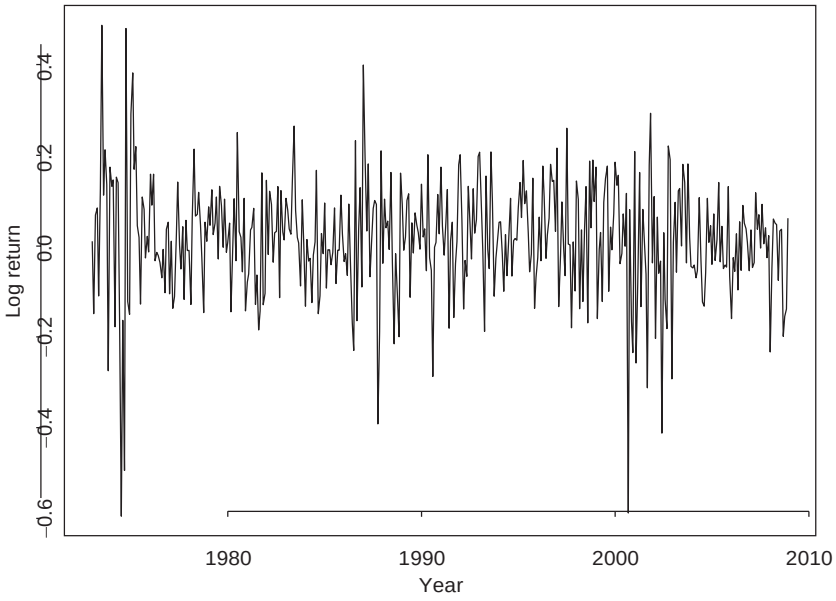


Figure 3.1 Time plot of monthly log returns of Intel stock from January 1973 to December 2008.

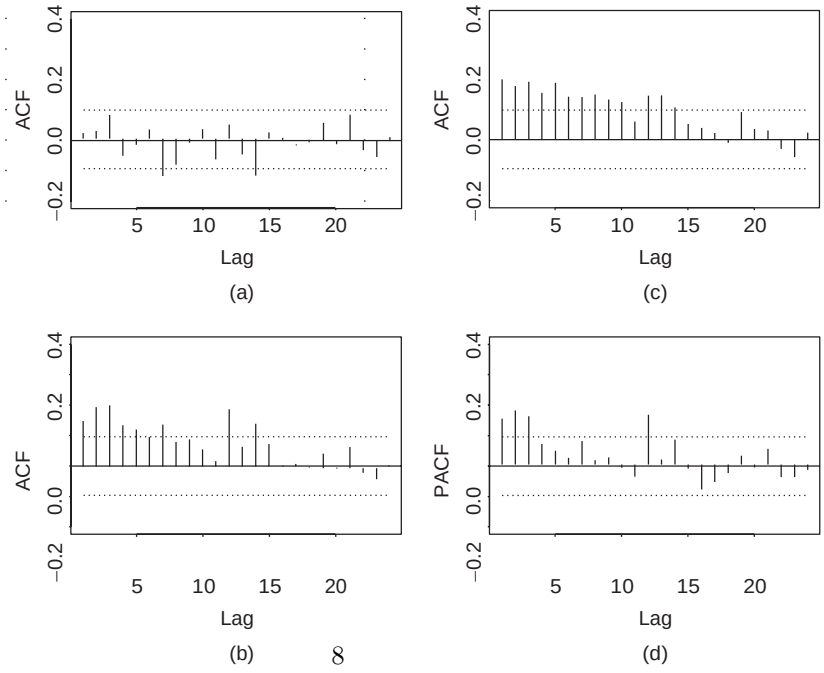


Figure 3.2 Sample ACF and PACF of various functions of monthly log stock returns of Intel Corporation from January 1973 to December 2008: (a) ACF of the log returns, (b) ACF of the squared log returns, (c) ACF of the absolute log returns, and (d) PACF of the squared log returns.

Lecture 27: Kalman Filter (Optional)

1 Announcement

The material covered in these lecture notes will not be included in the final exam.

2 State Space Models

State space model provides a flexible approach in time series analysis. They are used when the quantity of interest is not perfectly observed, but we only get access to a noisy version of it. Consider the univariate time series y_t satisfying

$$\begin{aligned} y_t &= \mu_t + e_t \\ \mu_{t+1} &= \mu_t + \eta_t, \end{aligned} \tag{1}$$

where $\{e_t\}$ and $\{\eta_t\}$ are two independent Gaussian white noise series and $t = 1, \ldots, T$. The initial value μ_1 may be given or follows a known distribution, and is independent of $\{e_t\}$ and $\{\eta_t\}$ for $t > 0$. In the equation above, μ_t is a pure random walk with initial value μ_1, and y_t is an observed version of μ_t with added noise e_t. In the literature μ_t is referred to as the trend of the series, which is *not* directly observable, and y_t is the *observed* data with observational noise e_t. Notice that the dynamic dependence of y_t is governed by that of μ_t, because $\{e_t\}$ is not serially correlated.

The model specified in Eq. (1) can be used to analyze the realized volatility of an asset price (see Example 11.1 in Tsay's book for details). We think of μ_t as the *true* log-volatility of the asset price, and y_t as the logarithm of realized volatility. The true log-volatility is not directly observed, but evolves over time following a random-walk model. The observed volatility y_t instead is constructed from high-frequency transactions data. For instance, the daily realized volatility used can be obtained using the sum of squares of intraday 10-minute log returns measured in percentage (typically, no overnight returns or the first 10-minute intra-day returns are used, and we refer to Chapter 3 of Tsay's book for more details about realized volatility). This construction is subject to market microstructure noises. Such noise captures a variety of frictions inherent in the trading process: bid–ask bounces, discreteness of price changes, differences in trade sizes or informational content of price changes, gradual response of prices to a block trade, inventory control effects, etc... The standard deviation of e_t denotes the scale used to measure the impact of market microstructure noises.

The model in Eq. (1) is a linear Gaussian state-space model. The variable μ_t is called the state of the system at time t and is not directly observed. The equation for y_t provides the link between the data y_t and the (hidden) state μ_t and is called the *observation equation* with measurement error e_t. The equation for μ_t governs the time evolution of the state variable and is the *state equation* (or *state transition equation*) with innovation η_t.

2.1 Statistical Inference

The objective of the analysis is to infer properties of the state μ_t from the data $\{y_t\ t = 1, \ldots, T\}$ and the model. There are three types of inferences that are commonly discussed in the literature: *filtering, prediction*, and *smoothing*. Let $\mathcal{F}_t = \{y_1, \ldots, y_t\}$ be the information available at time t (inclusive), and assume that the model is known, including all parameters. Briefly, the three types of inferences can be described as follows:

- *Filtering*. Filtering means to recover the state variable μ_t given $\mathcal{F}_t$, that is to remove the measurement errors from data.
- *Prediction*. Prediction means to forecast μ_{t+h} or y_{t+h} for $h > 0$ given $\mathcal{F}_t$, where t is the forecast origin.
- *Smoothing*. Smoothing is to estimate μ_t given $\mathcal{F}_T$, where $T > t$.

For a more precise description of the inference, let us first introduce some notation. Let $\mu_{t|j} = \mathbb{E}[\mu_t|\mathcal{F}_j]$ and $\Sigma_{t|j} = Var(\mu_t|\mathcal{F}_j)$ be, respectively, the conditional mean and conditional variance of the actual (hidden) state μ_t given $\mathcal{F}_j$. Similarly, let us use $y_{t|j}$ to denote the conditional mean of y_t given $\mathcal{F}_j$. Moreover, let $v_t = y_t - y_{t|t-1}$ be the 1-step ahead forecast error for the observation, and $V_t = Var(v_t|\mathcal{F}_{t-1})$ the corresponding conditional variance. From Eq. (1), we obtain that

$$y_{t|t-1} := \mathbb{E}[y_t|\mathcal{F}_{t-1}] = \mathbb{E}[\mu_t + e_t|\mathcal{F}_{t-1}] = \mathbb{E}[\mu_t|\mathcal{F}_{t-1}] = \mu_{t|t-1},$$

where above we have used that $\{e_t\}$ is a white noise series. Consequently, we obtain that

$$v_t = y_t - y_{t|t-1} = y_t - \mu_{t|t-1},$$

and

$$\begin{aligned} V_t &= Var(y_t - \mu_{t|t-1}|\mathcal{F}_{t-1}) = Var(\mu_t + e_t - \mu_{t|t-1}|\mathcal{F}_{t-1}) \\ &= Var(\mu_t - \mu_{t|t-1}|\mathcal{F}_{t-1}) + Var(e_t|\mathcal{F}_{t-1}) = \Sigma_{t|t-1} + \sigma_e^2, \end{aligned} \tag{2}$$

where we have used that, conditional on $\mathcal{F}_{t-1}$, e_t is independent of $\mu_t - \mu_{t|t-1}$. It can also be easily seen that

$$\begin{aligned} \mathbb{E}[v_t] &= \mathbb{E}\left[\mathbb{E}\left[v_t|\mathcal{F}_{t-1}\right]\right] = \mathbb{E}\left[\mathbb{E}\left[y_t - y_{t|t-1}|\mathcal{F}_{t-1}\right]\right] = \mathbb{E}\left[y_{t|t-1} - y_{t|t-1}\right] = 0 \\ Cov(v_t, y_j) &= \mathbb{E}[v_t y_j] = \mathbb{E}\left[\mathbb{E}\left[v_t y_j|\mathcal{F}_{t-1}\right]\right] = \mathbb{E}\left[y_j \mathbb{E}\left[v_t|\mathcal{F}_{t-1}\right]\right] = 0, \qquad j < t. \end{aligned}$$

Hence, the 1-step ahead forecast error v_t is uncorrelated (and hence independent given that the system is Gaussian) with y_j, for $j < t$.

We next state, without proof, a theorem in multivariate statistics that will be useful to derive the Kalman filter equations

Theorem 2.1. *Suppose that $\mathbf{x}, \mathbf{y}, \mathbf{z}$ are three random vectors such that their joint distribution is multivariate normal. Let μ_x, μ_y, μ_z denote, respectively, the mean of the vector x,y, and z. Assume that the covariance matrices of $\mathbf{x}, \mathbf{y}, \mathbf{z}$ denoted, respectively, by $\mathbf{\Sigma_{xx}}, \mathbf{\Sigma_{yy}}, \mathbf{\Sigma_{zz}}$ are nonsingular, and that $Cov(\mathbf{y}, \mathbf{z}) := \mathbf{\Sigma_{yz}} = 0$. Then,*

- $\mathbb{E}[\mathbf{x}|\mathbf{y}] = \mu_{\mathbf{x}} + \mathbf{\Sigma_{xy}}\mathbf{\Sigma_{yy}^{-1}}(\mathbf{y} - \mu_{\mathbf{y}})$

- $Var\left[\mathbf{x}|\mathbf{y}\right] = \mathbf{\Sigma_{xx}} - \mathbf{\Sigma_{xy}}\mathbf{\Sigma_{yy}^{-1}}\mathbf{\Sigma_{yx}}$
- $\mathbb{E}\left[\mathbf{x}|\mathbf{y},\mathbf{z}\right] = \mathbb{E}\left[\mathbf{x}|\mathbf{y}\right] + \mathbf{\Sigma_{xz}}\mathbf{\Sigma_{zz}^{-1}}(\mathbf{z} - \mu_{\mathbf{z}})$
- $Var\left[\mathbf{x}|\mathbf{y},\mathbf{z}\right] = Var(\mathbf{x}|\mathbf{y}) - \mathbf{\Sigma_{xz}}\mathbf{\Sigma_{zz}^{-1}}\mathbf{\Sigma_{zx}}$.

3 Kalman Filter

The goal of the Kalman filter is to update the knowledge of the hidden state variable μ_t *recursively* once a new observation y_t becomes available. In other words, knowing the conditional distribution of μ_t given $\mathcal{F}_{t-1}$ and the new observation y_t, we want to obtain the conditional distribution of μ_t given $\mathcal{F}_t$. The fundamental insight by Kalman was in exploiting the Gaussian property of the system, to obtaining a formula which uses the predicted mean $\mu_{t|t-1}$ and the new observation y_t to obtain $\mu_{t|t}$, and the predicted variance $\Sigma_{t|t-1}$ to obtain the updated variance $\Sigma_{t|t}$. Hence, these computations are recursively done and *do not require, at any point in time, to revisit the whole history of observations* $\mathcal{F}_t = \{y_1, y_2, \ldots, y_t\}$ to make the updates. Because $\mathcal{F}_t = \{\mathcal{F}_{t-1}, v_t\}$, giving y_t and $\mathcal{F}_{t-1}$ is equivalent to the information set generated by v_t and $\mathcal{F}_{t-1}$. Hence, it suffices to consider the joint conditional distribution of $(\mu_t, v_t)^\top$ given $\mathcal{F}_{t-1}$ before we apply the theorem above.

The conditional distribution of v_t given $\mathcal{F}_{t-1}$ is Gaussian with zero mean and variance given by V_t. Moreover, the distribution of μ_t given $\mathcal{F}_{t-1}$ is Gaussian with mean $\mu_{t|t-1}$ and variance $\Sigma_{t|t-1}$. Next, we need to compute the conditional covariance between μ_t and v_t given $\mathcal{F}_{t-1}$. Using the definition, we obtain

$$\begin{aligned}
Cov(\mu_t, v_t|\mathcal{F}_{t-1}) &= \mathbb{E}\left[\mu_t v_t\middle|\mathcal{F}_{t-1}\right] = \mathbb{E}\left[\mu_t(y_t - \mu_{t|t-1})\middle|\mathcal{F}_{t-1}\right] \\
&= \mathbb{E}\left[\mu_t(\mu_t + e_t - \mu_{t|t-1})\middle|\mathcal{F}_{t-1}\right] \\
&= \mathbb{E}\left[\mu_t(\mu_t - \mu_{t|t-1})\middle|\mathcal{F}_{t-1}\right] + \mathbb{E}\left[\mu_t e_t\middle|\mathcal{F}_{t-1}\right] \\
&= \mathbb{E}\left[\mu_t(\mu_t - \mu_{t|t-1})\middle|\mathcal{F}_{t-1}\right] - \underbrace{\mathbb{E}\left[\mu_{t|t-1}(\mu_t - \mu_{t|t-1})\middle|\mathcal{F}_{t-1}\right]}_{=0} + \mathbb{E}\left[\mu_t e_t\middle|\mathcal{F}_{t-1}\right] \\
&= Var(\mu_t|\mathcal{F}_{t-1}) + \mathbb{E}[e_t]\mu_{t|t-1} \\
&= Var(\mu_t|\mathcal{F}_{t-1})
\end{aligned}$$

We can now apply the theorem conditional on $\mathcal{F}_{t-1}$, by setting $x = \mu_t$, $y = (y_1, y_2, \ldots, y_{t-1})$, and $z = v_t$ in the theorem above, to obtain

$$\begin{aligned}
\mu_{t|t} &= \mu_{t|t-1} + \frac{\Sigma_{t|t-1}}{V_t} v_t := \mu_{t|t-1} + K_t v_t \\
\Sigma_{t|t} &= \Sigma_{t|t-1} - \frac{\Sigma_{t|t-1}^2}{V_t} := \Sigma_{t|t-1}(1 - K_t),
\end{aligned}$$

where $K_t = \frac{\Sigma_{t|t-1}}{V_t}$ is referred to as the Kalman gain and captures the information content of the observation. It quantifies the contribution of the new shock v_t to the estimation of the hidden state variable μ_t. This also corresponds to the regression coefficient of the hidden state μ_t on v_t. The Kalman gain is the fundamental insight of Kalman, and provides a means to update the

filtered estimate of μ_t by correcting the predicted value $\mu_{t|t-1}$ with the information power of the new observation (the information power is quantified by the Kalman gain).

We can then use our filtered estimate of μ_t given $\mathcal{F}_t$ to predict μ_{t+1}. This is achieved via

$$\begin{aligned}\mu_{t+1|t} &= \mathbb{E}\left[\mu_t + \eta_t \middle| \mathcal{F}_t\right] = \mathbb{E}\left[\mu_t \middle| \mathcal{F}_t\right] = \mu_{t|t} \\ \Sigma_{t+1|t} &= Var(\mu_{t+1}\big|\mathcal{F}_t) = Var(\mu_t\big|\mathcal{F}_t) + Var(\eta_t) = \Sigma_{t|t} + \sigma_\eta^2.\end{aligned}$$

Above, we have used the fact that the state noise term η_t is independent of the filtration $\mathcal{F}_t$, which only contains information about the noise terms $\eta_1, \eta_2, \ldots, \eta_{t-1}$. Once the new data y_{t+1} is observed, we can then repeat the procedure above to update our knowledge of μ_{t+1}. Putting altogether, we obtain the Kalman filter equations. Assume that μ_1 is initially distributed as a Gaussian distribution with mean $\mu_{1|0}$ and variance $\Sigma_{1|0}$, where both parameters are given. We then have the following sufficient statistics for the incremental computation of the filter:

$$\begin{aligned}v_t &= y_t - \mu_{t|t-1} \\ V_t &= \Sigma_{t|t-1} + \sigma_e^2 \\ K_t &= \frac{\Sigma_{t|t-1}}{V_t} \\ \mu_{t+1|t} &= \mu_{t|t-1} + K_t v_t \\ \Sigma_{t+1|t} &= \Sigma_{t|t-1}(1 - K_t) + \sigma_\eta^2,\end{aligned}$$

for $t = 1, \ldots, T$.

4 Properties of the Forecast Error and Maximum Likelihood estimation (not included in the final exam)

The properties of the forecast errors v_t are very useful in many applications. Given the initial values $\Sigma_{1|0}$ and $\mu_{1|0}$ which are independent of y_t, the Kalman filter enables to compute v_t recursively as a linear function of $y_1, y_2, \ldots, y_t$. Specifically, we have

$$v_1 = y_1 - \mu_{1|0} \tag{3}$$

$$v_2 = y_2 - \mu_{2|1} = y_2 - \mu_{1|0} - K_1(y_1 - \mu_{1|0}) \tag{4}$$

$$v_3 = y_3 - \mu_{3|2} = y_3 - \mu_{1|0} - K_2(y_2 - \mu_{1|0}) - K_1(1 - K_2)(y_1 - \mu_{1|0}), \tag{5}$$

and so on. We can write this transformation in a matrix form as follows

$$\mathbf{v} = \mathbf{K}(\mathbf{y} - \mu_{\mathbf{1|0}}\mathbf{1_T}), \tag{6}$$

where $\mathbf{v} = (v_1, v_2, \ldots, v_T)'$, $\mathbf{y} = (y_1, y_2, \ldots, y_T)'$, $\mathbf{1}$ is the T-dimensional vector of ones, and $\mathbf{K}$ is a lower triangular matrix

$$\mathbf{K} = \begin{bmatrix} 1 & 0 & 0 & \ldots & 0 \\ k_{21} & 1 & 0 & \ldots & 0 \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ k_{31} & k_{32} & 0 & \ldots & 0 \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ k_{T1} & k_{T2} & k_{T3} & \ldots & 1, \end{bmatrix}$$

where $k_{i,i-1} = -K_{i-1}$ and $k_{ij} = -(1-K_{i-1})(1-K_{i-2})\dots(1-K_{j+1})K_j$ for $i = 2,\dots,T$, and $j = 1,\dots,i-2$. Notice that the Kalman gain does not depend on $\mu_{1|0}$ or the data $\{y_1, y_2, \dots, y_t\}$. It depends on $\Sigma_{1|0}, \sigma_e^2$ and σ_η^2. This transformation facilitates the maximum likelihood estimation procedure. Importantly, $v_1, v_2, \dots, v_t$ are mutually independent under the Gaussian assumption. To see this, consider the joint probability density function of the data

$$p(y_1, y_2, \dots, y_T) = p(y_1) \prod_{j=2}^{T} p(y_j | \mathcal{F}_{j-1})$$

Eq. (6) indicates that the determination of the Jacobian $\mathbf{K}$ of the transformation is one (the determinant of a lower triangular matrix is the product of its diagonal entries). Moreover, $\mu_{1|0}$ is given, thus $p(v_1) = p(y_1)$. Hence, the joint probability density function of $\mathbf{v}$ is given by

$$p(\mathbf{v}) = p(y_1)\Pi_{j=2}^{T} p(y_j | \mathcal{F}_{j-1}) = p(v_1) \prod_{j=2}^{T} p(v_j | \mathcal{F}_{j-1}) = \prod_{j=1}^{T} p(v_j),$$

where we have used the property shown in last lecture that $Cov(v_t, y_j) = 0$ for all $j < t$. This independent property simplifies the maximum likelihood estimation procedure of the variance of the noise terms. First,

$$p(y_1, y_2, \dots, y_T | \sigma_e, \sigma_\eta) = p(y_1) \prod_{t=2}^{T} p(v_t | \mathcal{F}_{t-1}, \sigma_e, \sigma_\eta),$$

where y_1 is Gaussian with mean $\mu_{1|0}$ and variance V_1, and v_t is Gaussian with mean 0 and variance V_t. Assuming $\mu_{1|0}$ and $\Sigma_{1|0}$ to be known and taking logarithms, we have

$$\log L(\sigma_e, \sigma_\eta) = -\frac{T}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^{T} \log(V_t) + \frac{v_t^2}{V_t}.$$

The above likelihood function can be evaluated recursively via the Kalman filter. Many software packages perform state space estimation via a Kalman filter algorithm.